PSP (PYTHON FOR PROBLEM SOLVING) — IA-1 EXAM KIT

Exam: Wednesday 29 October 2026 (per student; no separate date announcement was in the Slack dumps).

Generated: 2026-10-28 · **Batch:** S1 B'26-CS B (PSP B)

Sources: portal export `PSP-questions.md` (100 assignments, 9 quizzes, 14 lecture windows, Slack sheets, 47-question workbook — captured 23 Oct), live portal pages captured 28 Oct (47/67 assignments solved, 3/9 quizzes), Slack dumps `academic-announcement-26.md` / `psp-26.md` (to 23 Oct), syllabus announcement 14 Oct.

Everything below is from those sources or standard Python — nothing invented.

PART 0 — EXAM CARD (read first, 3 minutes)

Syllabus — VERBATIM from the 14 Oct announcement ("Syllabus for IA-1 PSP")

#	Announcement line
L1	Lecture - 1 Outline (Flowcharts)
L2	Lecture - 2 Outline (Variable, Datatype, Typecasting)
L3	Lecture - 3 Outline (Input, Arithmetic operator)
L4	Lecture - 4 Outline (if else)
L5	Lecture - 5 Outline (if else ladder)
L6	Lecture - 6 Outline (Nested if else)
L7	Lecture - 7 Outline (Introduction to Function)
L8	Lecture - 8 Outline (Function with if else)
L9	Lecture - 9 Outline (Logical operators, Function with nested if-else)
L10	Lecture - 10 Outline (Function with recursive thinking)
L11	Lecture - 11 Outline (For loop)
L12	Lecture - 12 Outline (While loop)

What is confirmed / not confirmed

- **Confirmed:** syllabus = L1–L12 above; your real state: 47/67 assignments solved, 3/9 quizzes, Contest 2 partially solved.
- **NOT in any dump (through 23 Oct):** an announced paper pattern, marks, duration, or venue for PSP IA-1. Students asked for the "finalized IA1 syllabus and paper pattern for all 5 subjects" on 17 Oct; only syllabuses were posted.
- Sibling facts (real, from the same channel): Math-1 IA-1 was announced as **a written paper**; AP IA-1 ran as a **S-VYASA written paper** (5×3M + 3×5M choice, 2 h, no calculator); PSP and other contests ran **on Athena, 9:00–11:00 AM**.

Protocol — be ready for BOTH shapes of paper

If it is a written paper (most likely):

1. Write full programs, not pseudocode: correct indentation, `:` after `if` / `for` / `while` / `def`, `print(...)` with commas where asked.

2. Read-aloud rule: before writing, say the exact output string in quotes. Spelling/case is marks (`"Yes"` not `"YES"` unless the statement says YES).

3. Order: do L11/L12 loop-writing and L4–L6 conditionals first (they are your 100 recent assignments), functions last (they take longest to write cleanly).

4. Every loop question: run the sample input on paper once, line by line, before you move on.

If it is an MCQ/online test on Athena:

1. Trace-before-answer: write the variable table for loops (`i`, `x`) — that is how every quiz question in this course was written.

2. Check whether the paper states negative marking. If it does not, attempt everything; if it does, still attempt anything you can half-solve — but follow the printed rule exactly.

3. Never leave a blank on a written section.

Desk order (both shapes): 60-second morning card (Part 6) → loop patterns (Part 2, types 9–13) → condition ladder patterns (types 4–6) → solutions you have not re-read (Part 3).

PART 1 — CONCEPT COURSE, L1 → L12

L1 — Flowcharts

- **What:** a diagram of the program: oval = start/end, parallelogram = input/output, rectangle = process, diamond = decision, arrows = flow.
- **Exam form:** "draw a flowchart for…" or "what shape is the decision box?" — answer: **diamond** for decisions, **rectangle** for assignments, **parallelogram** for input/output, **oval** for terminal.
- **Gotcha:** arrows must show a loop going back, and every diamond has exactly two labelled exits (Yes/No or True/False).
- **Tested in:** PSP Contest 1 syllabus announcement (Flowchart was question 1 of the syllabus list, 1 Sep).

L2 — Variables, datatypes, typecasting

```
x = 10            # int

pi = 3.14         # float

name = "Arush"    # str

flag = True       # bool

n = int("42")     # str -> int (typecast)

f = float("3.5")  # str -> float

s = str(99)       # int -> str

type(x)           # <class 'int'>
```

- **Gotchas:** `input()` ALWAYS returns a string — `n = input()` then `n + 1` crashes (`str + int`). Integer division `//`, remainder `%`. `int("3.7")` crashes — use `float` first.

- **Tested in:** quizzes "Print function…Python Variable…Type() function" (2 Sep), assignments Swap Two Variables, Find The Square, Integer Type.

L3 — Input and arithmetic operators

```
a = int(input())

b = int(input())

print(a + b, a - b, a * b, a / b, a // b, a % b, a ** 2)
```

- **Gotchas:** precedence is `**` > `* / // %` > `+ -`. `/` always float (`7/2 = 3.5`), `//` floors. Two inputs on ONE line: `l, r = map(int, input().split())`.
- **Tested in:** Swap Two Variables (60/60), Rectangle Area, Energy Consumption, Celsius to Fahrenheit, Convert Minute to Hour, Compound Interest, Contest-2 "Find the remainder".

L4 — if–else

```
if condition:

    # runs when True

else:

    # runs when False
```

- **Gotchas:** the `:` is required; indentation defines the block (4 spaces); `==` compares, `=` assigns; condition can be a comparison directly (`if n % 2 == 0:`).
- **Tested in:** Even or Odd Checker, Who is Greater?, Weather Report, AM_PM, Identify Square or Rectangle, Profit or Loss; quiz "Conditional Statements, Python if-else" (9 Sep, you scored 8/8).

L5 — if–elif–else ladder

```
if score >= 90:

    grade = "A+"

elif score >= 75:

    grade = "A"

elif score >= 60:

    grade = "B"

else:

    grade = "C"
```

- **Gotchas:** only ONE branch runs; `elif` not `else if`; order matters — put the strictest/highest condition first; last branch can be bare `else`.
- **Tested in:** Grade Calculator, The Number Detective, FizzBuzz II, Income Tax, Electricity Bill, Triangle Type, Library fine, Salary, BMI; Contest-2 "The Gatekeeper" and "Function Based Calculator"; quiz "If-elif-else ladder" (21 Sep, 0/8 — unattempted).

L6 — Nested if–else

```
if a + b > c:

    if a + c > b:

        if b + c > a:

            print("Triangle valid")

        else:

            print("Invalid")

else:

    print("Invalid")
```

- **Gotchas:** every inner `if` needs its own `:` and indentation level; `and` / `or` (L9) can flatten nesting — but the syllabus wants you to be able to read BOTH forms.
- **Tested in:** Can It Be a Triangle?, Leap Year Checker, Roller Coaster, Accommodation Decision, Find the Smallest/Biggest; quizzes "Python Function, Nested if-else" (23 Sep, 0/14).

L7 — Introduction to functions

```
def average(a, b, c):

    return (a + b + c) / 3



print(average(80, 90, 70))
```

- **What:** `def name(params): body` — define once, call anywhere after it. `return` sends a value out; `print` only shows it.
- **Gotchas:** the function does NOTHING until called; `return` with no value gives `None`; a function without `return` cannot be used in an expression.
- **Tested in:** Calculate Percentage Using Function, Percentage-1/-2, Average of Three; workbook Units 1–7.

L8 — Function with if–else

```
def get_grade(m1, m2, m3):

    if m1 < 40 or m2 < 40 or m3 < 40:

        return "Fail"

    pct = (m1 + m2 + m3) / 300 * 100

    if pct >= 90:

        return "A+"

    # ...
```

- **Gotchas:** `return` exits immediately — no code after it runs; decide return TYPE (string vs number) from the statement, not from how you tested.
- **Tested in:** Subject Grades, Smart Conditional Calculator, Grade Report (Contest 2 — in Part 3); quiz "Python Function…Arguments and parameter, Return" (28 Sep, 8/8 — your best score).

L9 — Logical operators + functions with nested if–else

```
if a > b and b > c:      # both must be True

if x == 1 or x == 2:     # either True

if not flag:             # flips True/False
```

- **Gotchas:** `and` / `or` / `not` (words — never `&&` / `||`); short-circuit: `x != 0 and 10/x > 1` never divides by zero when `x == 0`; precedence: `not` > `and` > `or`.
- **Tested in:** Voter Eligibility, Greater or Equal, NST Library, Museum Booking, Greater of Four; Contest-2 Gatekeeper; assignment sets due 5 Oct (logical operators).

L10 — Function with recursive thinking

```
def factorial(n):

    if n <= 1:        # BASE CASE — without it: infinite recursion / crash

        return 1

    return n * factorial(n - 1)



def sum_to(n):

    if n <= 0:

        return 0

    return n + sum_to(n - 1)
```

- **What:** a function that calls itself with a SMALLER input, until a base case stops it.
- **Gotchas:** every recursion needs a base case; each call multiplies stack frames (deep recursion hits the limit); trace with a call tree on paper.
- **Tested in:** workbook Unit 12 (Recursion Made Easy), sum_to / sign / factorial patterns in Slack sheets; "Factorial-Loops" and "Is It Prime?" carry Recursion in their topic tags (Part 3 has loop versions — both forms examinable).

L11 — For loop

```
for i in range(1, n + 1):   # 1..n inclusive

    print(i * i)

for i in range(n):          # 0..n-1

    ...

for ch in "abc":            # iterate a string

    ...
```

- **Gotchas:** `range(a, b)` stops at `b-1`; `range(n)` starts at 0; accumulating needs `total = 0` BEFORE the loop (a `total = 0` inside prints the last value only); print-on-same-line vs newline: `print(x, end="")`.
- **Tested in:** Sum 1 to N, Multiplication Table, Print all divisors, Multiples of 3 and 5, Squares I/II, Count Divisible by 3 or 5, Fibonacci, Odd Factors, First Number Divisibly by 7 (Part 3).

L12 — While loop

```
n = int(input())

while n > 0:

    digit = n % 10      # take last digit

    n //= 10            # drop last digit
```

- **What:** repeats while the condition stays True — use when you don't know how many iterations (digit counting, reversing, searching).
- **Gotchas:** you must CHANGE something the condition depends on, or it loops forever; `break` exits the loop, `continue` skips to the next check; `while True` + `break` is legal.
- **Tested in:** Count the Digits, Reverse a Number, Digit Sum, First+Last Digit, Even/Odd digit counts, Largest/Smallest Digit, Armstrong, Palindrome number, All Digits Same, Check if Rare, Perfect Number (all in Part 3); quizzes "While loop" (12 Oct, 0/10), "Control flow, Break, Continue" (16 Oct, 0/8).

PART 2 — PATTERN LIBRARY (the 13 archetypes every question in this course is made of)

Copy the skeleton, fill the blanks. Question names are the real ones from your portal.

TYPE 1 — Print pattern (L1/print) — *Mirror Triangle, Inverted Triangle, Numbers Triangle, Hollow Square, No Time to Newline*

```
n = 5

for i in range(1, n + 1):

    print("*" * i)                 # left-aligned triangle

for i in range(n, 0, -1):

    print("*" * i)                 # inverted

for i in range(1, n + 1):

    print(" " * (n - i) + "*" * i)  # mirror (right-aligned)
```

TYPE 2 — Input + conversion formula (L3) — *Celsius→Fahrenheit, Rectangle Area, Energy, Minute→Hour, Compound Interest, BMI*

```
c = float(input())

print(c * 9 / 5 + 32)          # exact string as sample says

h, m = divmod(total_min, 60)   # or: h = total_min // 60; m = total_min % 60
```

TYPE 3 — Two-branch classify (L4) — *Even/Odd, Profit/Loss, Weather, AM/PM, Square/Rectangle*

```
if n % 2 == 0:

    print("Even")

else:

    print("Odd")
```

TYPE 4 — Ladder with thresholds (L5) — *Grade Calculator, Income Tax, Electricity Bill, BMI, Library Fine, Salary*

```
if x < a:        ...

elif x < b:      ...   # always ascending thresholds

elif x < c:      ...

else:            ...
```

Rule: write the boundaries in strictly increasing order; test the boundary value itself (exactly 75, exactly 40).

TYPE 5 — Nested validation (L6) — *Can It Be a Triangle, Leap Year, Roller Coaster, Accommodation*

```
if a + b > c and a + c > b and b + c > a:   # or 3 nested ifs (syllabus L6 style)
    print("YES")
else:
    print("NO")


# leap year: divisible by 4, but century years only if divisible by 400
if year % 400 == 0 or (year % 4 == 0 and year % 100 != 0):
    print("Leap")
```

TYPE 6 — Function + decision (L8) — *Subject Grades, Smart Calculator, Grade Report*

```
def solve(...):
    if <fail condition>:
        return <fail value>
    # compute
    return <answer>
# no input()/print() in the function unless the statement says so
```

TYPE 7 — Logical eligibility (L9) — *Voter Eligibility, Gatekeeper, NST Library, Museum Booking*

```
if u == "admin" and p == "password123":
    print("Login successful")
elif u == "admin" or p == "password123":   # exactly one matches
    print("Check username or password")
else:
    print("Invalid credentials")
```

TYPE 8 — Recursion (L10) — *sum_to, factorial, sign, recursion in workbook Unit 12*

```
def f(n):

    if n < 2:            # base case FIRST

        return ...

    return ... f(n - 1)
```

TYPE 9 — For-loop accumulate (L11) — *Sum 1..N, divisors, multiples, table, squares, Fibonacci*

```
total = 0                    # BEFORE the loop — classic bug if inside

for i in range(1, n + 1):

    if condition(i):

        total += i

print(total)
```

TYPE 10 — For-loop build a string/list, then print (L11) — *Squares II, Count Divisible by 3 or 5, Fibonacci, Odd Factors*

```
out = []

for i in range(1, n + 1):

    if cond(i):

        out.append(str(i))

print(" ".join(out))          # one line, space-separated
```

TYPE 11 — While digit-extraction (L12) — *Count digits, Reverse, Digit sum, First+last, Even/odd counts, Largest digit, Add 1 to each digit, Armstrong*

```
n = int(input())

while n > 0:

    d = n % 10        # last digit

    ...               # use d

    n //= 10          # drop it
```

One template, 8 questions. Variants: count (`c += 1`), sum (`s += d`), reverse (`r = r * 10 + d`), max (`if d > m: m = d`), first digit (strip until `n < 10`), add-1 (`nd = (d + 1) % 10` — 9 becomes 0).

TYPE 12 — While + break early-exit (L12, L9) — *First Number Divisibly by 7, Check if All Digits are Same, Check if Rare*

```
found = None

for i in range(l, l + 7):        # at most 7 candidates — even if R is 10^18

    if i % 7 == 0:

        found = i

        break

print(found if found else "NOT FOUND")
```

TYPE 13 — Number theory with loop (L11/L12) — *Is It Prime?, Perfect Number, Palindrome number*

```
# prime: check only up to sqrt(n)

i = 2

while i * i <= n:

    if n % i == 0:

        break

    i += 1

# palindrome: reverse and compare, or compare str to str[::-1] if strings allowed
```

PART 3 — SOLVED COUNTER (your 23 unsolved questions, all solved and sample-verified)

53/53 sample checks pass (every sample case in every statement was run — verified 28 Oct). Statements quoted verbatim where it matters (exact output strings, constraints).

C-1. The Gatekeeper — Contest 2, medium, 0/80 (if-elif-else ladder)

> "The correct username is `admin`. The correct password is `password123`." Print `Login successful` if both correct; `Check username or password` if exactly one correct; `Invalid credentials` if both wrong.

```
u = input().strip()

p = input().strip()

ok_u = (u == "admin")

ok_p = (p == "password123")

if ok_u and ok_p:

    print("Login successful")

elif ok_u or ok_p:

    print("Check username or password")

else:

    print("Invalid credentials")
```

Samples: `admin/password123 → Login successful`; `admin/wrongpass → Check username or password`. *(Note: the statement's em-dashes are punctuation — samples show plain strings.)*

C-2. Newton and Rounding — Contest 2, hard, 0/120 (arithmetic operators)

> Round n to nearest multiple of 10. Ends with 0 → same. Last digit 1–4 → lower. 6–9 → higher. **Last digit 5 → always LOWER.**

```
n = int(input())

d = n % 10

if d <= 5:

    print(n - d)        # 0,1,2,3,4,5 all go DOWN (5 is the trap)

else:

    print(n + (10 - d))
```

Samples: `5 → 0` · `113 → 110` · `2359 → 2360`. Also checked: `0 → 0`, `10 → 10`.

C-3. Grade Report — Contest 2, medium, 0/80 (functions)

> Complete `def get_grade(mark1, mark2, mark3)`. Any subject < 40 → return `Fail`. Else percentage of 300: ≥90 → `A+`, 75–<90 → `A`, 60–<75 → `B`, 40–<60 → `C`. Return only — no input/print.

```
def get_grade(mark1, mark2, mark3):

    if mark1 < 40 or mark2 < 40 or mark3 < 40:

        return "Fail"

    pct = (mark1 + mark2 + mark3) / 300 * 100

    if pct >= 90:

        return "A+"

    if pct >= 75:

        return "A"

    if pct >= 60:

        return "B"

    return "C"
```

Samples: `95,92,88 → A+` (91.7%) · `80,90,70 → A` (80%) · `75,35,80 → Fail`.

C-4. Count Numbers Divisible by 3 or 5 — 0/20 (for loop)

> Print all i in 1..N divisible by 3 or 5 **but not both** (XOR). `3≤N≤100`. Sample: `15 → 3 5 6 9 10 12`.

```
n = int(input())

out = []

for i in range(1, n + 1):

    if (i % 3 == 0) ^ (i % 5 == 0):     # XOR: exactly one of them

        out.append(str(i))

print(" ".join(out))
```

C-5. Squares from 1 to N (Part I) — 0/20 (for loop)

> Print squares $1^2..N^2$, **one per line**. `1≤N≤20`.

```
n = int(input())

for i in range(1, n + 1):

    print(i * i)
```

C-6. Squares from 1 to N (Part II) — 0/20 (for loop)

> Same squares **on one line, space-separated**. Sample: `5 → 1 4 9 16 25`.

```
n = int(input())

print(" ".join(str(i * i) for i in range(1, n + 1)))
```

(If generators are banned in your environment: build a list with `append` first — TYPE 10.)

C-7. Factorial-Loops — 0/20 (loops/recursion)

> N! for `0≤N≤20`. `0! = 1`.

```
n = int(input())

f = 1

for i in range(2, n + 1):

    f *= i

print(f)
```

Samples: `5 → 120`, `10 → 3628800`, `0 → 1`.

C-8. Check if a Number is a Palindrome — 0/20 (while/recursion)

> "reads same backwards and forwards". Sample: `54145 → Yes`, `12345 → No`.

```
n = int(input())

s = str(n)

print("Yes" if s == s[::-1] else "No")
```

Loop version (if strings blocked, as in "Count the Digits"): reverse the number with TYPE 11 and compare `r == original`.

C-9. Digit Sum using Loop — 0/20 (function only)

> `sumOfDigits(n)` must **return** the sum. `28393 → 25`.

```
def sumOfDigits(n):

    s = 0

    while n > 0:

        s += n % 10

        n //= 10

    return s
```

C-10. Count the Digits — 0/20 (while)

> **Blocked in backend:** string literals, list literals, `str()` / `list()` conversions, indexing/iteration/string methods on `input()`. So pure arithmetic. `28232 → 5`.

```
n = int(input())

c = 0

while n > 0:

    c += 1

    n //= 10

print(c)
```

C-11. Reverse a Number — 0/20 (while)

> `536446 → 644635`; `100 → 1` (no trailing zeros kept).

```
n = int(input())

r = 0

while n > 0:

    r = r * 10 + n % 10

    n //= 10

print(r)
```

C-12. Sum of First and Last Digits — 0/20 (while)

> `14368 → 9` (1 + 8).

```
n = int(input())

last = n % 10

while n >= 10:

    n //= 10          # n is now the first digit

print(n + last)
```

Edge (no sample given): single digit 7 → first = last = 7 → literal reading gives **14**. Keep the literal formula; if the judge disagrees on that one edge, it is the only unverifiable case in this file.

C-13. Find the Count of Even and Odd Digits — 0/20 (while)

> `295892 → 3 3` (even count first, then odd, space-separated).

```
n = int(input())

e = o = 0

while n > 0:

    d = n % 10

    if d % 2 == 0:

        e += 1

    else:

        o += 1

    n //= 10

print(e, o)
```

C-14. Find the Largest Digit in a Number — 0/40 (while)

> `2839 → 9`. **`0≤N≤10^9` — N can be 0** (sample-free edge: print 0).

```
n = int(input())

m = 0

while n > 0:

    d = n % 10

    if d > m:

        m = d

    n //= 10

print(m)
```

(Works for N = 0: loop never runs, prints 0.)

C-15. Is It Prime? — 0/20 (loops)

> Print exactly `YES` / `NO`. `1` is NOT prime. `7 → YES`, `10 → NO`.

```
n = int(input())

if n < 2:

    print("NO")

else:

    i = 2

    prime = True

    while i * i <= n:

        if n % i == 0:

            prime = False

            break

        i += 1

    print("YES" if prime else "NO")
```

C-16. Add 1 to Each Digit — 0/20 (loops)

> "adding 1 to digit 9 makes it 0". `765392 → 876403`.

```
n = int(input())

res = 0

mul = 1

while n > 0:

    d = (n % 10 + 1) % 10     # 9 -> 0, others -> +1

    res += d * mul

    mul *= 10

    n //= 10

print(res)
```

Edge: `9 → 0` (leading zeros vanish when printed as an integer — matches "print an integer").

C-17. Fibonacci Series of a given input — 0/20 (loops)

> First N terms, space-separated, starting **0 1 1 2 3**. `5 → 0 1 1 2 3`. `1≤N≤20`.

```
n = int(input())

a, b = 0, 1

out = []

for _ in range(n):

    out.append(str(a))

    a, b = b, a + b

print(" ".join(out))
```

C-18. Find Odd Factors — 0/20 (for loop)

> All odd factors of N in increasing order. `21 → 1 3 7 21`. `1≤N≤10^5`.

```
n = int(input())

out = []

for i in range(1, n + 1):

    if n % i == 0 and i % 2 == 1:

        out.append(str(i))

print(" ".join(out))
```

(1 is always included — the sample confirms it.)

C-19. Check if Even Sum is Divisible by 2 — 0/20 (while/loop)

> Sum all EVEN numbers in [L, R], print `Yes` if that sum % 2 == 0 else `No`. **`1≤L≤R≤10^18` — you CANNOT loop the range.**

```
l, r = map(int, input().split())

first = l if l % 2 == 0 else l + 1

last  = r if r % 2 == 0 else r - 1

if first > last:

    s = 0

else:

    k = (last - first) // 2 + 1          # count of evens

    s = k * (first + last) // 2          # arithmetic-series sum

print("Yes" if s % 2 == 0 else "No")
```

Sample: `23 29 → Yes` (24+26+28 = 78). Also checked: no evens in range → sum 0 → `Yes`; L=R=10^18 → `Yes`.

C-20. Check for Perfect Number — 0/20 (while)

> Sum of PROPER divisors (excluding N) equals N. `6 → True` (1+2+3). Print `True` / `False`.

```
n = int(input())

s = 0

i = 1

while i * i <= n:

    if n % i == 0:

        if i != n:

            s += i

        other = n // i

        if other != i and other != n:

            s += other

    i += 1

print("True" if s == n else "False")
```

Checked: `6 → True`, `28 → True`, `10 → False`.

C-21. First Number Divisibly by 7 — 0/20 (for + break)

> Smallest multiple of 7 in [L, R], else `NOT FOUND`. **`R` up to 10^18 — but only 7 candidates can matter.**

```
l, r = map(int, input().split())

ans = "NOT FOUND"

for i in range(l, min(r, l + 6) + 1):

    if i % 7 == 0:

        ans = str(i)

        break

print(ans)
```

Samples: `23 29 → 28` · `23 27 → NOT FOUND` · exact `7 7 → 7` · huge range ending 10^18 → `999999999999999999` (10^18 ≡ 1 mod 7).

C-22. Check if All Digits are Same — 0/40 (while + break)

> `77777 → Yes`, `22888 → No`.

```
n = int(input())

first = n % 10

ok = True

while n > 0:

    if n % 10 != first:

        ok = False

        break

    n //= 10

print("Yes" if ok else "No")
```

C-23. Check if Rare — 0/40 (while + break)

> "rare if all of its digits are divisible by X". Input `N X`. `28 2 → True`.

```
n, x = map(int, input().split())

ok = True

while n > 0:

    if (n % 10) % x != 0:

        ok = False

        break

    n //= 10

print("True" if ok else "False")
```

Checked: `28 3 → False`.

PART 4 — DRILL INDEX (what exists, what is done, what is left)

4.1 Your 9 quizzes (portal state, captured 28 Oct) — 3/9 solved

Quiz	Window / date	Score	Revise with
Print function, Comments, Variable, Primitive, `type()`	2 Sep	**0/12**	L2 · assignments: Swap Two Variables, Find The Square, Integer Type
Input, Arithmetic operators, precedence	7 Sep	2/26	L3 · Rectangle Area, Energy, C→F, Minute→Hour
Conditional Statements, if-else	9 Sep	**8/8 ✓**	L4 — your strongest area
If-elif-else ladder	21 Sep	**0/8**	L5 · Grade Calculator, Income Tax, Electricity Bill, Triangle Type

Quiz	Window / date	Score	Revise with
Python Function, Nested if-else	23 Sep	**0/14**	L6–L7 · Can It Be a Triangle?, Leap Year, Subject Grades
Function, Nested if-else, Arguments, Return	28 Sep	**8/8 ✓**	L8 — keep this level
Iteration, Recursion, Range	12 Oct	**0/14**	L10–L11 · TYPE 8 + TYPE 9/10 patterns
While loop	12 Oct	**0/10**	L12 · all of Part 3 C-9…C-14
Control flow, Loops, Break, Continue	16 Oct	**0/8**	L11–L12 · TYPE 12 (Div7, All Same, Rare)

Question text for the unattempted quizzes is NOT in any file — the portal only shows questions after starting a test (forbidden by the rules of this project). Revise by pattern, not by past paper.

4.2 Assignment sets solved (the 47 you have done — quick re-read list by date)

- **8 Sep (L2):** Print Mirror Right Angle Triangle · Print Inverted Triangle · Print Numbers Right-Angled Triangle · Print Hollow Square · No Time to Newline *(all 10/20)*
- **12 Sep (L3):** Swap Two Variables (60/60) · Find The Square · Rectangle Area Calculator · Calculate and Print Age · Energy Consumption · C→F Conversion · Minute→Hour Format
- **14 Sep (L4):** Average of Three · Compound Interest –1 · Even or Odd · Integer Type · Who is Greater? · Weather Report · AM_PM · Identify Square or Rectangle
- **21 Sep (L4–L5):** Absolute Difference · Profit or Loss · Voter Eligibility II · Grade Calculator · The Number Detective · Greater or Equal
- **26 Sep (L5):** FizzBuzz II · Income Tax (40/40) · Electricity Bill (partial 30/60) · Triangle Type · All Three Equal · Find the Biggest! · Library fine · Calculate salary · Check Profit or Loss · BMI-1
- **28 Sep (L6):** Can It Be a Triangle? · Leap Year (40/40) · Roller Coaster · Accommodation Decision · Tax by Age & Income · Find the Smallest!
- **3–5 Oct (L7–L9):** Percentage Using Function · Percentage-1/-2 · Greater/Smaller ×2 · Greater of Three · Subject Grades · Bridge Between Villages · Electricity Bill-2 · Smart Conditional Calculator · C→F-2 · Leap Year or Not · Great Dragon · Function Based Calculator · NST Library (60/60) · Movie Mania · Will Ramandeep Sir Fly –2 · Bus Route · Museum Booking · Greater of Four
- **9 Oct (Contest 2):** Find the remainder (40/40) · The Frame (40/40) · Function Based Calculator (80/80) · Calculate Final Price (40/80) — **backlog: The Gatekeeper, Newton and Rounding, Grade Report → Part 3**
- **10–19 Oct (L10–L12):** Palindrome with Digit Validation · NewForces Ratings (60/60) · Sum in Range L..R · Multiplication Table · Print divisors · Multiples of 3 and 5 · Find The Even Numbers · Sum 1 to N · Sum of Even/Odd Digits (40/40) · Smallest Digit (40/40) · Armstrong II (40/40)
- **Not yet done (→ Part 3 has all):** Count Divisible 3 or 5 · Squares I & II · Factorial-Loops · Palindrome number · Digit Sum fn · Count the Digits · Reverse · First+Last · Even/Odd counts · Largest Digit · Is It Prime? · Add 1 to Each Digit · Fibonacci · Odd Factors · Even Sum %2 · Perfect Number · Div7 · All Digits Same · Check if Rare

4.3 Workbook — "Python Functions: Practice Workbook" (Slack, 19 Oct; 47 questions, 150 marks, 13 units)

Units: 1 What is a function (5) · 2 Define & call (10) · 3 `return` (10) · 4 `print` (5) · 5 print vs return (15) · 6 Take a value out (10) · 7 Send values in (10) · 8 Decisions/loops inside functions (15) · 9 Functions calling functions + scope (10) · 10 Functions working together (15) · 11 Function reference (15) · 12 Recursion made easy (15) · 13 Final challenge (15).

Full text of all 47 questions is in `export/PSP-questions.md` (lines ~8010–8630) and its PDF. If you do only one thing from this list: Units 5, 8, 12 (print-vs-return, loops-in-functions, recursion) — they map exactly to L7–L10.

4.4 Slack standing directives (real, from #psp-26)

- **2 questions/day minimum** from the two Codeforces sheets (If-Else & Math, 63 problems; Loops, 34 problems) — make your own copy (File → Make a copy) before ticking.
- **12-problem Codeforces code list** shared 22 Oct; **Ashwin's answer key posted 27 Oct** in-channel: `977 A, 978 B, 988 A, 996 A, 731 A, 832 A, 1800 A, 910 A, 115 A, 977 B, 1113 A, 948 A` (A/B suffix = which problem of each pair).
- Workbook PDF link is in the export's sources section.

4.5 What the exam can and cannot draw from

Source	In-file?	Count
Assignment questions (full text + status)	yes	100 (47 solved by you)
Contest 2 full text	yes	7
In-class lecture questions (per lecture)	yes	14 windows
Quiz schedules + your scores	yes	9
Quiz question text (unattempted)	**no** — needs Start Test (forbidden)	4 quizzes
Quiz-contest MCQs (PSP Contest 1, Wifi Demo, Mock Test)	**no** — same reason	4
Workbook	yes, full text	47
Codeforces sheets	titles + ratings only, no statements	97 + 12

PART 5 — GAPS (honest)

1. **Paper pattern, marks, duration, venue for PSP IA-1: not announced** anywhere in the dumps through 23 Oct, and Slack could not be re-harvested tonight (session expired — QR re-login needed). If a classmate has a newer screenshot of the pattern, trust it over this file's assumptions.

2. **Quiz question text for the 4 unattempted quizzes does not exist in any capture** — starting the test to read them would consume an attempt, which this project never does.

3. **Single-digit edge in C-12 (First+Last digit)** has no sample on the portal — literal answer 14, flagged above.

4. **Exact sample output formatting** (trailing spaces, blank lines) was normalised to whitespace-equal comparison during verification; when in doubt on the paper, copy the sample's spacing character-for-character.

5. Physically absent data cannot be manufactured: no past IA-1 PSP paper exists in any source (IA-1 has not happened yet).

PART 6 — 60-SECOND MORNING CARD (screenshot this)

1. **`:` after `if` / `elif` / `else` / `for` / `while` / `def`. 4-space indent.**

2. **`input()` is a string** → `int(input())` unless told otherwise; two values: `a, b = map(int, input().split())`.

3. **Digit loop = one template:** `while n > 0: d = n % 10; n //= 10` → count/sum/reverse/max/first/odd-even/add-1 all from it.

4. **Accumulator starts BEFORE the loop** (`total = 0` outside).

5. **`range(1, n+1)` is 1..n; `range(n)` is 0..n-1; `range(n, 0, -1)` counts down.**

6. **Print exact strings:** `Yes/No`, `True/False`, `YES/NO` — copy from the statement, never invent case.

7. **Prime:** loop to `i*i <= n`; **1 is not prime**; **0! = 1**.

8. **Round-half-down:** last digit 5 goes DOWN (Newton and Rounding).
9. **Function-only questions:** write `def` + `return`, no `input()` / `print()` unless asked.
10. **Huge ranges (10^18):** never loop them — arithmetic formula (even-sum) or ≤7 candidates (div-by-7).
11. **Recursion:** base case FIRST, then `return … f(n-1)`.
12. **break** exits, **continue** skips — diamond flowchart = decision, parallelogram = input/output.

Sources: `export/PSP-questions.md` (23 Oct portal export), live portal pages captured 28 Oct (PSP B: 47/67 assignments, 3/9 quizzes), `materials/slack-questions/academic-announcement-26.md` (syllabus, 14 Oct), `psp-26.md`, `general-announcements-26.md`, `student-progression-batch-b.md`. All 23 solutions in Part 3 were executed against their portal sample cases before publication: 53/53 pass.

EVERY PSP question (portal, till now)

Generated: 2026-11-06T17:45:52 by `scrape.py --export PSP`

For the AI tutor: teach these in due-date order. Status and XP are the student's real portal state. Question text is included in a code block wherever the tracker captured it; questions without text are listed at the end.

Print function in python, Comments in python, Python Variable, Python Primitive ... [Post Class] — due 2026-09-08 21:00

Print Mirror Right Angle Triangle — 10/20 XP — SOLVED — deadline 2026-09-08 21:00

Topics: Print function in python

URL: https://my.newtonschool.co/playground/code/sc5dsftdd5ak

Back

Print Mirror Right Angle Triangle

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Print Mirror Right Angle Triangle

Easy

10/20

Time Limit: 2, Memory Limit: 256000

Write a Python program to print the mirror image of a right-angled triangle using asterisks (*). The triangle should have 5 rows. Each row should:

Contain stars in increasing order (1 in the first row, 2 in the second, and so on)

Include spaces on the left side so that the stars are right-aligned.

See the example below for clarity.

Input

You do not need to take any input.

Output

Print the mirror right-angled triangle as shown in the example.

Example

Output

```
    *
    *

   **
   **

  ***
  ***
```

```
 ****
 ****


*****
*****
```

Print Inverted Triangle — 10/20 XP — SOLVED — deadline 2026-09-08 21:00

Topics: Print function in python

URL: https://my.newtonschool.co/playground/code/yby7ujad18f7

Back

Print Inverted Triangle

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Print Inverted Triangle

Easy

10/20

Time Limit: 2, Memory Limit: 256000

Write a Python program to print an inverted right-angled triangle using asterisks (*).

The triangle should have 5 rows.

The first row should contain 5 stars.

Each subsequent row should have one less star than the previous row.

Input

You do not need to take any input.

Output

Print the inverted right-angled triangle as shown in the example.

Example

Output:

*****

*****

****

****

```
***
***


**
**


*
*
```

Print Numbers Right-Angled Triangle — 10/20 XP — SOLVED — deadline 2026-09-08 21:00

Topics: Print function in python

URL: https://my.newtonschool.co/playground/code/ts5h6mew954q

Back

Print Numbers Right-Angled Triangle

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Print Numbers Right-Angled Triangle

Easy

10/20

Time Limit: 2, Memory Limit: 256000

Write a Python program to print a right-angled triangle made of numbers.

The triangle should have 5 rows.

The first row should contain the number 1.

The second row should contain 1 2.

The third row should contain 1 2 3, and so on.

See the example below for clarity.

Note: There should be a space between each number in a row.

Input

You do not need to take any input.

Output

Print the right-angled triangle of numbers as shown in the example.

Example

Output

```
1
```

```
1


1


2
1 2


1


2


3
1 2 3


1


2


3


4
1 2 3 4


1


2


3


4


5
1 2 3 4 5
```

Print Hollow Square — 10/20 XP — SOLVED — deadline 2026-09-08 21:00

Topics: Print function in python

URL: https://my.newtonschool.co/playground/code/0ku6oqeccpe3

Back

Print Hollow Square

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Print Hollow Square

Easy

10/20

Time Limit: 2, Memory Limit: 256000

Write a Python program to print a hollow square pattern using asterisks (*).

The square should have 5 rows and 5 columns.

The stars should form the border of the square.

The inside of the square should be empty (filled with spaces).

See the example below for clarity.

Input

You do not need to take any input.

Output

Print the hollow square pattern as shown in the example.

Example

Output

```
*****
*****


*   *
*   *


*   *
```

```
*   *


*   *

*   *


*****

*****
```

No Time to Newline — 10/20 XP — SOLVED — deadline 2026-09-08 21:00

Topics: Endline and formatting

URL: https://my.newtonschool.co/playground/code/rlpkygvrjcpc

Back

No Time to Newline

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

No Time to Newline

Easy

10/20

Time Limit: 2, Memory Limit: 256000

Write a program to print the below quote in a single line.

Quote

1.Do not watch the clock

2.Do what it does keep going

Your task is to print only the line number 1 of the quote.

Line number 2 of the quote will automatically appear in the output section when you run your code.

Input

You do not need to take any input. You need to print only line number 1 of the Quote.

Output

The output will be the quote in a single line as per below example.

```
Example

Input:



Output:

Do not watch the clock Do what it does keep going
```

Python Input, Input() function, Input() for different datatypes, Python operator ... [In Class] — due 2026-09-12 16:20

Swap Two Variables — 60/60 XP — SOLVED — deadline 2026-09-12 16:20

Topics: Input() function

URL: https://my.newtonschool.co/playground/code/il2gc70cns3y

Back

Swap Two Variables

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Swap Two Variables

Hard

2x

60/60

Time Limit: 2, Memory Limit: 256000

Write a program to swap two integers a and b, using a third variable.

Input

User Task

You need not handle input or output. Your task is to swap the numbers stored in variables a and b.

Custom Input

The first line of input contains an integer a.

The second and final line contains an integer b.

Output

You need not to output anything.

Custom Output

The driver code will print two space-separated numbers representing the swapped values.

Constraints

0

≤

a

≤

1

0

3

0≤a≤10

3

0

≤

b

≤

1

0

3

0≤b≤10

3

Example

Input

10

20

Output

20 10

Explanation

Given the two integer numbers 10 and 20, after the swap the numbers will be 20 and 10.

Hence, the output is 20 10.

Python Input, Python operators, Input() function, Input() for different datatype ... [Post Class] — due 2026-09-12 17:00

Find The Square — 20/20 XP — SOLVED — deadline 2026-09-12 17:00

Topics: Python Arithmetic operators

URL: https://my.newtonschool.co/playground/code/tjil7yari3ay

Back

Find The Square

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Find The Square

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Write a program to take a number X as input and print its square.

Input

The first and only line of input contains a single integer X.

Output

Print an integer, representing the square of X.

Constraints

0

≤

X

≤

5

0≤X≤5

The input number will always be an integer.

Example

Input

4

Output

16

Explanation

```
The given number is 4, and its square will be 4 * 4 = 16.
```

Rectangle Area Calculator — 20/20 XP — SOLVED — deadline 2026-09-12 17:00

Topics: Python Arithmetic operators

URL: https://my.newtonschool.co/playground/code/987hc9looztb

Back

Rectangle Area Calculator

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Rectangle Area Calculator

Easy

2x

20/20

Time Limit: 2, Memory Limit: 128000

Write a program that calculates the area of a rectangle given its length and width.

The program should prompt the user to enter the length and width of the rectangle, and then calculate and print the area.

Input

The first line of input contains a single integer, representing the length of rectangle.

The second line of input contains a single integer, representing the width of rectangle.

Output

Print the answer in following format:

"Area of the rectangle is: [calculated area]"

Constraints

1

≤

L

e

n

g

t

h

,

W

i

d

t

h

≤

100

1≤Length,Width≤100

Length and width will be positive numbers.

Example

Input

5

3

Output

Area of the rectangle is: 15

Calculate and Print Age — 20/20 XP — SOLVED — deadline 2026-09-12 17:00

Topics: Python Arithmetic operators

URL: https://my.newtonschool.co/playground/code/dpqdzb37tiuh

Back

Calculate and Print Age

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Calculate and Print Age

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Write a Python program that takes two integers as input: the birth year and the current year.

Your task is to calculate the age by subtracting the birth year from the current year, and print the result.

Input

The first line of input contains a single integer, the birth year.

The second line also contains a single integer, the current year.

Output

Print an integer, the calculated age.

Constraints

1950 <= Birth Year <= Current Year <= 2025

Example

Input

2000

2024

Output

24

Explanation

Brith Year = 2000

Current Year = 2024

```
Age = 2024 - 2000 = 24


Input

2006

2025

Output

19

Explanation

Brith Year = 2006

Current Year = 2025

Age = 2025 - 2006 = 19
```

Energy Consumption Calculation — 20/20 XP — SOLVED — deadline 2026-09-12 17:00

Topics: Python Arithmetic operators

URL: https://my.newtonschool.co/playground/code/k8718dpafb9h

Back

Energy Consumption Calculation

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Energy Consumption Calculation

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given two integers: units representing the total electricity consumed and cost representing the cost of each unit.

Your task is to write a Python program that calculates the total electricity cost and prints the result.

Input

The first and only line of input contains two space-separated integers, representing units and cost respectively.

Output

Print an integer representing the total electricity cost.

Constraints

0

≤

u

n

i

t

s

≤

1000

0≤units≤1000

1

≤

c

o

s

t

≤

1000

1≤cost≤1000

Example

Input

20 10

Output

200

Explanation

Since 20 units of electricity was consumed and the cost per unit is 10. The total cost would be 200.

Celsius to Fahrenheit Conversion — 20/20 XP — SOLVED — deadline 2026-09-12 17:00

Topics: Precedence and Associativity

URL: https://my.newtonschool.co/playground/code/haikjdpg2f4e

Back

Celsius to Fahrenheit Conversion – 2

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Celsius to Fahrenheit Conversion – 2

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given an integer, representing the temperature in celsius.

Your task is to convert it to Fahrenheit, using the below given formula:

Fahrenheit = (Celsius × 9 / 5) + 32

You must complete the function celsius_to_fahrenheit(celsius) that returns the equivalent temperature in Fahrenheit.

Function Signature:

def celsius_to_fahrenheit(celsius):

It takes a single parameter, celsius, the temperature in Celsius.

Input

User Task

As this is a functional problem, there is no need to handle input from the user. Your task is to complete the function celsius_to_fahrenheit(celsius), which takes single parameter, celsius, the temperature in celsius.

Custom Input

The first and only line of input contains a single integer, celsius, representing the temperature in celsius.

Output

Return the corresponding temperature in Fahrenheit (as a float).

```
Constraints

−100 ≤ celsius ≤ 100

Example

Input

0

Output

32.0

Explanation

Input Celsius = 0

Fahrenheit = (0 × 9/5) + 32 = 0 + 32 = 32.0

So the output is 32.0.


Input

37

Output

98.6

Explanation

Input Celsius = 37

Fahrenheit = (37 × 9/5) + 32 = 98.6

So the output is 98.6.
```

Convert Minute to Hour and Minute Format — 20/20 XP — SOLVED — deadline 2026-09-12 17:00

Topics: Python Arithmetic operators

URL: https://my.newtonschool.co/playground/code/1cgxc2by8uqq

Back

Convert Minute to Hour and Minute Format

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Convert Minute to Hour and Minute Format

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Write a program to convert total minutes into hours and remaining minutes.

Print the hours and minutes in the below given format:

Time: {x} hour(s) and {y} minute(s)

where {x} will be replaced by hours and {y} by minutes.

Input

The first and only line of input contains a single integer, representing the total minutes.

Output

Print the hours and minutes in the below given format:

Time: {x} hour(s) and {y} minute(s)

where {x} will be replaced by hours and {y} by minutes.

Constraints

0 ≤ total minutes ≤ 103

Example

Input

135

Output

Time: 2 hour(s) and 15 minute(s)

```
Input

60

Output

Time: 1 hour(s) and 0 minute(s)
```

PSP LAb [In Class] — due 2026-09-14 10:52

Average of Three — 20/20 XP — SOLVED — deadline 2026-09-14 10:52

Topics: Input() for different datatypes

URL: https://my.newtonschool.co/playground/code/4kb4el5cnb5z

Back

Average of Three

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Average of Three

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Write a program that takes three numbers as input and prints their average.

Note: The result should be rounded to two decimal places.

Input

The first and only line of input contains three space-separated numbers (integer or float).

Output

Print the average of the three numbers rounded to two decimal places.

Constraints

−

100

≤

a

,

b

,

c

≤

100

−100≤a,b,c≤100

```
Each number will be a valid integer or float.

Example

Input

10 20 30

Output

20.0

Explanation

Average = (10 + 20 + 30) / 3 = 60 / 3 = 20.0



Input

3.3 1.2 1

Output

1.83

Explanation

To calculate the average, we add all the values:

3.3 + 1.2 + 1 = 5.5



Now, divide the total by the number of values:

5.5 ÷ 3 = 1.8333333333333333



Since the question requires the answer to be rounded to two decimal places, we round
1.8333333333333333 to 1.83.



Final Answer: 1.83
```

Compound Interest Calculation in Python -1 — 20/20 XP — SOLVED — deadline 2026-09-14 10:52

Topics: Python Arithmetic operators

URL: https://my.newtonschool.co/playground/code/jlzmjfrkte1s

Back

Compound Interest Calculation in Python -1

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Compound Interest Calculation in Python -1

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Write a program that calculates the compound interest.

The program should take principal amount (P), rate of interest (r), and time period (t) as inputs.
Use the formula:

$$
C
$$

$$
I
$$

=

(

$$
P
$$

*

(

1

+

(

$$
r
$$

/

100

)

)

t

)

−

P

CI=(P∗(1+(r/100))

t

)−P

Note: Display the result rounded to one decimal point.

Input

The first and only line of input contains three space-separated numbers, representing the principal amount (P), rate of interest (r), and time period (t) respectively

Note: The principal amount (P) and the time period (t) are provided as integers, while the rate of interest (r) is given as a floating-point number.

Output

The program should print the calculated compound interest.

Note: Display the result rounded to one decimal point.

Constraints

1

≤

P

≤

1

0

3

1≤P≤10

3

1

≤

r

≤

1

0

2

1≤r≤10

2

1

≤

t

≤

10

1≤t≤10

Example

Input:

1000 5.0 3

Output:

157.6

Explanation:

principal amount: 1000

rate of interest (as a percentage): 5

the time period (in years): 3

Compound interest: 157.63, calculated using formula CI = (P * (1 + (r / 100)) ^ t) - P

Conditional Statements, Python if- else [In Class] — due 2026-09-14 16:54

Even or Odd Checker — 20/20 XP — SOLVED — deadline 2026-09-14 16:54

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/nh90a28g9oyx

Back

Even or Odd Checker

Total XP

6,643

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Even or Odd Checker

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Write a Python program that checks if a given number is even or odd.

Input

The input contains a single integer number

Output

The program should print a single word either – Even or Odd according to number.

Example

Input:

7

Output:

Odd

Input:

6

Output:

```
Even
```

Integer Type — 20/20 XP — SOLVED — deadline 2026-09-14 16:54

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/5k61xy0qc0u0

Back

Integer Type

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Integer Type

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given a non-zero integer num. Your task is to check whether the number is Positive or Negative.

Input

The first and only line contains an integer num.

Output

Print exactly one word, either Positive or Negative, depending on the value of the input.

Constraints

−

30

≤

n

u

m

≤

30

−30≤num≤30

n

u

m

```
≠

0

num

□

=0

Example

Input:

5

Output:

Positive


Input:

-9

Output:

Negative
```

Who is Greater? — 20/20 XP — SOLVED — deadline 2026-09-14 16:54

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/606bjkv3zvdm

Back

Who is Greater?

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Who is Greater?

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given two distinct integers a and b. Your task is to compare two numbers a and b, and print the comparison in the following format:

print "a is greater than b", if a is greater than b, or

print "b is greater than a", if b is greater than a

Note: You are guaranteed that a ≠ b.

Input

A single line containing two space-separated integers a and b.

Output

Print exactly one line in one of the following formats:

a is greater than b – if a is greater than b

b is greater than a – if b is greater than a

Note: You must use the actual values of a and b in the output.

```
Constraints

-102 ≤ a, b ≤ 102


a ≠ b

Example


Input:


12 5


Output:


12 is greater than 5









Input:


-3 10


Output:


10 is greater than -3
```

Conditional Statements, Python if- else [Post Class] — due 2026-09-14 21:05

Weather Report — 20/20 XP — SOLVED — deadline 2026-09-14 21:05

Topics: Python if- else, If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/krjnhzw8l2lp

Back

Weather Report

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

Try AI Plan Mode

QUESTION

Weather Report

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Write a program that takes an integer current temperature as input, and determines the type of weather using the below conditions:

Temp < 0 : Freezing Weather

Temp 0–9 : Very Cold

Temp 10–19 : Cold

Temp 20–29 : Normal

Temp 30–39 : Hot

Temp >=40 : Very Hot

Input

A line containing an integer number representing the temperature.

Output

Print the weather according to the temperature as The weather today is classified as: answer

Example

Input

29

Output

```
The weather today is classified as: Normal

Explanation

According to the given question the temperature 20–29 are considered Normal



Input

9

Output

The weather today is classified as: Very Cold

Explanation

According to the given question the temperature 0–9 are considered Very Cold
```

AM_PM — 20/20 XP — SOLVED — deadline 2026-09-14 21:05

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/9fd6gabsf4f0

Back

AM_PM

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

AM_PM

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Write a program that takes the hour (0–23) in 24-hour format as input and prints whether it is AM or PM.

Note:

The time is considered AM if the hour is between 0 and 11 (inclusive).

The time is considered PM if the hour is between 12 and 23 (inclusive).

Input

First line will contain an integer hour (0–23) as input.

Output

The program should print the appropriate answer(AM or PM) based on the hour.

Constraints

0 ≤ hour ≤ 23

Example

Example 1

Input:

10

Output:

```
AM


Example 2

Input:

13


Output:

PM
```

Identify Square or Rectangle — 20/20 XP — SOLVED — deadline 2026-09-14 21:05

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/5oyx5ukgsyae

Back

Identify Square or Rectangle

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Identify Square or Rectangle

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given two integers representing the length and breadth of a rectangle.

You need to print "Square" if the given rectangle is square as well, otherwise print "Rectangle".

Input

The first and only line of input contains two space-separated integers, representing the length and breadth of the rectangle respectively.

Output

Print "Square" or "Rectangle" based on the dimensions.

Constraints

1

≤

l

e

n

g

t

h

,

b

r

e

a

d

t

h

≤

100

1≤length,breadth≤100

Example

Input

5 5

Output

Square

Input

10 9

Output

Rectangle

Python Input, Python operators, Input() function, Input() for different datatype ... [In Class] — due 2026-09-21 11:10

Absolute Difference of Two Numbers — 20/20 XP — SOLVED — deadline 2026-09-21 11:10

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/cymrq6g9ua89

Back

Absolute Difference of Two Numbers

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Absolute Difference of Two Numbers

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Rakshit has two numbers, and he wants to find how far apart they are on the number line — that is, their absolute difference.

Your task is to find and print the absolute difference of the two given numbers.

Input

The first line of input contains two space-separated integers a and b.

Output

Print the absolute difference of a and b in the output.

Constraints

1

≤

a

,

b

≤

1000

1≤a,b≤1000

Example

Input

8 19

```
Output

11

Explanation

The absolute difference between 8 and 19 is | 8 - 19 | = 11.
```

Check if Profit or Loss — 20/20 XP — SOLVED — deadline 2026-09-21 11:10

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/f7muynmik3if

Back

Check if Profit or Loss

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Check if Profit or Loss

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given two integers:

CP — the Cost Price of an item

SP — the Selling Price of the item

Your task is to determine whether the transaction resulted in a Profit or a Loss.

Note: There will be no test case where SP = CP.

Input

The first line contains two space-separated integers CP and SP.

Output

Print:

Profit if the selling price is greater than the cost price.

Loss if the selling price is less than the cost price.

Constraints

```
1 ≤ CP, SP ≤ 109

Example

Input



100 150

Output


Profit

Explanation




Since SP > CP, the transaction resulted in a profit.




Input


200 120

Output


Loss

Explanation




Since SP < CP, the transaction resulted in a loss.
```

Voter Eligibility II — 20/20 XP — SOLVED — deadline 2026-09-21 11:10

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/rm8oeu6qxg9g

Back

Voter Eligibility II

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Voter Eligibility II

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given an integer input, representing the age of a person.

You task is to check whether the person is eligible to vote or not.

A person is eligible to vote if their age is greater than or equal to 18.

Print Yes if the person can vote, otherwise print No.

Input

The first and only line of input contains a single integer, representing the age.

Output

Print Yes if the person can vote, otherwise print No.

Constraints

0

≤

a

g

e

≤

100

```
0≤age≤100

Example

Input

20

Output

Yes


Input

15

Output

No
```

If- elif-else ladder [In Class] — due 2026-09-21 16:57

Grade Calculator — 10/20 XP — SOLVED — deadline 2026-09-21 16:57

Topics: If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/zdxcy5xxzbt4

Back

Grade Calculator

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Grade Calculator

Easy

10/20

Time Limit: 2, Memory Limit: 256000

Write a Python program that takes a student's score as input and prints their grade based on the following criteria:

Grading Criteria:

If the score is 90 or above, the grade is A.

If the score is 80 or more but less than 90, the grade is B.

If the score is 70 or more but less than 80, the grade is C.

If the score is 60 or more but less than 70, the grade is D.

If the score is less than 60, the grade is F.

Input

The one and only line containing a single integer representing the student's score.

Output

Print in a line containing the grade corresponding to the score.

```
Constraints

1<=score<=100

Example

Sample 1:


Input:

85


Output:

Grade B


Explanation:

The score is 85, which falls in the range of 80 or more but less than 90. So, the grade is B.




Sample 2:


Input:

60


Output:

Grade D


Explanation:

The score is 60, which is 60 or more but less than 70. Therefore, the grade is D.
```

The Number Detective — 10/20 XP — SOLVED — deadline 2026-09-21 16:57

Topics: If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/8n06m3da1116

Back

The Number Detective

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

The Number Detective

Easy

10/20

Time Limit: 2, Memory Limit: 256000

Given a number n, write a Python program to check whether the number n is positive, negative, or zero.

Print Positive if n is greater than 0

Print Negative if n is less than 0

Print Zero if n is equal to 0

Input

The one and only line contains a single integer n.

Output

Print whether the number is Positive, Negative, or Zero.

Constraints

−1000 ≤ number ≤ 1000

Example

Sample Input 1:

```
13

Sample Output 1:

Positive

Explanation:



Since 13 is greater than 0, that's why the output is Positive number.









Sample Input 2:

-13

Sample Output 2:

Negative

Explanation:



Since -13 is less than 0, that's why the output is Negative number.
```

Greater or Equal — 10/20 XP — SOLVED — deadline 2026-09-21 16:57

Topics: If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/ewx4lamuv3as

Back

Greater or Equal

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Greater or Equal

Easy

10/20

Time Limit: 2, Memory Limit: 256000

You are given two integers a and b.

Your task is to compare the two numbers and print:

a is greater than b if the value of a is greater than the value of b

b is greater than a if the value of b is greater than the value of a

a and b are equal if both values are equal

Note: Replace a and b with their actual input values in the output.

Input

The first line contains two space-separated integers a and b.

Output

Print exactly one line in one of the following formats:

a is greater than b – if the value of a is greater than the value of b

b is greater than a – if the value of b is greater than the value of a

a and b are equal – if both values are equal

Note: Replace a and b with their actual input values in the output.

Constraints

1 ≤ a , b ≤ 100

1≤a,b≤100

Example

Example 1

Input

12 5

Output

12 is greater than 5

Explanation

Among the two given numbers, 12 is greater than 5.

Example 2

Input

1 10

Output

10 is greater than 1

Explanation

Among the two given numbers, 10 is greater than 1.

Example 3

```
Input

7 7

Output

7 and 7 are equal

Explanation

Both numbers have the same value.
```

Python Input, Python operators, Input() function, Input() for different datatype ... [In Class] — due 2026-09-26 10:43

FizzBuzz II — 20/20 XP — SOLVED — deadline 2026-09-26 10:43

Topics: If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/c64zs11c5ukr

Back

FizzBuzz II

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

FizzBuzz II

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Arjun has recently started learning about divisibility rules. His friend gave him a simple problem to test his understanding.

Given an integer n, your task is to help Arjun print:

"FizzBuzz" if n is divisible by both 3 and 5.

"Fizz" if n is divisible by 3 only.

"Buzz" if n is divisible by 5 only.

Otherwise, print the number n itself.

Input

A single integer n — the number to be evaluated.

Output

Print:

"FizzBuzz" if divisible by both 3 and 5

"Fizz" if divisible only by 3

```
"Buzz" if divisible only by 5

Else, print the number itself



Constraints

-10^9 <= n <= 10^9

Example

1. Sample Input:

15

Sample Output:

FizzBuzz

Explanation: 15 is divisible by 3 and 5, hence the output as per the condition is "FizzBuzz".


2. Sample Input:

5


Sample Output:

Buzz


Explanation: 5 is divisible by 5, hence the output as per the condition is "Buzz".


3. Sample Input:

3

Sample Output:

Fizz

Explanation: 3 is divisible by 3, hence the output as per the condition is "Fizz".
```

Income Tax Calculator — 40/40 XP — SOLVED — deadline 2026-09-26 10:43

Topics: If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/nr7d9k6xdb5a

Back

Income Tax Calculator

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Income Tax Calculator

Medium

2x

40/40

Time Limit: 2, Memory Limit: 128000

Write a program to calculate the income tax for an individual based on their annual income in Indian Rupees (₹) provided as input. The tax is calculated according to the following slabs:

Income Range (₹)	Tax Calculation
Up to ₹2,50,000	No tax.
₹2,50,001 to ₹5,00,000	5% tax on amount between ₹2,50,001 to ₹5,00,000
₹5,00,001 to ₹10,00,000	₹12,500 plus 10% tax on amount between ₹5,00,001 to ₹10,00,000
Above ₹10,00,000	₹62,500 plus 20% tax on amount above ₹10,00,000.

The program should read the annual income from the user, calculate the tax accordingly, and output the tax amount upto 1 decimal places.

Note: We can round the output to 1 decimal place using print(round(tax,1)).

Input

A single integer representing the annual income in rupees.

Output

Print the tax amount as a float number up to 1 decimal places.

Constraints

1

≤

Income

≤

1

0

9

1≤Income≤10

9

Example

input

200000

output

0.0

Explanation: As income is less than 250000, there will not be any tax that the individual is paying, that's why the output is 0.0.

input

550000

output

17500.0

Explanation:

Total income = 550000

1. No tax upto 250000

2. 5% tax in between 250000 to 500000 = 250000*0.05 = 12500.0

3. 10% tax in between 500000 to 1000000,

there for remaining taxable income = 550000 – 500000 = 50000

10% of 50000 = 50000 * 0.1 = 5000.0

Total Tax to be paid = 0.0 + 12500.0 + 5000.0 = 17500.0

Calculate Electricity Bill — 30/60 XP — SOLVED — deadline 2026-09-26 10:43

Topics: If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/5r6xhlhs40lc

Back

Calculate Electricity Bill – 2

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Calculate Electricity Bill – 2

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

An electricity board calculates monthly power usage bills based on the number of units consumed using a slab-wise tariff system. Each slab charges a different rate per unit as follows:

First 100 units: ₹4 per unit

Next 100 units (i.e., units 101–200): ₹5 per unit

Next 100 units (i.e., units 201–300): ₹6 per unit

Next 100 units (i.e., units 301–400): ₹7 per unit

Units above 400: ₹8 per unit

After calculating the base bill, a 10% surcharge is added to the total bill.

You are required to implement the function calculate_bill(units) that calculates the total electricity bill including the surcharge, and returns a float rounded to 1 decimal place using the built-in round() function.

Function Signature:

def calculate_bill(units): which takes single parameter: units: represents the number of electricity units consumed.

Input

User Task

As this is a functional problem, there is no need to handle input from the user. Your task is to complete the function calculate_bill(units), which takes single parameter: units: represents the number of electricity units consumed.

Custom Input

A Single line containing an integer representing the number of units.

Output

Return a float – the total electricity bill including the 10% surcharge, rounded to 1 decimal place using round().

Example

Input

1

Output

Electricity Bill: Rs. 4.4

Explanation

Units are less than 100 therefore total cost is unit charge + surcharge 4 X 1 + 4 X 0.1 = 4.4

Input

100

Output

Electricity Bill: Rs. 440.0

Explanation

Units are equal to 100 therefore total cost is unit charge + surcharge 4 X 100 + 400 X 0.1 = 440.0

Input

340

Output

Electricity Bill: Rs. 1958.0

```
Explanation

For the first 100 units(i.e from 1 to 100) cost will be: 100*4 = 400

For next 100 units(i.e from 101 to 200) cost will be: 100*5 = 500

For next 100 units(i.e from 201 to 300) cost will be: 100*6 = 600

For remaining 40 units(i.e from 301 to 340) cost will be: 40*7 = 280

Total Cost excluding surcharge = 400 + 500 + 600 + 280 = 1780

Final Bill including 10% surcharge on Total cost = 1780 + (1780*0.1) = 1780 + 178 = 1958
```

If- elif-else ladder [In Class] — due 2026-09-26 15:30

Triangle Type Checker — 20/20 XP — SOLVED — deadline 2026-09-26 15:30

Topics: Nested if-else, If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/qc9ncrcoex1x

Back

Triangle Type Checker

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Triangle Type Checker

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given the lengths of the three sides of a triangle. Your task is to determine and print the type of triangle formed.

Use the following rules:

Equilateral: All three sides are equal

Isosceles: Exactly two sides are equal

Scalene: All sides are different

Input

A single line contains three space-separated integers representing the lengths of the three sides of a triangle.

Output

Print one of the following based on the type of triangle:

Equilateral

Isosceles

Scalene

```
Note: Your output must match the cases exactly as shown above. Outputs are case-sensitive.

Constraints

1 ≤ a, b, c ≤ 100



The input will always represent a valid triangle.

Example

input

3 4 3



output

Isosceles



Explanation:

Two sides of a triangle are equal -> Isosceles
```

Check if All Three Numbers are Equal — 20/20 XP — SOLVED — deadline 2026-09-26 15:30

Topics: Nested if-else

URL: https://my.newtonschool.co/playground/code/m27koht9p2md

Back

Check if All Three Numbers are Equal

Total XP

6,673

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Check if All Three Numbers are Equal

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given three integers a, b, and c.

Your task is to determine whether all three numbers are equal.

Print:

Numbers are equal if all three numbers have the same value.

Numbers are not equal otherwise.

Input

The first line contains three space-separated integers a, b, and c.

Output

Print exactly one line:

Numbers are equal if all three numbers are equal.

Numbers are not equal otherwise.

Constraints

1

≤

a

,

b

,

c

≤

100

1≤a,b,c≤100

Example

Example 1

Input

5 5 5

Output

Numbers are equal

Explanation

All three numbers have the same value.

Example 2

Input

5 5 7

Output

Numbers are not equal

Explanation

The three numbers do not all have the same value.

Find the Biggest ! — 20/20 XP — SOLVED — deadline 2026-09-26 15:30

Topics: Nested if-else, Python if- else, If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/0jmhf4wwbnt8

Back

Find the Biggest !

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Find the Biggest !

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Comparing numbers is a common task in programming. Your goal is to determine the largest of three given numbers.

Your task:

Given three integers A, B, and C, print the largest among them.

Input

The first and only line of input contains three space-separated integers, A, B, and C respectively.

Output

Print an integer, the largest number among the given three numbers.

Constraints

0

≤

A

,

B

,

C

≤

100

0≤A,B,C≤100

```
Example

Input

1 2 3

Output

3


Input

5 3 1

Output

5


Input

10 9 10

Output

10
```

If- elif-else ladder [Post Class] — due 2026-09-26 16:00

Library fine — 10/20 XP — SOLVED — deadline 2026-09-26 16:00

Topics: If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/x8aarbvbrgnw

Back

Library fine

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

Try AI Plan Mode

QUESTION

Library fine

Easy

10/20

Time Limit: 2, Memory Limit: 256000

Calculate the fine for a late return of a library book based on the following details:

For the first 3 days, the fine is Rs. 10.

For the next 4 to 10 days, the fine is Rs. 20.

For any period exceeding 10 days, the fine is Rs.100.

Input

A line containing an integer representing the number of late days.

Output

Print a line containing the total fine amount.

Example

Input

10

Output

20

Explanation

Fine for 10 late days is 20 Rs according to the question.

```
Input

20

Output

100

Explanation

Fine for 10 late days is 100 Rs according to the question.
```

Calculate salary — 10/20 XP — SOLVED — deadline 2026-09-26 16:00

Topics: If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/8u4fn097c5xk

Back

Calculate salary

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Calculate salary

Easy

10/20

Time Limit: 2, Memory Limit: 256000

Calculate the gross salary of an employee based on their basic salary according to the following criteria:

basic salary <= 10,000, HRA = 20% and DA = 80%.

20,000 >= basic salaries > 10,000 , the HRA = 25% and DA = 90%.

basic salary > 20,000, HRA = 30% and DA = 95%.

Hint: Gross Salary = Basic Salary + HRA + DA

Input

A line containing an integer representing the basic salary of the employee.

Output

Print the total gross salary as The gross salary of the employee is: answer

Example

Input

10001

Output

The gross salary of the employee is: 21502.15

```
Explanation

By using the formula answer is 21502.15
```

Check Profit or Loss — 10/20 XP — SOLVED — deadline 2026-09-26 16:00

Topics: Python if- else, If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/gdnv1arw8xa9

Back

Check Profit or Loss

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Check Profit or Loss

Easy

10/20

Time Limit: 2, Memory Limit: 256000

You are given two integers, cost and sell, representing the cost price and the selling price of a product.

Your task is to determine whether the transaction results in a profit, a loss, or no profit and no loss.

Also, you need to calculate the corresponding amount.

Input

The first line of input contains a single integer, representing cost.

The second line also contains a single integer, representing sell.

Output

The program should print two lines.

The first line should contain one of the words Profit, Loss, or No Profit No Loss.

The second line should contain an integer representing the profit or loss amount, and it should be 0 in case of no profit and no loss.

Constraints

1

≤

c

o

s

t

≤

1000

1≤cost≤1000

1

≤

s

e

l

l

≤

1000

1≤sell≤1000

Example

Input

150

100

Output

Loss

50

Input

100

100

Output

No Profit No Loss

0

BMI Calculator-1 — 10/20 XP — SOLVED — deadline 2026-09-26 16:00

Topics: If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/yvepr94fgd09

Back

BMI Calculator-1

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

BMI Calculator-1

Easy

10/20

Time Limit: 2, Memory Limit: 256000

You are asked to determine a person's Body Mass Index (BMI) based on their weight and height.

The BMI is calculated using the following formula:

BMI = weight / (height_in_meters x height_in_meters)

Here,

weight is given in kilograms

height is given in centimeters and must be converted to meters

You must round the BMI value to exactly 1 decimal place using the built-in round() function.

Based on the rounded BMI value, print the corresponding BMI category:

If BMI is less than 18.5, print Underweight

If BMI is between 18.5 and 24.9 (inclusive), print Normal weight

If BMI is between 25 and 29.9 (inclusive), print Overweight

If BMI is 30 or greater, print Obese

Input

The input consists of two lines as follows:

The first line contains an integer weight — the person's weight in kilograms.

The second line contains an integer height — the person's height in centimeters.

Output

Print two lines:

First line: the BMI value rounded to one decimal place using round()

Second line: the corresponding BMI category.

Constraints

1 ≤ weight ≤ 300

50 ≤ height ≤ 250

Example

Input

60

165

Output

22.0

Normal weight

Explanation:

BMI = 60/1.65*1.65 = 22.0

22.0 is in range 18.5 to 24.9

Category = Normal weight

Python Input, Python operators, Conditional Statements, Input() function, Input(... [In Class] — due 2026-09-28 10:38

Can It Be a Triangle ? — 10/20 XP — SOLVED — deadline 2026-09-28 10:38

Topics: Nested if-else

URL: https://my.newtonschool.co/playground/code/zmbd9rh1pinh

Back

Can It Be a Triangle ?

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Can It Be a Triangle ?

Easy

10/20

Time Limit: 2, Memory Limit: 256000

You are given the lengths of three sides. Your task is to check whether these three sides can form a valid triangle.

Input

The first and only line of input contains three space-separated integers, representing the lengths of three sides.

Output

Print Yes if the sides form a valid triangle.

Print No otherwise.

Constraints

1

≤

S

i

d

e

L

e

n

g

t

h

$\leq$

100

$1 \leq SideLength \leq 100$

Example

Input

20 10 13

Output

Yes

Explanation

All combinations of side pairs sum to more than the third side. So, a triangle is possible.

Input

2 5 3

Output

No

Explanation

2 + 3 is not greater than 5, so a triangle cannot be formed.

Leap Year Checker — 40/40 XP — SOLVED — deadline 2026-09-28 10:38

Topics: Nested if-else, If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/lds6jda0joac

Back

Leap Year Checker

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Leap Year Checker

Medium

2x

40/40

Time Limit: 2, Memory Limit: 256000

You are given an integer representing a year. Your task is to determine whether it is a leap year or not.

Input

The input contains a single integer: year — the year to check

Output

Print a single line as follows:

Leap year if the year is a leap year

Not a leap year otherwise

Constraints

1 ≤ year ≤ 10^9

Example

Example 1:

Input:

2000

Output:

Leap year

Explanation:

2000 is divisible by 400 ⇒

2000

mod

400

=

0

2000mod400=0

So, it's a leap year.

Example 2:

Input:

1900

Output:

Not a leap year

Explanation:

1900 is divisible by 100 ⇒

1900

mod

100

=

0

1900mod100=0

```
but not divisible by 400 ⇒

1900


mod


400

≠

0

1900mod400

□

=0


So, it's not a leap year.
```

Roller Coaster Ride — 20/40 XP — SOLVED — deadline 2026-09-28 10:38

Topics: Nested if-else, If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/8a67fz5d44vl

Back

Roller Coaster Ride

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Roller Coaster Ride

Medium

20/40

Time Limit: 2, Memory Limit: 256000

A theme park wants to check if Alice can ride their roller coaster. You are given the height (in cm) and age (in years) of Alice.

Eligibility and ticket rules are as follows:

Height Requirements:

Less than 120 cm → Not Allowed

120 – 140 cm → Must ride with Adult

Above 140 cm → Can ride alone

Age Restrictions:

Less than 5 years → Not Allowed

5 – 12 years → Must ride with Adult

Above 12 → Can ride alone

Ticket Price Calculation:

If Allowed Alone → ₹300

Allowed with Adult → ₹200

Your task is to print whether Alice is allowed to ride or not. If Alice is allowed to ride, print the ticket price of ride in the next line.

Note: Alice's eligibility is determined by both height and age. If either height or age imposes a stricter restriction, that restriction takes precedence.

For example, if her age requires her to ride with an adult, she cannot ride alone even if her

height allows it.

Similarly, if her age or height disqualifies her from riding, she is not allowed to ride regardless of the other factor.

Input

The first line of input contains a single integer, representing height.

The second line also contains a single integer, representing age.

Output

The output should contain one or two lines:

On the first line, print Alice's riding status:

Not Allowed – if Alice is not allowed to ride.

Must ride with Adult – if Alice can ride only with an adult.

Can ride alone – if Alice can ride by herself.

If Alice is allowed to ride (either with an adult or alone), print the ticket price on the second line (without ₹ sign).

Constraints

50

≤

h

e

i

g

h

t

≤

300

50≤height≤300

1

≤

a

g

e

≤

```
100

1≤age≤100

Example

Input

145

14

Output

Can ride alone

300

Explanation

Height of Alice is greater than 140 cm and also the age is above 12.

Therefore, Alice can ride alone and the ticket price will be ₹300.


Input

110

4

Output

Not Allowed
```

Python Function, Nested if-else [Post Class] — due 2026-09-28 20:35

Accommodation Decision — 10/20 XP — SOLVED — deadline 2026-09-28 20:35

Topics: Nested if-else

URL: https://my.newtonschool.co/playground/code/iue8zuqmsvel

Back

Accommodation Decision

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Accommodation Decision

Easy

10/20

Time Limit: 2, Memory Limit: 256000

A student wants to check whether they will stay On-Campus or Off-Campus based on the rules provided in the NST rulebook.

The rules for accommodation are:

If the student is a girl, she will definitely stay On-Campus, irrespective of the year.

If the student is a boy:

In first year → he will stay On-Campus

In second year → he will stay Off-Campus

Given the student's gender and year of study, determine their accommodation type.

Input

The input consists of two lines:

The first line contains a string gender, representing the gender of the student:

G for girl

```
B for boy

The second line contains an integer year, representing the year of study:

1 for first year

2 for second year

Output

Print a single line, either:

On-Campus or Off-Campus, according to the rules.

Constraints

The value of gender will be either "G" or "B".

The value of year will be either 1 or 2.

Example

Input

G

2

Output

On-Campus


Input

B

2

Output

Off-Campus
```

Calculate Tax Based on Age and Income — 10/20 XP — SOLVED — deadline 2026-09-28 20:35

Topics: Nested if-else

URL: https://my.newtonschool.co/playground/code/gggefmp1dam2

Back

Calculate Tax Based on Age and Income

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Calculate Tax Based on Age and Income

Easy

10/20

Time Limit: 2, Memory Limit: 256000

You are given two values:

age – the age of a person

income – the annual income of the person (in rupees)

Your task is to calculate the amount of tax that the person needs to pay.

The tax rate depends on both the person's age and annual income.

Tax Slabs

Income Bracket	Tax Rate (Age < 18)	Tax Rate (Age ≥ 18)
Income ≤ 10,00,000	5%	10%
10,00,000 < Income ≤ 30,00,000	15%	20%
Income > 30,00,000	25%	30%

Print the tax amount that must be paid.

Input

The first line contains two space-separated integers age and income.

Output

Print a single integer representing the tax amount.

Constraints

$1 \leq age \leq 120$

$0 \leq income \leq 109$

Example

Example 1

Input

25 800000

Output

80000

Explanation The person's age is at least 18 and the income falls in the first tax slab. Therefore, the applicable tax rate is 10%. Tax = 10% of 800000 = 80000.

Example 2

Input

15 2000000

Output

300000

Explanation The person's age is less than 18 and the income falls in the second tax slab. Therefore, the applicable tax rate is 15%. Tax = 15% of 2000000 = 300000.

Example 3

Input

40 5000000

Output

1500000

Explanation The person's age is at least 18 and the income falls in the third tax slab. Therefore, the applicable tax rate is 30%. Tax = 30% of 5000000 = 1500000.

Find the Smallest ! — 10/20 XP — SOLVED — deadline 2026-09-28 20:35

Topics: Nested if-else, If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/d0irgxswik71

Back

Find the Smallest !

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Find the Smallest !

Easy

10/20

Time Limit: 2, Memory Limit: 256000

Just like finding the largest value, identifying the smallest number is another common task in programming.

Your task:

Given three integers A, B, and C, print the smallest among them.

Input

The first and only line of input contains three space-separated integers, A, B, and C respectively.

Output

Print an integer, the smallest number among the given three numbers.

Constraints

0

≤

A

,

B

,

C

≤

100

0≤A,B,C≤100

Example

```
Input

1 2 3

Output

1


Input

5 3 1

Output

1


Input

10 9 10

Output

9
```

Nested if-else, Python Function [In Class] — due 2026-10-03 10:59

Calculate Percentage Using Function — 10/20 XP — SOLVED — deadline 2026-10-03 10:59

Topics: Return statement

URL: https://my.newtonschool.co/playground/code/87ji7g9e8qin

Back

Calculate Percentage Using Function

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Calculate Percentage Using Function

Easy

10/20

Time Limit: 2, Memory Limit: 256000

You are required to complete the function calculate_percentage(marks, total), which calculates the percentage score of a student.

The function receives two integer parameters:

marks – the number of marks obtained

total – the total number of marks

It should return the percentage score as an integer.

Note:

This is a functional problem. You do not need to take any input or print anything.

Just complete the function calculate_percentage(marks, total) which returns the percentage score.

Percentage should be returned as an integer.

Input

The function calculate_percentage receives two space-separated integers as parameters:

marks – the marks obtained.

total – the total marks.

You do not need to take input from the user. The values will be passed to the function directly.

Output

Return a single integer – the percentage score. You do not need to print anything. Just return the integer percentage value from the function calculate_percentage().

Constraints

1 ≤ total ≤ 300

1 ≤ marks ≤ total

Example

Input 1

45 50

Output 1

90

Explanation

Marks obtained = 45

Total marks = 50

Percentage = (45 × 100) // 50 = 4500 // 50 = 90

So, the function should return 90.

Input 2

72 80

Output 2

90

```
Explanation

Marks obtained = 72

Total marks = 80

Percentage = (72 × 100) // 80 = 7200 // 80 = 90

So, the function should return 90.
```

Percentage-1 — 10/20 XP — SOLVED — deadline 2026-10-03 10:59

Topics: Arguments and parameter

URL: https://my.newtonschool.co/playground/code/93a2jn9drf3p

Back

Percentage-1

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Percentage-1

Easy

10/20

Time Limit: 2, Memory Limit: 256000

Alice appeared for an exam in one subject. The exam is out of total marks. She wants to know her percentage score.

A predefined function PrintPercentage(total_marks, obtained_marks) is available. It takes two integer arguments:

total_marks – total marks of the exam

obtained_marks – marks obtained by Alice

The function prints the percentage (up to 2 decimal places).

Note: You do not need to define this function, just take the inputs and call this function and it will automatically print the percentage.

Input

A single line contains two space separated integers total marks and marks obtained by Alice.

Output

Use the predefined function PrintPercentage() and just call the function.

Constraints

1<= total_marks <= 100

```
0<= marks<= total_marks

Example

Input:

10 8



Output:

80.00




Explanation:

The input line is split into two integers: total_marks = 10 and obtained_marks = 8.


The predefined function PrintPercentage(10, 8) is called.


It calculates:

(8/10)×100=80.00

The function prints 80.00 as the final percentage.
```

Percentage-2 — 10/20 XP — SOLVED — deadline 2026-10-03 10:59

Topics: Arguments and parameter

URL: https://my.newtonschool.co/playground/code/4vcout7nqz0o

Back

Percentage-2

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Percentage-2

Easy

10/20

Time Limit: 2, Memory Limit: 256000

Alice appeared for an exam in a single subject. The exam has a certain total number of marks, and she wants to know her percentage score.

A predefined function GetPercentage(total_marks, obtained_marks) is available. It takes two integer arguments:

total_marks – the maximum marks of the exam

obtained_marks – the marks obtained by Alice

The function returns the percentage (as a float, rounded to two decimal places).

Note: You do not need to define this function. Just call it and print the returned value.

Input

A single line containing two space-separated integers: total_marks and obtained_marks

Output

Print the percentage returned by GetPercentage() function.

Constraints

1<= total_marks <= 100

```
0<= marks<= total_marks

Example

Input:

10 8



Output:

80.00




Explanation:

The input line is split into two integers: total_marks = 10 and obtained_marks = 8.


The predefined function PrintPercentage(10, 8) is called.


It calculates:

(8/10)×100=80.00

print 80.00 as the final percentage.
```

Python Function, Nested if-else, Arguments and parameter, Return statement [In Class] — due 2026-10-03 15:02

Basic Greater or Smaller — 20/20 XP — SOLVED — deadline 2026-10-03 15:02

Topics: Python Function, Nested if-else, Python if- else

URL: https://my.newtonschool.co/playground/code/poxmks7bvp48

Back

Basic Greater or Smaller

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Basic Greater or Smaller

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given two functions:

greater(a, b) → prints the greater of two integers a and b.

smaller(a, b) → prints the smaller of two integers a and b.

Note: You should call these functions as required

Write a program that takes:

A single character input: "G" for greater or "S" for smaller

Two space-separated integers

Task:

If the character is "G", print the greater of the two numbers .

If the character is "S", print the smaller of the two numbers .

```
Input

The first line contains choice: A single character, either "G" or "S"

The second line contains a, b: Two space-separated integers

a!=b

Output

Print the greater or smaller of the two numbers as per input constraints.




Constraints

a, b are integers between -10⁴ and 10⁴

The first input is guaranteed to be "G" or "S"

Example

Input

G

4 9


Output

9


Explanation

"G" means print the greater number.

Between 4 and 9, the greater is 9, so the output is 9
```

Basic Greater or Smaller-2 — 20/20 XP — SOLVED — deadline 2026-10-03 15:02

Topics: Python Function, Nested if-else, Python if- else

URL: https://my.newtonschool.co/playground/code/xb23mpkn7y8m

Back

Basic Greater or Smaller-2

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Basic Greater or Smaller-2

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Write a program that takes:

A single character input: "G" for greater or "S" for smaller

Two space-separated integers

Task:

If the character is "G", print the greater of the two numbers using the greater() function.

If the character is "S", print the smaller of the two numbers using the smaller() function.

You are supposed to implement two functions:

greater(a, b) → Gives greater of two integers a and b.

smaller(a, b) → Gives smaller of two integers a and b.

Note: You should implement and then call these functions as required.

```
Input

The first line contains choice: A single character, either "G" or "S"

The second line contains a, b: Two space-separated integers

Output

Print the greater or smaller of the two numbers using the given functions.


Constraints

a, b are integers between -10⁴ and 10⁴

The first input is guaranteed to be "G" or "S"

Example

Input

G

4 9


Output

9


Explanation

"G" means print the greater number.

Between 4 and 9, the greater is 9, so the output is 9
```

Greater of Three — 20/20 XP — SOLVED — deadline 2026-10-03 15:02

Topics: Python Function, Nested if-else, Python if- else

URL: https://my.newtonschool.co/playground/code/hyytojgv0yno

Back

Greater of Three

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Greater of Three

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given three integers a, b, and c.

A built-in function greater(x, y) is available to you. It returns the greater of the two values x and y.

Your task is to find the greatest among the three given numbers using the greater() function.

Note: You must use the provided greater() function. Do not compare all three numbers directly.

Input

The first line contains three space-separated integers a, b, and c.

Output

Print a single integer — the greatest among the three numbers.

Constraints

1 ≤ a, b, c ≤ 103

```
Example

Example 1

Input

10 25 15

Output

25


Example 2

Input

5 2 8

Output

8
```

Subject Grades — 20/20 XP — SOLVED — deadline 2026-10-03 15:02

Topics: Python Function, Nested if-else, Python if- else

URL: https://my.newtonschool.co/playground/code/xjpuonh5dgrn

Back

Subject Grades

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Subject Grades

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given the marks obtained in three subjects as integers: A, B, and C.

Your task is to call the predefined function GetGrade(marks) for each subject’s marks. The function returns the grade (a string) corresponding to the given marks.

Print the grade of each subject on a new line, in the order of A, B, and C.

Note: You do not need to define the GetGrade() function. It is already provided.

Input

The first line of input contains a single integer, representing A.

The second line contains an integer B.

The last line of input contains an integer C.

Output

You have to print the grade for each subject on a new line.

Constraints

0

≤

A

≤

100

0≤A≤100

0

≤

B

≤

100

0≤B≤100

0

≤

C

≤

100

0≤C≤100

Example

Input

99

84

45

Output

A

B

E

Explanation

The marks for the three subjects are 99, 84, and 45 respectively.

The grade for marks = 99, is A.

The grade for marks = 84, is B.

```
The grade for marks = 45, is E.
```

Python Function, Nested if-else, Arguments and parameter, Return statement [Post Class] — due 2026-10-03 16:00

Bridge Between Villages -2 — 20/20 XP — SOLVED — deadline 2026-10-03 16:00

Topics: Python Function, Nested if-else, Python logical Operators, Return statement, Arguments and parameter

URL: https://my.newtonschool.co/playground/code/wuioa3508kse

Back

Bridge Between Villages -2

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Bridge Between Villages -2

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

In a rural district, three villages — Raga, Sora, and Tula — are planning to connect to each other with straight bridges. The government has provided the proposed lengths for these bridges as follows:

Raga to Sora: a kilometers

Sora to Tula: b kilometers

Tula to Raga: c kilometers

You are required to implement the function is_triangle(a, b, c) that returns "YES" or "NO" depending on whether the given bridge lengths can form a valid triangle.

Function Signature:

def is_triangle(a, b, c): which takes three parameters: a: represents proposed length from Raga to Sora, b: represents proposed length from Sora to Tula, c: represents proposed length from Tula to Raga

Input

User Task

As this is a functional problem, there is no need to handle input from the user. Your task is to

complete the function is_triangle(a, b, c), which takes three parameters: a: represents proposed length from Raga to Sora, b: represents proposed length from Sora to Tula, c: represents proposed length from Tula to Raga

Custom Input

Input consist of single line that contains three space-separated positive integers:

a b c

Output

Return YES if the construction plan can go ahead, otherwise return NO.

Constraints

1 ≤ a, b, c ≤ 104

Example

Input:

6 7 10

Output:

YES

Explanation

The bridge lengths are (6, 7, 10) which satisfy

6 + 7 > 10 (13 > 10), 7 + 10 > 6 (17 > 6), and 10 + 6 > 7 (16 > 7).

Since all pairs of bridges can connect to enclose a region, the output is "YES".

Input:

3 5 8

Output:

NO

Explanation

The bridge lengths are(3, 5, 8) not follows the below condition

```
3 + 5 > 8 → 8 > 8 (False).

5 + 8 > 3 -> 13 > 3 (True)

3 + 8 > 5 -> 11 > 5 (True)

Since one condition is False, bridges cannot enclose a region; the output is "NO".
```

Calculate Electricity Bill - 2 — 20/20 XP — SOLVED — deadline 2026-10-03 16:00

Topics: Return statement, Arguments and parameter

URL: https://my.newtonschool.co/playground/code/5r6xhlhs40lc

Back

Calculate Electricity Bill – 2

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Calculate Electricity Bill – 2

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

An electricity board calculates monthly power usage bills based on the number of units consumed using a slab-wise tariff system. Each slab charges a different rate per unit as follows:

First 100 units: ₹4 per unit

Next 100 units (i.e., units 101–200): ₹5 per unit

Next 100 units (i.e., units 201–300): ₹6 per unit

Next 100 units (i.e., units 301–400): ₹7 per unit

Units above 400: ₹8 per unit

After calculating the base bill, a 10% surcharge is added to the total bill.

You are required to implement the function calculate_bill(units) that calculates the total electricity bill including the surcharge, and returns a float rounded to 1 decimal place using the built-in round() function.

Function Signature:

def calculate_bill(units): which takes single parameter: units: represents the number of electricity units consumed.

Input

User Task

As this is a functional problem, there is no need to handle input from the user. Your task is to complete the function calculate_bill(units), which takes single parameter: units: represents the number of electricity units consumed.

Custom Input

A Single line containing an integer representing the number of units.

Output

Return a float – the total electricity bill including the 10% surcharge, rounded to 1 decimal place using round().

Example

Input

1

Output

Electricity Bill: Rs. 4.4

Explanation

Units are less than 100 therefore total cost is unit charge + surcharge 4 X 1 + 4 X 0.1 = 4.4

Input

100

Output

Electricity Bill: Rs. 440.0

Explanation

Units are equal to 100 therefore total cost is unit charge + surcharge 4 X 100 + 400 X 0.1 = 440.0

Input

340

Output

Electricity Bill: Rs. 1958.0

```
Explanation

For the first 100 units(i.e from 1 to 100) cost will be: 100*4 = 400

For next 100 units(i.e from 101 to 200) cost will be: 100*5 = 500

For next 100 units(i.e from 201 to 300) cost will be: 100*6 = 600

For remaining 40 units(i.e from 301 to 340) cost will be: 40*7 = 280

Total Cost excluding surcharge = 400 + 500 + 600 + 280 = 1780

Final Bill including 10% surcharge on Total cost = 1780 + (1780*0.1) = 1780 + 178 = 1958
```

Smart Conditional Calculator — 20/20 XP — SOLVED — deadline 2026-10-03 16:00

Topics: Return statement, Arguments and parameter

URL: https://my.newtonschool.co/playground/code/e66cy6dbjxgb

Back

Smart Conditional Calculator

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Smart Conditional Calculator

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given two integers. Your task is to implement a smart calculator that chooses the arithmetic operation based on the parity (even/odd nature) of the two numbers.

Rules:

If both numbers are even, return their sum.

If both numbers are odd, return their product.

If one number is even and the other is odd, return the absolute difference between the two (i.e., subtract the smaller from the larger).

🔧 Task

Write the function definition for smart_calculator(a, b). This means:

✅ You must write the function header (starting with def)

✅ You must also write the function body with correct logic

❌ Do not write any code for input or output (e.g., no input() or print() statements)

Input

User Task

As this is a functional problem, there is no need to handle input from the user. Your task is to complete the function definition smart_calculator(a, b), which takes two parameters:

a: an integer

b: an integer

Custom Input

A Single line containing two space-separated integers:

a and b, representing the two numbers to process.

Output

Return single integer representing the result of the operation.

Constraints

$-10^4 \leq a, b \leq 10^4$

Example

Input

4 8

Output

12

Explanation

Both numbers are even → 4 + 8 = 12

Celsius to Fahrenheit Conversion - 2 — 20/20 XP — SOLVED — deadline 2026-10-03 16:00

Topics: Return statement, Arguments and parameter

URL: https://my.newtonschool.co/playground/code/haikjdpg2f4e

Back

Celsius to Fahrenheit Conversion – 2

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Celsius to Fahrenheit Conversion – 2

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given an integer, representing the temperature in celsius.

Your task is to convert it to Fahrenheit, using the below given formula:

Fahrenheit = (Celsius × 9 / 5) + 32

You must complete the function celsius_to_fahrenheit(celsius) that returns the equivalent temperature in Fahrenheit.

Function Signature:

def celsius_to_fahrenheit(celsius):

It takes a single parameter, celsius, the temperature in Celsius.

Input

User Task

As this is a functional problem, there is no need to handle input from the user. Your task is to complete the function celsius_to_fahrenheit(celsius), which takes single parameter, celsius, the temperature in celsius.

Custom Input

The first and only line of input contains a single integer, celsius, representing the temperature in celsius.

Output

Return the corresponding temperature in Fahrenheit (as a float).

```
Constraints

-100 ≤ celsius ≤ 100

Example

Input

0

Output

32.0

Explanation

Input Celsius = 0

Fahrenheit = (0 × 9/5) + 32 = 0 + 32 = 32.0

So the output is 32.0.


Input

37

Output

98.6

Explanation

Input Celsius = 37

Fahrenheit = (37 × 9/5) + 32 = 98.6

So the output is 98.6.
```

Python if- else, Python logical Operators [In Class] — due 2026-10-05 10:55

Leap Year or Not ? — 20/40 XP — SOLVED — deadline 2026-10-05 10:55

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/mpbp2fcg56oh

Back

Leap Year or Not ?

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Leap Year or Not ?

Medium

20/40

Time Limit: 2, Memory Limit: 256000

A leap year occurs every 4 years, with some exceptions. But you don’t need to worry about the logic — we’ve already written the leap year checker function for you.

Your task is to use the pre-defined function and print whether the given year is a leap year or not.

Your task:

Take an integer input year

Call the function is_leap(year)

If it returns True, print Leap Year

If it returns False, print Not a Leap Year

Note: Use only the pre-defined functions provided as comments in the starter code.

Input

The first and only line of input contains a single integer, representing the year.

Output

If the input year is leap year, print "Leap Year", else print "Not a Leap Year".

Constraints

0

≤

y

e

a

r

≤

3

∗

1

0

3

0≤year≤3∗10

3

Example

Input

0

Output

Leap Year

Input

2003

Output

Not a Leap Year

Great Dragon Tournament. — 20/40 XP — SOLVED — deadline 2026-10-05 10:55

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/eni755r2q5o3

Back

Great Dragon Tournament.

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Great Dragon Tournament.

Medium

20/40

Time Limit: 2, Memory Limit: 256000

In the Great Dragon Tournament, every dragon is judged based on two attributes:

Power (how strong the dragon’s attack is)

Stamina (how long the dragon can endure a fight)

The judges decide each dragon’s category based on these two values.

BALANCED – If power is equal to stamina.

OVERPOWERED – If power is at least twice the stamina.

UNBREAKABLE – If stamina is at least twice the power.

MIGHTY – If power is greater than stamina, but less than double.

ENDURING – If stamina is greater than power, but less than double.

Your task is to complete the function dragon_category(power, stamina) that returns the correct category of the dragon based on the given values of power and stamina.

Note: You don't need to print anything, just return the final output from the function. You don't have to take any input.

Input

The first line contains an integer power.

The second line contains an integer stamina.

Note: The input section shown is only for debugging purposes. You don’t need to explicitly take input in your code. It simply illustrates the format in which inputs should be provided when creating your own test cases.

Output

The correct category of the dragon.

Constraints

1

≤

p

o

w

e

r

≤

1000

1≤power≤1000

1

≤

s

t

a

m

i

n

a

≤

1000

1≤stamina≤1000

Example

Input

500

500

Output

BALANCED

Explanation

Here, the power of dragon is equal to stamina, i.e. 500.

Therefore, the correct category of dragon is BALANCED.

Function Based Calculator. — 20/40 XP — SOLVED — deadline 2026-10-05 10:55

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/16km8rga31m7

Back

Function Based Calculator.

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

Try AI Plan Mode

QUESTION

Function Based Calculator.

Medium

2x

80/80

Time Limit: 2, Memory Limit: 256000

You are building a calculator — but this time, you're not allowed to use direct arithmetic operators.

All the arithmetic operations have already been defined as functions in the code for you.

Your Task:

Read two integers and one operator symbol from the input

Based on the operator, call the appropriate function

Print the result returned by that function

If the operator is invalid (other than ' + ' , ' - ' , ' * ', , ' // ', ' % ') for this question , print Invalid Operator

Note: Do not use arithmetic operators directly in your code. Use only the pre-defined functions provided as comments in the starter code.

Input

The first line of input contains two space-separated integers, representing A and B.

The second line contains a string op, representing the operation to perform.

Output

A single line — the result of the operation, or Invalid Operator if the symbol is not supported.

Note: Use pre-defined functions.

Constraints

–

1

0

3

≤

A

,

B

≤

1

0

3

−10

3

≤A,B≤10

3

B

≠

0

B

=0, for op = '/' or '%'

Example

Input

20 10

+

Output

30

Input

```
5 3

^

Output

Invalid Operator
```

Python if- else, Python logical Operators [In Class] — due 2026-10-05 17:03

NST Library — 60/60 XP — SOLVED — deadline 2026-10-05 17:03

Topics: Python Function, Nested if-else, Python if- else

URL: https://my.newtonschool.co/playground/code/4jzfl2d9wjpa

Back

NST Library

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

NST Library

Hard

2x

60/60

Time Limit: 2, Memory Limit: 256000

You are tasked with implementing a library rental system that calculates the rental fee for a single book based on specific criteria.

The rental fee is calculated according to the following rules:

Rental Duration:

If the rental duration is between 1 and 3 days, the daily fee is 1 Rupee per day.

If the rental duration is between 4 and 7 days, the daily fee is 0.75 Rupees per day.

If the rental duration is 8 days or more, the daily fee is 0.50 Rupees per day.

If the rental duration is 0 days or negative, output Invalid rental duration.

Book Type:

1 for New Release (add a surcharge of 2 Rs to the total fee).

2 for Classic (apply a discount of 10% on the total fee).

3 for Children’s Book (no additional charge or discount).

Membership Status:

1 for Premium (apply an additional 10% discount on the total fee).

2 for Basic (no changes are applied).

Note: The rental fee should be calculated in the given order only. Also, typecast the final amount to integer before printing.

Input

The first line contains an integer days representing the number of days the book was rented.

The second line contains an integer book representing the type of book.

The third line contains an integer status representing the membership status.

Output

Print a single line containing either the calculated rental amount or the message Invalid rental duration.

Constraints

−

10

≤

d

a

y

s

≤

10

−10≤days≤10

1

≤

b

o

o

k

≤

3

1≤book≤3

1

≤

s

t

a

t

u

s

≤

2

1≤status≤2

Example

Input

4

1

1

Output

4

Explanation:

The daily fee is 0.75 rupees for rentals between 4 and 7 days.

For 4 days, the base fee is calculated as 4 × 0.75 = 3 rupees.

Since the book type is a New Release, a surcharge of 2 rupees is added, bringing the total to 5 rupees.

With a Premium membership, a discount of 10% applies.

The discount on 5 rupees is 0.5, so the adjusted amount becomes 4.5 rupees.

After typecasting the final value to an integer, the result becomes 4.

Input

-1

2

2

Output

```
Invalid rental duration
```

Python if- else, Python logical Operators [Post Class] — due 2026-10-05 20:35

Movie Mania — 20/20 XP — SOLVED — deadline 2026-10-05 20:35

Topics: Python Function, Nested if-else

URL: https://my.newtonschool.co/playground/code/cnbbqdvnjjr6

Back

Movie Mania

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Movie Mania

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Four movies were released: Spiderman, Superman, Batman, and Aquaman in four distinct years given as four space-separated integers: year1, year2, year3, and year4 respectively.

Movie

Release year

Spiderman year1

Superman year2

Batman year3

Aquaman year4

You are required to print the name of the movie that was released earliest of the four.

Note-1: Refer Input section, output section and example test cases.

Note-2: Use of built-in mathematical functions is strictly prohibited in this question.

Input

The first and only line of input contains four distinct space separated integers year1, year2, year3, and year4 representing the years of release of the corresponding movies given in the problem statement.

Output

Print a single line representing the name of the movie which was released earliest.

Constraints

The values

year1

,

year2

,

year3

,

year4

year1,year2,year3,year4 represent four distinct years.

Each year satisfies the condition

1

≤

year

i

≤

4000

1≤year

i

≤4000 for

i

∈

{

1

,

2

,

3

,

```
4

}

i∈{1,2,3,4}.

Example

Input:

2014 2015 2019 1989



Output:

Aquaman



Explanation:

1989, the 4th number, is the minimum and the earliest year among the given years, and hence the
movie corresponding to it, Aquaman, is the answer.
```

Will Ramandeep Sir Fly? - 2 — 20/20 XP — SOLVED — deadline 2026-10-05 20:35

Topics: Python Function, Nested if-else

URL: https://my.newtonschool.co/playground/code/7lor16gzent8

Back

Will Ramandeep Sir Fly? – 2

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Will Ramandeep Sir Fly? – 2

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Ramandeep Sir is known for his highly selective flying decisions after past flight troubles. He follows a strict set of rules before agreeing to board a plane.

Here are his conditions:

If the trip is international, Ramandeep Sir will not fly, regardless of any other conditions.

If the trip is domestic, and the flight duration is less than or equal to 1.5 hours, he will fly.

If the trip is domestic and the duration is more than 1.5 hours, he will not fly if:

His seat is a middle seat, or

The airline is IngidoX

In all other cases, he will fly.

Your task is to determine whether Ramandeep Sir will fly given the flight details.

Input

The input consists of four lines:

The first line contains a floating-point number d — the duration of the flight in hours.

The second line contains a string seat_type — one of "window", "middle", or "aisle".

The third line contains a string airline — the airline name (case-sensitive, no spaces).

The fourth line contains a string trip_type — either "domestic" or "international".

Output

Print a single line as following:

Ramandeep Sir will fly — if he agrees to board the flight.

Ramandeep Sir will not fly — otherwise.

Note: Output must exactly match the above (case-sensitive, no extra spaces).

Constraints

0.5 ≤ d ≤ 10.0

0.5≤d≤10.0

seat_type ∈ {"window", "middle", "aisle"}

airline is a non-empty string with no whitespace

trip_type ∈ {"domestic", "international"}

Example

Example 1:

Input:

```
2.0

middle

IngidoX

international


Ouput:

Ramandeep Sir will not fly


Example 2:

Input:

1.0

middle

IngidoX

domestic


Ouput:

Ramandeep Sir will fly
```

NST Bus Route Optimization — 20/20 XP — SOLVED — deadline 2026-10-05 20:35

Topics: Python Function, Nested if-else

URL: https://my.newtonschool.co/playground/code/2kxfo1p28ay2

Back

NST Bus Route Optimization

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

NST Bus Route Optimization

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

The bus driver at NST Campus is planning his daily pickup route. He needs to pick up students from two hostels: Hostel 1 and Hostel 2, and return to the NST campus afterward. To save fuel, he wants to choose the route that results in the minimum total distance traveled.

There are three roads available:

A road of length d1 connects the NST campus and Hostel 1

A road of length d2 connects the NST campus and Hostel 2

A road of length d3 directly connects Hostel 1 and Hostel 2

The driver can travel on any road any number of times, and in any direction. He must start at NST campus, pick up students from both hostels, and return back to the campus, while minimizing the total distance.

Please refer to the following image for further clarity:

Click to enlarge

Input

A single line contains three integers:

d1 — the distance between NST campus and Hostel 1

d2 — the distance between NST campus and Hostel 2

d3 — the distance between Hostel 1 and Hostel 2

Output

Print a single integer — the minimum total distance the bus must travel.

Constraints

1 ≤ d_1, d_2, d_3 ≤ 10

8

Example

Example 1:

Input

4 6 2

Output

12

```
Explanation:

Try all possible loops:

Campus → Hostel 1 → Hostel 2 → Campus: 4 + 2 + 6 = 12

Campus → Hostel 2 → Hostel 1 → Campus: 6 + 2 + 4 = 12

Campus → Hostel 1 → Campus → Hostel 2 → Campus: 4 + 4 + 6 + 6 = 20, not minimal


So, minimum = 12




Example 2:

Input

2 2 8


Output

8
```

Museum Ticket Booking System — 20/20 XP — SOLVED — deadline 2026-10-05 20:35

Topics: Python Function, Nested if-else, Python if- else

URL: https://my.newtonschool.co/playground/code/ty9vjaz6iw3s

Back

Museum Ticket Booking System

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Museum Ticket Booking System

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are tasked with implementing the function calculateTicketPrice for a Museum Ticket Booking system.

The function should calculate the ticket price based on the following rules:

The base ticket price is $12.

If the visitor is under 12 years old, apply a discount of $5.

If the visit is on a weekend, add a surcharge of $3.

If the visitor has a student card, apply an additional discount of $2.

The function calculateTicketPrice accepts the following parameters:

age (number) – The age of the visitor.

isWeekend (boolean) – true if the visit is on a weekend, otherwise false.

hasStudentCard (boolean) – true if the visitor has a student card, otherwise false.

Your task is to return the final ticket price after applying all applicable discounts and surcharges.

Input

The first line contains an integer age — the age of the visitor.

The second line contains a boolean isWeekend — true if the visit is on a weekend, otherwise false.

The third line contains a boolean hasStudentCard — true if the visitor has a student card, otherwise false.

Output

Return a single integer representing the final ticket price.

Constraints

1

≤

a

g

e

≤

100

1≤age≤100

Example

Example

Input

23

true

false

Output

15

Explanation

Base ticket price = 12.

```
The visitor is 23 years old, so no age discount is applied.

The visit is on a weekend, so 3 is added to the ticket price.

The visitor does not have a student card, so no student discount is applied.



Therefore, the final ticket price is 15.
```

Greater of Four — 20/20 XP — SOLVED — deadline 2026-10-05 20:35

Topics: Python Function, Nested if-else, Python if- else

URL: https://my.newtonschool.co/playground/code/ympu0ctxfxs1

Back

Greater of Four

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Greater of Four

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given four integers a, b, c, and d.

A built-in function greater(x, y) is available to you. It returns the greater of the two values x and y.

Your task is to find the greatest among the four given numbers using the greater() function.

Note: You must use the provided greater() function. Do not compare all four numbers directly.

Input

The first line contains four space-separated integers a, b, c, and d.

Output

Print a single integer — the greatest among the four numbers.

Constraints

1 ≤ a, b, c, d ≤ 103

```
Example

Example 1

Input

10 25 15 24

Output

25


Example 2

Input

5 2 8 7

Output

8
```

PSP Contest 2 [Contest] — due 2026-10-09 11:45

Find the remainder — 40/40 XP — SOLVED — deadline 2026-10-09 11:45

Topics: Python Arithmetic operators

URL: https://my.newtonschool.co/playground/code/wrtaigg345ws

Back

Find the remainder

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Find the remainder

Easy

2x

40/40

Time Limit: 2, Memory Limit: 256000

You are given two numbers a and b. Find the remainder when a is divided by b.

Input

The first line contains integer a.

The second line contains integer b.

Output

Print the remainder when a is divided by b.

Constraints

0

≤

a

≤

200

0≤a≤200

1

≤

b

≤

200

```
1≤b≤200




Example

Input

9

7

Output

2

Explanation

When we divide 9 by 7, we get 2 as remainder.

Therefore, ans = 2.
```

The Frame — 40/40 XP — SOLVED — deadline 2026-10-09 11:45

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/bb3jyl2rf4tp

Back

The Frame

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

The Frame

Easy

2x

40/40

Time Limit: 2, Memory Limit: 256000

In the annual design festival of a small town, a young engineer is given four wooden pieces to create the frame of a special display panel.

Each piece is straight and has a fixed length. The engineer is not allowed to cut, break or bend any of the pieces. All four pieces must be used exactly once.

The display panel is supposed to have a perfectly balanced shape. This means that its four sides must have exactly the same length, and the four pieces must be arranged so that they form a proper square.

The engineer has received the lengths of the four available pieces. Determine whether the four pieces can be arranged to form the required shape.

Input

The input contains a single line with four integers a, b, c, and d, representing the lengths of the four wooden pieces.

$1 \le a, b, c, d \le 10$

All four pieces must be used.

No piece can be cut, bent, broken, or modified.

Output

Print YES if the four pieces can be arranged to form a square.

Otherwise, print NO.

Constraints

1 ≤ a , b , c , d ≤ 10

1≤a,b,c,d≤10

All four pieces must be used exactly once.

The pieces cannot be modified in any way.

Example

Example 1

Input:

1 2 3 4

Output:

NO

Explanation:

The four pieces have different lengths, so they cannot form desired shape.

Example 2

Input:

5 5 5 5

```
Output:

YES


Explanation:

All four pieces have the same length. Therefore, they can be arranged as the desired shape.
```

Function Based Calculator. — 80/80 XP — SOLVED — deadline 2026-10-09 11:45

Topics: Python if- else

URL: https://my.newtonschool.co/playground/code/16km8rga31m7

Back

Function Based Calculator.

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

Try AI Plan Mode

QUESTION

Function Based Calculator.

Medium

2x

80/80

Time Limit: 2, Memory Limit: 256000

You are building a calculator — but this time, you're not allowed to use direct arithmetic operators.

All the arithmetic operations have already been defined as functions in the code for you.

Your Task:

Read two integers and one operator symbol from the input

Based on the operator, call the appropriate function

Print the result returned by that function

If the operator is invalid (other than ' + ' , ' - ' , ' * ', , ' // ', ' % ') for this question , print Invalid Operator

Note: Do not use arithmetic operators directly in your code. Use only the pre-defined functions provided as comments in the starter code.

Input

The first line of input contains two space-separated integers, representing A and B.

The second line contains a string op, representing the operation to perform.

Output

A single line — the result of the operation, or Invalid Operator if the symbol is not supported.

Note: Use pre-defined functions.

Constraints

−

1

0

3

≤

A

,

B

≤

1

0

3

−10

3

≤A,B≤10

3

B

≠

0

B

□

=0, for op = '/' or '%'

Example

Input

20 10

+

Output

30

Input

```
5 3

^

Output

Invalid Operator
```

Calculate Final Price — 40/80 XP — SOLVED — deadline 2026-10-09 11:45

Topics: Datatypes & Variables, Functions (DSA), Python Arithmetic operators

URL: https://my.newtonschool.co/playground/code/8eaek8eib11u

Back

Calculate Final Price

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Calculate Final Price

Medium

40/80

Time Limit: 2, Memory Limit: 256000

Riya is shopping for a new product. The product has an original price, but she receives a discount on it. After applying the discount, a tax is added to the discounted price.

Given the original price, discount percentage, and tax percentage, define and implement the following function to calculate the final price:

```
def final_price(price, discount, tax):
```

The function should calculate the final price and return the result. The returned value must then be stored and printed outside the function.

Steps to be followed:

Use the function signature def final_price(price, discount, tax):.

The function must return the final price.

Do not print the result inside the function.

Take input and call the function outside the function.

Store the returned value in a variable and print it outside the function.

Input

The input consists of three lines:

The first line contains a integer number price, representing the original price.

The second line contains a integer number discount, representing the discount percentage.

The third line contains a integer number tax, representing the tax percentage.

Output

Print a single floating number representing the final price.

Constraints

1 ≤ price ≤ 100000

0 ≤ discount ≤ 100

0 ≤ tax ≤ 100

Example

Input 1:

1000

10

5

Output 1:

945.0

Explanation:

The original price is 1000.

```
After a 10% discount, the price becomes 900.

A 5% tax is then added to this discounted price.

The final price is 945.0.


Input 2:


2000

20

10


Output 2:


1760.0


Explanation:


The original price is 2000.

After a 20% discount, the price becomes 1600.

A 10% tax is then added to this discounted price.

The final price is 1760.0
```

The Gatekeeper — 0/80 XP — NOT SOLVED — deadline 2026-10-09 11:45

Topics: If- elif-else ladder

URL: https://my.newtonschool.co/playground/code/1dja9ovexna6

Back

The Gatekeeper

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

The Gatekeeper

Medium

0/80

Time Limit: 2, Memory Limit: 256000

Write a Python program to simulate a simple login system.

Login Details:

The correct username is admin.

The correct password is password123.

The program should take two inputs from the user:

username

password

Based on the input, the program should print:

Login successful – if both the username and password are correct.

Check username or password – if only one of them is correct.

Invalid credentials – if both are incorrect.

Input

The first line contains a string representing the username.

The second line contains a string representing the password.

Output

Print a single line containing one of the following message based on the entered credentials:

Login successful – if both the username and password are correct.

Check username or password – if only one of them is correct.

Invalid credentials – if both are incorrect.

Example

Sample 1:

Input:

admin

password123

Output:

Login successful

Explanation:

Both the entered username (admin) and password (password123) match the correct credentials.

Hence, the program prints Login successful.

Sample 2:

Input:

admin

wrongpass

Output:

Check username or password

```
Explanation:


The username is correct (admin), but the password (wrongpass) is incorrect.

Since only one of the credentials is correct, the output is Check username or password.
```

Newton and Rounding — 0/120 XP — NOT SOLVED — deadline 2026-10-09 11:45

Topics: Python Arithmetic operators

URL: https://my.newtonschool.co/playground/code/6h902kkag1my

Back

Newton and Rounding

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Newton and Rounding

Hard

0/120

Time Limit: 2, Memory Limit: 256000

Newton has a non-negative integer n. He wants to round it to the nearest multiple of 10.

If n already ends with 0, it is already rounded.

If the last digit is 1–4, move to the lower multiple of 10.

If the last digit is 6–9, move to the higher multiple of 10.

If the last digit is 5, always move to the lower multiple of 10.

Your task is to find and print the number Newton will round to.

Input

A single integer n — the number that Newton has.

Output

Print the result of rounding n.

Constraints

0

≤

n

≤

5000

0≤n≤5000

Example

Input

5

Output

0

Explanation

Normally, 5 could be rounded to 0 or 10, but as per the problem rule, when n = 5, always output 0.

Input

113

Output

110

Explanation

113 is closer to 110 than 120, so the rounded number is 110.

Input

2359

Output

2360

Explanation

2359 is closer to 2360 than 2350, so it rounds up.

Grade Report — 0/80 XP — NOT SOLVED — deadline 2026-10-09 11:45

Topics: Functions (DSA)

URL: https://my.newtonschool.co/playground/code/go5ltf2xj034

Back

Grade Report

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Grade Report

Medium

0/80

Time Limit: 2, Memory Limit: 256000

You are given marks in three subjects. Your task is to determine the student's grade.

Complete the following function:

```
def get_grade(mark1, mark2, mark3):
    # Write your code here
```

Rules

If marks in any subject are below 40, return Fail.

Otherwise, calculate the percentage out of 300.

Grade based on percentage:

90% and above → A+

75% to below 90% → A

60% to below 75% → B

40% to below 60% → C

Note

You only need to complete get_grade().

You do not need to take input or print anything.

The function should return the grade (A+, A, B, C, or Fail).

Input

The first line contains an integer mark1, representing marks obtained in the first subject.

The second line contains an integer mark2, representing marks obtained in the second subject.

The third line contains an integer mark3, representing marks obtained in the third subject.

Output

The output should contain the final grade returned by the get_grade(mark1, mark2, mark3) function.

The grade will be one of A+, A, B, C, or Fail.

Constraints

0 ≤ mark1, mark2, mark3 ≤ 100

Example

Example 1

Input

95

92

88

Output

A+

Explanation

The student scores at least 40 in all subjects. The total is 275 out of 300, giving a percentage of 91.7%. Therefore, the grade is A+.

Example 2

Input

80

90

70

Output

A

Explanation

The student passes all three subjects. The total is 240 out of 300, giving a percentage of 80.0%. Therefore, the grade is A.

Example 3

Input

```
75

35

80


Output


Fail


Explanation



The student scored below 40 in one subject (35), so the final grade is Fail.
```

Nested if-else [In Class] — due 2026-10-10 10:43

Palindrome with Digit Validation — 20/20 XP — SOLVED — deadline 2026-10-10 10:43

Topics: Nested if-else

URL: https://my.newtonschool.co/playground/code/0paxymnbok5k

Back

Palindrome with Digit Validation

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Palindrome with Digit Validation

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Write a program that take an integer number n and an integer digit d, then prints one of these messages based on the conditions below:

1. If n is a palindrome and its last digit is d:

    "Number is palindrome and last digit is d"

2. If n is a palindrome but its last digit is not d:

    "Number is palindrome and last digit is not d"

3. If n is not a palindrome:

    "Number is not a palindrome"

Note:

Use the provided isPalindrome(n) function returns True or False. Do not rewrite it.

A palindrome number is a number that reads the same backward as forward.

Examples: 121, 1331, 44444

Numbers that are not palindrome do not read the same backward.

Examples: 123, 4567, 1201

Input

The Input will be the two space-separated integers n and d.

Output

The output will be the string message as per the problem description.

Constraints

1 <= n <= 1000

1 <= d <= 9

Example

Input 1:

121 1

Output 1:

Number is palindrome and last digit is 1

Explanation:

121 is palindrome and last digit in 121 is d as well

so print "Number is palindrome and last digit is 1"

Input 2:

121 3

Output 2:

Number is palindrome and last digit is not 3

Explanation:

121 is palindrome but last digit in 121 is not d that is 3,

so print "Number is palindrome and last digit is not 3"

Input 3:

123 3

```
Output 3:

Number is not a palindrome


Explanation:

123 is not even palindrome so we don't check for last digit

just print Number is not a palindrome
```

NewForces Ratings Calculator — 60/60 XP — SOLVED — deadline 2026-10-10 10:43

Topics: Nested if-else

URL: https://my.newtonschool.co/playground/code/vm6i10bc0it6

Back

NewForces Ratings Calculator

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

NewForces Ratings Calculator

Hard

2x

60/60

Time Limit: 2, Memory Limit: 256000

In the Newforces contest, participants ratings can increase or decrease based on the contest type they participate in and whether they hack any solutions. The new rating is calculated based on following rules:

Rules based on Rating Criteria:

If the participant's rating is below 1200:

Div. 1 (Type 1): Rating increases by 200 points.

Div. 2 (Type 2): Rating increases by 80 points.

Educational (Type 3): Rating increases by 50 points.

If the participant's rating is between 1200 and 2000 (inclusive):

Div. 1 (Type 1): Rating increases by 100 points.

Div. 2 (Type 2): Rating remains the same.

Educational (Type 3): Rating decreases by 30 points.

If the participant's rating is above 2000:

Div. 1 (Type 1): Rating increases by 50 points.

Div. 2 (Type 2): Rating decreases by 30 points.

Educational (Type 3): Rating decreases by 100 points.

Rules based on whether the user has hacked any solution or not:

If the user has hacked any solution: Applies a further increase of 50 points to the rating.

Did Not Hack: No additional change is applied.

Note: Please check input format for more details

Input

The input consists of three lines:

r — an integer representing the current rating of the participant

t — an integer representing the contest type:

1 → Div. 1

2 → Div. 2

3 → Educational

h — an integer indicating hacking status:

1 → Hacked at least one solution

2 → Did not hack any solution

Output

Print a single integer — the new rating of the participant after considering both contest type and hacking bonus (if any).

Constraints

$1100 \le r \le 2100$

$t \in \{1, 2, 3\}$

$h \in \{1, 2\}$

Example

Example 1

Input:

1100

1

1

Output:

1350

```
Explanation:

Rating < 1200, contest is Div. 1 → +200

Hacked a solution → +50

Final rating = 1100 + 200 + 50 = 1350


------------------------------------------------------------


Example 2

Input:

1500

3

2


Output:

1470


Explanation:

Rating is between 1200 and 2000, contest is Educational → −30

Did not hack → no bonus

Final rating = 1500 − 30 = 1470
```

Recursion Mechanism, Iteration in Python, Range () function [In Class] — due 2026-10-12 10:50

Find the Sum of Numbers in Range L to R — 20/20 XP — SOLVED — deadline 2026-10-12 10:50

Topics: Recursion, Range () function

URL: https://my.newtonschool.co/playground/code/l3psf4gfskvg

Back

Find the Sum of Numbers in Range L to R

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Find the Sum of Numbers in Range L to R

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Write a program that takes two integers, L and R, as input and calculates the sum of all numbers from L to R inclusive.

Input

The first line of input contains a single integer L.

The second line of input contains a single integer R.

Output

Print an integer, representing the sum of all numbers from L to R inclusive.

Constraints

1

≤

L

≤

R

≤

100

1≤L≤R≤100

Example

Input

5

10

```
Output

45

Explanation

Numbers in range 5 to 10 are: 5, 6, 7, 8, 9, 10. And their sum is 45.


Input

5

5

Output

5
```

Recursion Mechanism, Iteration in Python, Range () function [In Class] — due 2026-10-17 11:03

Multiplication Table of N — 20/20 XP — SOLVED — deadline 2026-10-17 11:03

Topics: For Loop, Recursion

URL: https://my.newtonschool.co/playground/code/m5pmhpi8obpm

Back

Multiplication Table of N

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Multiplication Table of N

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Given a positive integer N, your task is to print the multiplication table of N from 1 to 10.

Each line should follow the format:

N x i = result

,where i ranges from 1 to 10, and result is the product of N and i.

Input

The first and only line of input contains a single integer N.

Output

Print 10 lines, where each line is in format:

N x i = result

,where i ranges from 1 to 10, and result is the product of N and i.

Constraints

1

≤

N

≤

10

1≤N≤10

Example

```
Input

5

Output

5 x 1 = 5

5 x 2 = 10

5 x 3 = 15

5 x 4 = 20

5 x 5 = 25

5 x 6 = 30

5 x 7 = 35

5 x 8 = 40

5 x 9 = 45

5 x 10 = 50
```

Print all divisors of N — 20/20 XP — SOLVED — deadline 2026-10-17 11:03

Topics: For Loop, Recursion

URL: https://my.newtonschool.co/playground/code/ebgggw9ce1f0

Back

Print all divisors of N

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Print all divisors of N

Easy

2x

20/20

Time Limit: 2, Memory Limit: 128000

You are given a positive integer N. Your task is to find and print all the positive divisors of N in increasing order.

Note: A divisor of N is a number d such that d divides N completely.

Input

The first and only line of input contains a single integer N.

Output

Print space-separated integers, representing the positive divisors of N in increasing order.

Constraints

1

≤

N

≤

1

0

5

1≤N≤10

5

```
Example

Input

10

Output

1 2 5 10

Explanation

A divisor of 10 is any number that divides 10 exactly, i.e., with no remainder.

Let’s check which numbers divide 10:

1 → 10 ÷ 1 = 10 → ✅ (no remainder)

2 → 10 ÷ 2 = 5 → ✅

3 → 10 ÷ 3 = 3.33… → ❌

4 → 10 ÷ 4 = 2.5 → ❌

5 → 10 ÷ 5 = 2 → ✅

6 to 9 → None divide 10 evenly

10 → 10 ÷ 10 = 1 → ✅
```

Multiples of 3 and 5 — 20/20 XP — SOLVED — deadline 2026-10-17 11:03

Topics: For Loop, Recursion

URL: https://my.newtonschool.co/playground/code/07vg13m6mszo

Back

Multiples of 3 and 5

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Multiples of 3 and 5

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to print all numbers between 1 and N (inclusive) that are divisible by both 3 and 5.

Input

The first and only line of input contains a single integer N.

Output

Print all numbers from 1 to N that are divisible by both 3 and 5, in increasing order.

The numbers should be printed on a single line, separated by a space.

Note: If there are no such numbers, you need not to print anything.

Constraints

1

≤

N

≤

1

0

6

1≤N≤10

```
6

Example

Input

30

Output

15 30

Explanation

For input 30, the numbers between 1 and 30 that are divisible by both 3 and 5 are: 15 and 30.


Input

5

Output


Explanation

It can be seen that upto 5, there is no number which is a multiple of 5 and 3 both.
```

Iteration in Python, Recursion Mechanism, Range () function, Intro and Types - L ... [Post Class] — due 2026-10-17 15:00

Find The Even Numbers — 20/20 XP — SOLVED — deadline 2026-10-17 15:00

Topics: Recursion, Range () function

URL: https://my.newtonschool.co/playground/code/8tluckwr59xq

Back

Find The Even Numbers

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Find The Even Numbers

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

Write a program that takes a positive integer N as input and prints all even numbers from 1 to N (inclusive), each on a separate line.

Input

The first and only line of input contains a single integer N, the number of bridges.

Output

Print all even numbers from 1 to N (inclusive), each on a separate line.

Constraints

1

≤

N

≤

20

1≤N≤20

Example

Input

3

Output

2

Input

```
6

Output

2

4

6
```

Sum from 1 to N — 20/20 XP — SOLVED — deadline 2026-10-17 15:00

Topics: For Loop, Recursion

URL: https://my.newtonschool.co/playground/code/lvr71f9ssooi

Back

Sum from 1 to N

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Sum from 1 to N

Easy

2x

20/20

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to calculate and print the sum of all integers from 1 to N (inclusive).

Input

The first and only line of input contains a single integer N.

Output

Print an integer, representing the sum of all integers from 1 to N (inclusive).

Constraints

1

≤

N

≤

50

1≤N≤50

Example

Input

10

Output

55

Explanation

The sum of number from 1 to 10 is equal to 55.

```
Input

20

Output

Total Energy: 210

Explanation

The sum of number from 1 to 20 is equal to 210.
```

Count Numbers Divisible by 3 or 5 — 0/20 XP — NOT SOLVED — deadline 2026-10-17 15:00

Topics: For Loop, Recursion

URL: https://my.newtonschool.co/playground/code/wqfz7z0xggkm

Back

Count Numbers Divisible by 3 or 5

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Count Numbers Divisible by 3 or 5

Easy

0/20

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to print all numbers in the range from 1 to N (inclusive) that are divisible by 3 or 5, but not both.

Input

The first and only line of input contains a positive integer N.

Output

Print all numbers in the range from 1 to N (inclusive) that are divisible by 3 or 5, but not both.

Constraints

3

≤

N

≤

100

3≤N≤100

Example

Input

15

Output

3 5 6 9 10 12

Explanation

From 1 to 15, numbers like 3, 5, 6, 9, 10, and 12 are divisible by 3 or 5, but not both.

```
15 is divisible by both, so it's excluded.
```

Squares from 1 to N (Part I) — 0/20 XP — NOT SOLVED — deadline 2026-10-17 15:00

Topics: For Loop

URL: https://my.newtonschool.co/playground/code/to18n5q10fsj

Back

Squares from 1 to N (Part I)

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Squares from 1 to N (Part I)

Easy

0/20

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to print the squares of all integers from 1 to N, one per line.

In other words, print:

1^2, 2^2, 3^2, ..., N^2, each on a separate line.

Input

The first and only line of input contains a single integer N.

Output

Print N lines, where the i-th line contains the square of integer i, where

1

≤

i

≤

N

1≤i≤N

Constraints

1 ≤ N ≤ 20

Example

Input

5

Output

```
1

4

9

16

25


Input

3

Output

1

4

9
```

Squares from 1 to N (Part II) — 0/20 XP — NOT SOLVED — deadline 2026-10-17 15:00

Topics: For Loop

URL: https://my.newtonschool.co/playground/code/qyf5g3girt88

```
Back

Squares from 1 to N (Part II)

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Squares from 1 to N (Part II)

Easy

0/20

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to print the squares of all integers from 1 to N,
on the same line, separated by a space.

Input

The first and only line of input contains a single integer N.

Output

A single line containing N space-separated integers, representing the squares of all integers from
1 to N.

Constraints

1 ≤ N ≤ 20

Example

Input

5

Output

1 4 9 16 25
```

Recursion Mechanism, Iteration in Python, Range () function [In Class] — due 2026-10-17 15:52

Factorial-Loops — 0/20 XP — NOT SOLVED — deadline 2026-10-17 15:52

Topics: Recursion, While loop

URL: https://my.newtonschool.co/playground/code/ehuog873narx

Back

Factorial-Loops

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Factorial-Loops

Easy

0/20

Time Limit: 2, Memory Limit: 128000

You are given a positive integer N. Your task is to compute and print the factorial of N.

Note: The factorial of a number N, denoted as N!, is the product of all positive integers less than or equal to N.

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, factorial of N

Constraints

0

≤

N

≤

20

0≤N≤20

Example

Input

5

Output

120

Explanation

```
5!= 5 * 4 * 3 * 2 * 1 = 120


Input

10

Output

3628800
```

Check if a Number is a Palindrome — 0/20 XP — NOT SOLVED — deadline 2026-10-17 15:52

Topics: Recursion, While loop

URL: https://my.newtonschool.co/playground/code/fads3shiyxzl

Back

Check if a Number is a Palindrome (Function)

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Check if a Number is a Palindrome (Function)

Medium

2x

0/40

Time Limit: 2, Memory Limit: 256000

Given a positive integer N, determine whether it is a palindrome.

Note: A number is called a palindrome if it reads same backwards and forwards.

Input

The first and only line of input contains a positive integer N.

Output

Print "Yes" if the given number is a palindrome, otherwise "No" (without quotes).

Constraints

1

≤

N

≤

1

0

9

1≤N≤10

9

Example

Input

```
54145

Output

Yes

Explanation

The given number reads same backwards and forwards. Thus it is a palindromic number.


Input

12345

Output

No
```

Digit Sum using Loop — 0/20 XP — NOT SOLVED — deadline 2026-10-17 15:52

Topics: Recursion, While loop

URL: https://my.newtonschool.co/playground/code/w61cb0h3z1fr

Back

Digit Sum using Loop

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Digit Sum using Loop

Easy

0/20

Time Limit: 2, Memory Limit: 128000

You are provided with a function sumOfDigits(n), where the integer n is passed as a parameter.

Input

You don't need to take the input.

Output

You need to return the sum of digits of n.

Constraints

1

≤

N

≤

1

0

9

1≤N≤10

9

Example

Input

28393

```
Output

25

Explanation

The digits of the given number are: 2, 8, 3, 9 and 3. And their sum is 25.
```

While loop [Post Class] — due 2026-10-17 19:35

Count the Digits — 0/20 XP — NOT SOLVED — deadline 2026-10-17 19:35

Topics: Recursion, While loop

URL: https://my.newtonschool.co/playground/code/uxi69ufxz200

Back

Count the Digits

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Count the Digits

Easy

0/20

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to find the count of digits in N. There will be no leading zeroes in the input number N. For example, inputs like 0023 or 0005 will not be given.

Note: The following things are blocked in the backend, so please don’t use them in your code:

String literals, list literals, and conversions using str() or list().

Indexing, iteration, or applying string methods on variables obtained from input().

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, the count of digits in the given number.

Constraints

1

≤

N

≤

1

0

```
9

1≤N≤10

9

There will be no leading zeroes in the input number N. For example, inputs
like 0023 or 0005 will not be given.

Example

Input

28232

Output

5

Explanation

There are total 5 digits in the given number that are: 2, 8, 2, 3 and 2.
```

Reverse a Number — 0/20 XP — NOT SOLVED — deadline 2026-10-17 19:35

Topics: While loop

URL: https://my.newtonschool.co/playground/code/u5nc58u56gp9

Back

Reverse a Number

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Reverse a Number

Easy

0/20

Time Limit: 2, Memory Limit: 128000

You are given a positive integer N. Your task is to reverse the given number.

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, representing the reversed number.

Constraints

1

≤

N

≤

1

0

9

1≤N≤10

9

Example

Input

536446

Output

644635

```
Input

100

Output

1
```

Sum of First and Last Digits of a Number — 0/20 XP — NOT SOLVED — deadline 2026-10-17 19:35

Topics: While loop

URL: https://my.newtonschool.co/playground/code/7qgquvnt2dlx

Back

Sum of First and Last Digits of a Number

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Sum of First and Last Digits of a Number

Easy

0/20

Time Limit: 2, Memory Limit: 128000

You are given a positive integer N. Your task is to calculate the sum of the first and last digits of the number.

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, the sum of the first and last digits of the number.

Constraints

1

≤

N

≤

1

0

9

1≤N≤10

9

Example

Input

14368

Output

9

```
Explanation

1 and 8 are the first and last digits of the given number respectively, and their sum is 9.
```

Find the Count of Even and Odd Digits — 0/20 XP — NOT SOLVED — deadline 2026-10-17 19:35

Topics: While loop

URL: https://my.newtonschool.co/playground/code/itxgoou9wvbd

Back

Find the Count of Even and Odd Digits

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Find the Count of Even and Odd Digits

Easy

0/20

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to separately find the count of its even digits and the count of its odd digits.

Input

The first and only line of input contains a positive integer N.

Output

Print two space-separated integers, representing the count of even and odd digits in the given number N.

Constraints

1

≤

N

≤

1

0

9

1≤N≤10

9

Example

Input

295892

Output

```
3 3

Explanation

The even digits in the given number are: 2, 8, and 2.

And the odd digits in the given number are: 9, 5, and 9.

Thus the count of even and odd digits will be 3 and 3 respectively.
```

Find the Largest Digit in a Number — 0/40 XP — NOT SOLVED — deadline 2026-10-17 19:35

Topics: Recursion, While loop

URL: https://my.newtonschool.co/playground/code/3nhaa69c3fr2

Back

Find the Largest Digit in a Number

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Find the Largest Digit in a Number

Medium

0/40

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to find the largest digit in the given number.

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, the largest digit in the given number.

Constraints

0

≤

N

≤

1

0

9

0≤N≤10

9

Example

Input

2839

Output

9

```
Explanation

The given number 2839 contains digits: 2, 8, 3 and 9, amongst them 9 is the largest digit.
```

While loop [In Class] — due 2026-10-19 10:52

Find the Sum of Even and Odd Digits — 40/40 XP — SOLVED — deadline 2026-10-19 10:52

Topics: While loop

URL: https://my.newtonschool.co/playground/code/8jp917i4ddjs

Back

Find the Sum of Even and Odd Digits

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Find the Sum of Even and Odd Digits

Medium

2x

40/40

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to separately find the sum of its even digits and the sum of its odd digits.

Input

The first and only line of input contains a positive integer N.

Output

Print two space-separated integers, representing the sum of even and odd digits in the given number N.

Constraints

1

≤

N

≤

1

0

9

1≤N≤10

9

Example

Input

295892

```
Output

12 23

Explanation

The even digits in the given number are: 2, 8, and 2.

And the odd digits in the given number are: 9, 5, and 9.

Thus the sum of even and odd digits will be 12 and 23 respectively.
```

Find the Smallest Digit in a Number — 40/40 XP — SOLVED — deadline 2026-10-19 10:52

Topics: Recursion, While loop

URL: https://my.newtonschool.co/playground/code/8alnm0d8n23c

Back

Find the Smallest Digit in a Number

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Find the Smallest Digit in a Number

Medium

2x

40/40

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to find the smallest digit in the given number.

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, the smallest digit in the given number.

Constraints

1

≤

N

≤

1

0

9

1≤N≤10

9

Example

Input

2839

Output

```
2

Explanation

The given number 2839 contains digits: 2, 8, 3 and 9, amongst them 2 is the smallest digit.
```

Armstrong or not II — 40/40 XP — SOLVED — deadline 2026-10-19 10:52

Topics: While loop

URL: https://my.newtonschool.co/playground/code/j7fkycqnxqvx

Back

Armstrong or not II

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Armstrong or not II

Medium

2x

40/40

Time Limit: 2, Memory Limit: 256000

You are given a number. Your task is to check if this number is an Armstrong number.

An Armstrong number is a number that is equal to the sum of its own digits, each raised to the power of the number of digits.

Input

First line will contain an Integer N.

Output

Print True if the number is Armstrong, otherwise print False.

Example

Input:

153

Output:

True

Explanation

The number 153 has 3 digits: 1, 5, and 3.

If we raise each digit to the power of 3 (the total number of digits) and sum them:

13 + 53 + 33 = 153

Since the sum is equal to the original number, 153 is an Armstrong number.

```
Input:

12

Output:

False

Explanation

The number 12 has 2 digits: 1 and 2.

If we raise each digit to the power of 2 (the total number of digits) and sum them:

12 + 22 = 5

Since the sum is not equal to the original number, 12 is not an Armstrong Number.
```

Loops, Control flow statements, Break, Continue, Python Function, Scope, Variabl ... [In Class] — due 2026-10-19 17:02

Is It Prime? — 0/20 XP — NOT SOLVED — deadline 2026-10-19 17:02

Topics: Loops, Recursion

URL: https://my.newtonschool.co/playground/code/6ysp7dm11g43

Back

Is It Prime?

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Is It Prime?

Easy

0/20

Time Limit: 2, Memory Limit: 128000

You are given a positive integer N. Your task is to determine whether N is a prime number. Print "YES" if the number is prime, otherwise print "NO".

Note: A prime number is a number greater than 1 that has no positive divisors other than 1 and itself.

Input

The first and only line of input contains a positive integer N.

Output

Print "YES" if N is a prime number.

Print "NO" otherwise.

Constraints

1

≤

N

≤

1

0

3

```
1≤N≤10

3

Example

Input

7

Output

YES

Explanation

7 is only divisible by 1 and 7 → Prime → Output "YES"


Input

10

Output

NO

Explanation

10 is divisible by 1, 2, 5, 10 → Not prime → Output "NO"
```

Add 1 to Each Digit — 0/20 XP — NOT SOLVED — deadline 2026-10-19 17:02

Topics: Loops

URL: https://my.newtonschool.co/playground/code/9kgf3qfr0tsp

Back

Add 1 to Each Digit

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Add 1 to Each Digit

Easy

0/20

Time Limit: 2, Memory Limit: 256000

Write a program to add 1 to each digit of the given number and print the result.

Note: Consider adding 1 to digit 9 makes it 0.

Input

The first and only line of input contains a positive integer N.

Output

Print an integer, the resultant number formed after adding 1 to each digit.

Constraints

1

≤

N

≤

1

0

9

1≤N≤10

9

Example

```
Input

765392

Output

876403
```

Fibonacci Series of a given input — 0/20 XP — NOT SOLVED — deadline 2026-10-19 17:02

Topics: Loops

URL: https://my.newtonschool.co/playground/code/u04pck04jkzy

Back

Fibonacci Series of a given input

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Fibonacci Series of a given input

Easy

0/20

Time Limit: 2, Memory Limit: 128000

You are given a positive integer N. Your task is to print the first N terms of the Fibonacci series.

Input

The first and only line of input contains a positive integer N.

Output

Print N space-separated integers, representing the first N terms of the Fibonacci series.

Constraints

1

≤

N

≤

20

1≤N≤20

Example

input:

5

output:

0 1 1 2 3

Control flow statements, Loops, Break, Continue [Post Class] — due 2026-10-21 11:00

Find Odd Factors — 0/20 XP — NOT SOLVED — deadline 2026-10-21 11:00

Topics: Loops, For Loop

URL: https://my.newtonschool.co/playground/code/mjgqoptuk9qj

Back

Find Odd Factors

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Find Odd Factors

Easy

0/20

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to print all the odd factors of N in increasing order.

Input

The first and only line of input contains a positive integer N.

Output

Print space-separated integers, the odd factors of the given number.

Constraints

1

≤

N

≤

1

0

5

1≤N≤10

5

Example

Input

21

Output

```
1 3 7 21
```

Check if Even Sum is Divisible by 2 — 0/20 XP — NOT SOLVED — deadline 2026-10-21 11:00

Topics: Loops, While loop

URL: https://my.newtonschool.co/playground/code/1v2ig3ylqca8

Back

Check if Even Sum is Divisible by 2

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Check if Even Sum is Divisible by 2

Easy

0/20

Time Limit: 2, Memory Limit: 256000

You are given two positive integers L and R representing a range. Your task is to calculate the sum of all even numbers in the range [L, R] (inclusive), and check whether this sum is divisible by 2.

Input

The first and only line of input contains two space-separated integers L and R.

Output

Print "Yes" if the sum of all even numbers in the range [L, R] is divisible by 2, else "No" (without quotes).

Constraints

1

≤

L

≤

R

≤

1

0

18

1≤L≤R≤10

18

Example

Input

```
23 29

Output

Yes

Explanation

The even numbers in the given range are: 24, 26 and 28. And the sum of the numbers is 78, which is
divisible by 2.
```

Check for Perfect Number — 0/20 XP — NOT SOLVED — deadline 2026-10-21 11:00

Topics: Loops, While loop

URL: https://my.newtonschool.co/playground/code/ijfi1qbt2wzt

Back

Check for Perfect Number

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Check for Perfect Number

Easy

0/20

Time Limit: 2, Memory Limit: 256000

Given an integer N, your task is to determine if N is a perfect number.

Note: A perfect number is a positive integer that is equal to the sum of its proper divisors (all positive divisors excluding the number itself). For example, 6 is a perfect number because 1 + 2 + 3 = 6.

Input

The first and only line of input contains a positive integer N.

Output

Print "True" if the number is a perfect number, otherwise "False" (without quotes).

Constraints

1

≤

N

≤

1

0

5

1≤N≤10

5

Example

Input

```
6

Output

True

Explanation

Divisors of 6 are 1, 2, 3 and 6. We are not required to include the number itself, thus the
remaining divisors are 1, 2, and 3, that sum up to 6.
```

First Number Divisibly by 7 — 0/20 XP — NOT SOLVED — deadline 2026-10-21 11:00

Topics: Loops, For Loop, Break

URL: https://my.newtonschool.co/playground/code/qi7iukfehbkg

Back

First Number Divisibly by 7

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

First Number Divisibly by 7

Easy

0/20

Time Limit: 2, Memory Limit: 256000

You are given two positive integers L and R representing a range. Your task is to find the smallest number divisible by 7 in the range [L, R] (inclusive). If no such number is found, print "NOT FOUND".

Input

The first and only line of input contains two space-separated integers L and R.

Output

Print the smallest number divisible by 7 in the range [L, R]. If no such number exist, print "NOT FOUND".

Constraints

1

≤

L

≤

R

≤

1

0

18

1≤L≤R≤10

18

Example

Input

```
23 29

Output

28

Explanation

28 is the only number divisible by 7 in the given range [23, 29].


Input

23 27

Output

NOT FOUND

Explanation

There is no number divisible by 7 in the given range [23, 27].
```

Check if All Digits are Same — 0/40 XP — NOT SOLVED — deadline 2026-10-21 11:00

Topics: While loop, Break

URL: https://my.newtonschool.co/playground/code/51trq43f8x5h

Back

Check if All Digits are Same

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Check if All Digits are Same

Medium

0/40

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to check whether all the digits in the number are the same.

Input

The first and only line of input contains a positive integer N.

Output

Print "Yes" if all the digits in the number are the same, else "No" (without quotes).

Constraints

1

≤

N

≤

1

0

9

1≤N≤10

9

Example

Input

77777

Output

Yes

```
Input

22888

Output

No
```

Check if Rare — 0/40 XP — NOT SOLVED — deadline 2026-10-21 11:00

Topics: While loop, Break

URL: https://my.newtonschool.co/playground/code/z86pihq0ztw8

Back

Check if Rare

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Check if Rare

Medium

0/40

Time Limit: 2, Memory Limit: 256000

A number is called rare if all of its digits are divisible by a given number X.

Given a positive integer N and another positive integer X, write a program to check whether N is a rare number with respect to X.

That is, check if every digit of N is divisible by X. If all digits satisfy this condition, print True; otherwise, print False.

Input

The first and only line of input contains two space-separated integer N and X.

Output

If all digits satisfy the given condition, print "True"; otherwise, print "False" (without quotes).

Constraints

1

≤

N

≤

1

0

9

1≤N≤10

9

1

≤

X

≤

9

1≤X≤9

Example

Input

28 2

Output

True

Explanation

The given value 2 divides both the digits of given number (2 and 8). Thus the given number is the rare number.

PSP IA1 Coding [Other] — due 2026-10-29 12:20

Alternating Sum — 0/40 XP — NOT SOLVED — deadline 2026-10-29 12:20

URL: https://my.newtonschool.co/playground/code/8g4rryer12lz

Back

Alternating Sum

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

Try AI Plan Mode

QUESTION

Alternating Sum

Easy

0/40

Time Limit: 2, Memory Limit: 256000

You are given a positive integer n. Go through the numbers from 1 to n one by one.

- If the number is odd, add it to the sum;
- if the number is even, subtract it from the sum.

In other words, find the value of 1 − 2 + 3 − 4 + ... up to n. Print the final sum.

Note: This is a write-from-scratch problem. You must read the input yourself using input() and print the answer using print(). Print only the answer, with no extra text.

Input

A single line containing one integer `n`.

Output

A single integer: the alternating sum of the numbers from 1 to `n`.

Constraints

- 1 ≤ n ≤ 10^5

Example

Example 1

Input:

5

Output:

3

Explanation: 1 - 2 + 3 - 4 + 5 = 3 (odd numbers are added, even numbers are subtracted).

Example 2

Input:

4

Output:

-2

Explanation: 1 - 2 + 3 - 4 = -2. The answer can be negative.

Count Numbers That Satisfy Both Conditions — 0/80 XP — NOT SOLVED — deadline 2026-10-29 12:20

Topics: P-Value

URL: https://my.newtonschool.co/playground/code/2ib3p8z4b3e6

Back

Count Numbers That Satisfy Both Conditions

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Count Numbers That Satisfy Both Conditions

Medium

0/80

Time Limit: 2, Memory Limit: 256000

You are given n integers. A number is counted only if it satisfies BOTH of these conditions at the same time:

It is divisible by 5.

It lies between 10 and 100, both 10 and 100 included.

Print how many of the n numbers are counted.

Note: This is a write-from-scratch problem. You must read the input yourself using input() and print the answer using print(). Print only the answer, with no extra text.

Input

The first line contains an integer `n`. Each of the next `n` lines contains one integer.

Output

A single integer: the count of numbers that satisfy both conditions.

Constraints

- 1 ≤ n ≤ 1000

- 0 ≤ each number ≤ 10^4

Example

Example 1

Input:

6

15

100

5

105

10

22

Output:

3

Explanation: 15, 100 and 10 satisfy both conditions. 5 is below 10, 105 is above 100, and 22 is not divisible by 5, so those three are not counted.

Calculate an Electricity Bill — 0/40 XP — NOT SOLVED — deadline 2026-10-29 12:20

Topics: If else ladder

URL: https://my.newtonschool.co/playground/code/89mmis3jn59u

Back

Calculate an Electricity Bill

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Calculate an Electricity Bill

Easy

0/40

Time Limit: 2, Memory Limit: 256000

Complete the function calculate_bill(units). It receives the number of electricity units consumed in a month and must return the total bill amount in rupees. The energy charge is calculated slab by slab, and each rate applies only to the units that fall inside that slab:

Units in the slab	Rate per unit
First 100 units (1 to 100)	₹5
Next 100 units (101 to 200)	₹7
Next 100 units (201 to 300)	₹10
Above 300 units	₹12

A fixed meter charge of ₹50 is always added to the energy charge, even when units is 0. Total bill = energy charge + ₹50.

Note: Do NOT take any input and do NOT print anything. The input is already read and the output is printed by the hidden code. Complete only the function calculate_bill(units) and return the total bill (a number). Do not format or round the value; the hidden code prints it with exactly 2 digits

after the decimal point.

Input

A single integer `units` (read by the hidden code and passed to your function as the argument `units`).

Output

The total bill returned by your function. The hidden code prints it with exactly 2 digits after the decimal point.

Constraints

- 0 ≤ units ≤ 10^4

Example

Example 1

Input:

80

Output:

450.00

Explanation: 80 × 5 = 400, plus ₹50 meter charge = 450.

Example 2

Input:

250

Output:

1750.00

Explanation: First 100 units × 5 = 500; next 100 units × 7 = 700; remaining 50 units × 10 = 500. Energy charge = 1700, plus ₹50 = 1750.

```
Example 3


Input:

350


Output:

2850.00


Explanation: 500 + 700 + 1000 for the first 300 units = 2200; remaining 50 units × 12 = 600. Energy
charge = 2800, plus ₹50 = 2850.
```

Count Numbers Divisible by 3 but Not by 2 — 0/120 XP — NOT SOLVED — deadline 2026-10-29 12:20

Topics: Intro and Types - Loops

URL: https://my.newtonschool.co/playground/code/x6h1sxbrcc00

Back

Count Numbers Divisible by 3 but Not by 2

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Count Numbers Divisible by 3 but Not by 2

Hard

0/120

Time Limit: 2, Memory Limit: 256000

You are given two integers L and R. Count how many integers in the range from L to R (both L and R included) are divisible by 3 but NOT divisible by 2. Print the count.

Note: This is a write-from-scratch problem. You must read the input yourself using input() and print the answer using print(). Print only the answer, with no extra text.

Input

The first line contains the integer `L`. The second line contains the integer `R`.

Output

A single integer: the count of numbers in the range `L` to `R` that are divisible by 3 and not divisible by 2.

Constraints

- 1 ≤ L ≤ R ≤ 10^5

Example

Example 1

```
Input:

1

20



Output:

3




Explanation: The multiples of 3 from 1 to 20 are 3, 6, 9, 12, 15, 18. The ones that are not
divisible by 2 are 3, 9 and 15.


Example 2




Input:

20

50


Output:

5


Explanation: The valid numbers are 21, 27, 33, 39 and 45.
```

Water Tank Filling Hours — 0/120 XP — NOT SOLVED — deadline 2026-10-29 12:20

Topics: The while loop

URL: https://my.newtonschool.co/playground/code/vkr1nyup53cp

Back

Water Tank Filling Hours

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Water Tank Filling Hours

Hard

0/120

Time Limit: 2, Memory Limit: 256000

Complete the function hours_to_fill(capacity). An empty water tank is filled by a pump. The hours are numbered 1, 2, 3, ... and in every hour the following happens, in this order:

Pumping: if the hour number is a multiple of 5, the pump is off for maintenance and no water is added. Otherwise the pump adds 30 litres if the current water level is below 100 litres, 20 litres if the level is 100 litres or more but below 200 litres, and 10 litres if the level is 200 litres or more.

Full check: if the water level has now reached capacity or gone above it, the tank is full and the filling stops in this hour.

Leak: if the tank is not full yet and the water level is 150 litres or more, 5 litres leak out at the end of the hour.

The tank is empty (0 litres) before hour 1. The function must return the hour number in which the tank becomes full.

Note: Do NOT take any input and do NOT print anything. The input is already read and the output is printed by the hidden code. Complete only the function hours_to_fill(capacity) and return the hour number (an int) in which the tank becomes full.

Input

A single integer `capacity` (read by the hidden code and passed to your function as the argument `capacity`).

Output

The integer returned by your function (printed by the hidden code).

Constraints

- 1 ≤ capacity ≤ 10^4

Example

Example 1

Input:

100

Output:

4

Explanation: Levels after hours 1 to 4: 30, 60, 90, 120. In hour 4 the level (120) reaches the capacity 100, so the answer is 4.

Example 2

Input:

200

Output:

11

```
Explanation: Levels at the end of hours 1 to 4: 30, 60, 90, 120. Hour 5 is maintenance, so the
level stays 120 (no leak because it is below 150). Hour 6: 140. Hour 7: 160, then a leak makes it
155. Hour 8: 175, then 170. Hour 9: 190, then 185. Hour 10 is maintenance, no water is added but
the leak makes it 180. Hour 11: 180 + 20 = 200, which reaches the capacity, so the answer is 11.
```

Convert Marks to a Grade — 20/40 XP — SOLVED — deadline 2026-10-29 12:20

Topics: If else ladder

URL: https://my.newtonschool.co/playground/code/c0v1zqrnogyi

Back

Convert Marks to a Grade

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

Try AI Plan Mode

QUESTION

Convert Marks to a Grade

Easy

20/40

Time Limit: 2, Memory Limit: 256000

Complete the function get_grade(marks). It receives the marks of a student (an integer from 0 to 100) and must return the grade as a string, using this table:

Marks	Grade
90 to 100	A
75 to 89	B
60 to 74	C
40 to 59	D
0 to 39	F

Note: Do NOT take any input and do NOT print anything. The input is already read and the output is printed by the hidden code. Complete only the function get_grade(marks) and return the grade as a string (exactly one capital letter: A, B, C, D or F).

Input

A single integer `marks` (read by the hidden code and passed to your function as the argument

`marks`).

Output

The grade string returned by your function (printed by the hidden code).

Constraints

- 0 ≤ marks ≤ 100

Example

Input:

82

Output:

B

Explanation: 82 lies in 75 to 89, so the grade is B.

Example 2

Input:

39

Output:

F

Explanation: 39 is below 40, so the grade is F.

Python Function, Scope, Variable Scopes [In Class] — due 2026-11-07 16:50

Check if a Number is a Palindrome (Function) — 0/40 XP — NOT SOLVED — deadline 2026-11-07 16:50

Topics: Python Function

URL: https://my.newtonschool.co/playground/code/fads3shiyxzl

Back

Check if a Number is a Palindrome (Function)

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Check if a Number is a Palindrome (Function)

Medium

2x

0/40

Time Limit: 2, Memory Limit: 256000

Given a positive integer N, determine whether it is a palindrome.

Note: A number is called a palindrome if it reads same backwards and forwards.

Input

The first and only line of input contains a positive integer N.

Output

Print "Yes" if the given number is a palindrome, otherwise "No" (without quotes).

Constraints

1

≤

N

≤

1

0

9

1≤N≤10

9

Example

Input

```
54145

Output

Yes

Explanation

The given number reads same backwards and forwards. Thus it is a palindromic number.


Input

12345

Output

No
```

Prime Numbers upto N (Function) — 0/40 XP — NOT SOLVED — deadline 2026-11-07 16:50

Topics: Python Function

URL: https://my.newtonschool.co/playground/code/94bsjtp8jnfk

Back

Prime Numbers upto N (Function)

Total XP

6,733

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Prime Numbers upto N (Function)

Medium

2x

0/40

Time Limit: 2, Memory Limit: 128000

Given an integer N, the task is to print all the prime numbers from 1 to N in ascending order.

Note: Each prime number should be printed on the same line, separated by a space.

Input

The first and only line of input contains a single integer

N

N.

Output

Print all the prime numbers up to

N

N.

Constraints

1

≤

N

≤

1

0

5

```
1≤N≤10

5

Example

Sample Input

30

Sample Output

2 3 5 7 11 13 17 19 23 29

Explanation

These are all the prime numbers between 1 and 30.
```

Find the Smallest and Largest Digit in a Number — 0/40 XP — NOT SOLVED — deadline 2026-11-07 16:50

Topics: Python Function

URL: https://my.newtonschool.co/playground/code/tj8ojiirzxa2

Back

Find the Smallest and Largest Digit in a Number

Total XP

6,703

Leaderboard Unlocked

Rank Increased

Run

Submit

QUESTION

Find the Smallest and Largest Digit in a Number

Medium

2x

0/40

Time Limit: 2, Memory Limit: 256000

You are given a positive integer N. Your task is to find the smallest and largest digit in the given number.

Input

The first line of input contains a single integer T, representing the number of test cases.

Each of the following T lines contains a single integer N, representing the number.

Output

Print T lines, each containing two integers.

The integers in the ith line represent the smallest and largest digit of the ith number, respectively.

Constraints

1

≤

T

≤

100

1≤T≤100

1

≤

N

```
≤

1

0

9

1≤N≤10

9

Example

Input

3

2839

231

91388

Output

2 9

1 3

1 9

Explanation

For 2839, the digits are 2, 8, 3, 9 → smallest is 2, largest is 9

For 231, the digits are 2, 3, 1 → smallest is 1, largest is 3

For 91388, the digits are 9, 1, 3, 8, 8 → smallest is 1, largest is 9
```

Quizzes / quiz contests (10) — schedules only

(Portal shows quiz QUESTIONS only after starting a test — starting tests is forbidden, so no question text exists for these.)

- Print function in python, Comments in python, Python Variable, Python Primitive ... - Revision — 2026-09-02 20:00 -> 2026-09-07 20:00 — xp 0/12
- Python Input, Python operators, Input() function, Input() for different datatype ... - Revision — 2026-09-07 17:00 -> 2026-09-12 17:00 — xp 2/26
- Conditional Statements, Python if- else - Revision — 2026-09-09 21:05 -> 2026-09-14 21:05 — xp 8/8
- If- elif-else ladder - Revision — 2026-09-21 16:00 -> 2026-09-26 16:00 — xp 0/8
- Python Function, Nested if-else - Revision — 2026-09-23 20:35 -> 2026-09-28 20:35 — xp 0/14
- Python Function, Nested if-else, Arguments and parameter, Return statement - Revision — 2026-09-28 16:00 -> 2026-10-03 16:00 — xp 8/8
- Iteration in Python, Recursion Mechanism, Range () function, Intro and Types - L ... - Revision — 2026-10-12 15:00 -> 2026-10-17 15:00 — xp 0/14
- While loop - Revision — 2026-10-12 19:35 -> 2026-10-17 19:35 — xp 0/10
- Control flow statements, Loops, Break, Continue - Revision — 2026-10-16 11:00 -> 2026-10-21 11:00 — xp 0/8

- Python Function, Scope, Variable Scopes - Revision — 2026-11-02 19:35 -> 2026-11-07 19:35 — xp 0/20

Questions below lecture videos (per lecture)

19wxczhsm2y8 [zqngjr790187]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments1

In-class Quizzes

1

Lecture Notes

QUIZ

Conditional Statements, Python if- else - Revision

Attempt duration - Sep 9th 2026 9:05 pm to Sep 14th 2026 9:05 pm

Attempted on Sep 12th 2026 12:32 am

View Solutions

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

Q1. Weather Report

Write a program that takes an integer current temperature as input, and determines the type of weather using the below conditions:

Temp < 0 : Freezing Weather

Temp 0-9 : Very Cold

Temp 10-19 : Cold

Temp 20-29 : Normal

Temp 30-39 : Hot

Temp >=40 : Very Hot

Read More

Your Solution

Q2. AM_PM

Write a program that takes the hour (0-23) in 24-hour format as input and prints whether it is AM or PM.

Note:

The time is considered AM if the hour is between 0 and 11 (inclusive).

The time is considered PM if the hour is between 12 and 23 (inclusive).

Read More

Your Solution

Q3. Identify Square or Rectangle

You are given two integers representing the length and breadth of a rectangle.

You need to print "Square" if the given rectangle is square as well, otherwise print "Rectangle".Read More

Your Solution

=== In-class Quizzes tab ===

In-class Quizzes

1

Lecture Notes

In-class Quizzes

These in-class quizzes are only for learning purposes and are not meant for evaluation.

Classroom Quiz - Conditional Statements, Python if- else

not attempted

Total Questions

3 Questions

Your Score

0/5

Classroom Quiz - Conditional Statements, Python if- else

Submit Quiz

QUESTION 1/3

Mark for review

Clear Selection

Which of the following is the correct syntax for an if-else statement in Python?

A

```
if(condition)
  statements;
else:
  statements;
```

B

```
if condition
{
  statements
}
else
{
  statements
}
```

C

```
if condition:
  statements
elif condition:
  statements
else:
  statements
```

D

```
if condition:
```

```
  statements
else:
  statements
```

Previous

Next

Newton SchoolQUESTION 2/3

Mark for review

Clear Selection

What will be the output of the following code?

```
  x = 7
  if x % 2 == 0:
      print(x*2)
  else:
      print(x*3)
```

A

None of the Above

B

14

C

21

D

7

Previous

Next

Newton SchoolQUESTION 3/3

Mark for review

Clear Selection

What will be the output of the following Python code?

```
if False:
    print("1")
else:
    print("2")
```

Previous

Next

Newton School

d4amrhe6jend [zqngjr790187]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments1

Lecture Notes

QUIZ

Control flow statements, Loops, Break, Continue - Revision

Attempt duration - Oct 16th 2026 11:00 am to Oct 21st 2026 11:00 am

Your submission will be marked as “LATE SUBMISSION” because you have crossed the given submission deadline

Attempt Now

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

Q1. Find Odd Factorseasy

You are given a positive integer N. Your task is to print all the odd factors of N in increasing order.Read More

Complete Now

Q2. Check if Even Sum is Divisible by 2easy

You are given two positive integers L and R representing a range. Your task is to calculate the sum of all even numbers in the range [L, R] (inclusive), and check whether this sum is divisible by 2.Read More

Complete Now

Q3. Check for Perfect Numbereasy

Given an integer N, your task is to determine if N is a perfect number.

Note: A perfect number is a positive integer that is equal to the sum of its proper divisors (all positive divisors excluding the number itself). For example, 6 is a perfect number because 1 + 2 + 3 = 6.Read More

Complete Now

Q4. First Number Divisibly by 7easy

You are given two positive integers L and R representing a range. Your task is to find the smallest number divisible by 7 in the range [L, R] (inclusive). If no such number is found, print "NOT FOUND".Read More

Complete Now

Q5. Check if All Digits are Samemedium

You are given a positive integer N. Your task is to check whether all the digits in the number are the same.Read More

Complete Now

Q6. Check if Raremedium

A number is called rare if all of its digits are divisible by a given number X.

Given a positive integer N and another positive integer X, write a program to check whether N is a rare number with respect to X.

That is, check if every digit of N is divisible by X. If all digits satisfy this condition, print True; otherwise, print False.Read More

Complete Now

dflwfh8lf150 [zqngjr790187]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments1

Lecture Notes

QUIZ

Print function in python, Comments in python, Python Variable, Python Primitive ... - Revision

Attempt duration - Sep 2nd 2026 8:00 pm to Sep 7th 2026 8:00 pm

Your submission will be marked as “LATE SUBMISSION” because you have crossed the given submission deadline

Complete Now

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

Q1. Print Mirror Right Angle Triangleeasy

Write a Python program to print the mirror image of a right-angled triangle using asterisks (*). The triangle should have 5 rows. Each row should:

Contain stars in increasing order (1 in the first row, 2 in the second, and so on)

Include spaces on the left side so that the stars are right-aligned.

See the example below for clarity.Read More

Your Solution

Q2. Print Inverted Triangleeasy

Write a Python program to print an inverted right-angled triangle using asterisks (*).

The triangle should have 5 rows.

The first row should contain 5 stars.

Each subsequent row should have one less star than the previous row.

Read More

Your Solution

Q3. Print Numbers Right-Angled Triangleeasy

Write a Python program to print a right-angled triangle made of numbers.

The triangle should have 5 rows.

The first row should contain the number 1.

The second row should contain 1 2.

The third row should contain 1 2 3, and so on.

See the example below for clarity.

Note: There should be a space between each number in a row.

Read More

Your Solution

Q4. Print Hollow Squareeasy

Write a Python program to print a hollow square pattern using asterisks (*).

The square should have 5 rows and 5 columns.

The stars should form the border of the square.

The inside of the square should be empty (filled with spaces).

See the example below for clarity.Read More

Your Solution

Q5. No Time to Newlineeasy

Write a program to print the below quote in a single line.

Quote

1.Do not watch the clock

2.Do what it does keep going

Your task is to print only the line number 1 of the quote.

Line number 2 of the quote will automatically appear in the output section when you run your code.

Read More

Your Solution

dulnk24cx9gi [zqngjr790187]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments1

Lecture Notes

QUIZ

Python Function, Nested if-else, Arguments and parameter, Return statement - Revision

Attempt duration - Sep 28th 2026 4:00 pm to Oct 3rd 2026 4:00 pm

Attempted on Sep 30th 2026 5:59 pm

View Solutions

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

Q1. Bridge Between Villages -2easy

In a rural district, three villages — Raga, Sora, and Tula — are planning to connect to each other with straight bridges. The government has provided the proposed lengths for these bridges as follows:

Raga to Sora: a kilometers

Sora to Tula: b kilometers

Tula to Raga: c kilometers

You are required to implement the function is_triangle(a, b, c) that returns "YES" or "NO" depending on whether the given bridge lengths can form a valid triangle.

Function Signature:

def is_triangle(a, b, c): which takes three parameters: a: represents proposed length from Raga to Sora,
b: represents proposed length from Sora to Tula, c: represents proposed length from Tula to Raga

Read More

Your Solution

Q2. Calculate Electricity Bill - 2easy

An electricity board calculates monthly power usage bills based on the number of units consumed using a slab-wise tariff system. Each slab charges a different rate per unit as follows:

First 100 units: ₹4 per unit

Next 100 units (i.e., units 101–200): ₹5 per unit

Next 100 units (i.e., units 201–300): ₹6 per unit

Next 100 units (i.e., units 301–400): ₹7 per unit

Units above 400: ₹8 per unit

After calculating the base bill, a 10% surcharge is added to the total bill.

You are required to implement the function calculate_bill(units) that calculates the total electricity bill including the surcharge, and returns a float rounded to 1 decimal place using the built-in round() function.

Function Signature:

def calculate_bill(units): which takes single parameter: units: represents the number of electricity units consumed.

Read More

Your Solution

Q3. Smart Conditional Calculatoreasy

You are given two integers. Your task is to implement a smart calculator that chooses the arithmetic operation based on the parity (even/odd nature) of the two numbers.

Rules:

If both numbers are even, return their sum.

If both numbers are odd, return their product.

If one number is even and the other is odd, return the absolute difference between the two (i.e., subtract the smaller from the larger).

🔧 Task

Write the function definition for smart_calculator(a, b). This means:

✅ You must write the function header (starting with def)

✅ You must also write the function body with correct logic

❌ Do not write any code for input or output (e.g., no input() or print() statements)

Read More

Your Solution

Q4. Celsius to Fahrenheit Conversion - 2easy

You are given an integer, representing the temperature in celsius.

Your task is to convert it to Fahrenheit, using the below given formula:

Fahrenheit = (Celsius × 9 / 5) + 32

You must complete the function celsius_to_fahrenheit(celsius) that returns the equivalent temperature in Fahrenheit.

Function Signature:

def celsius_to_fahrenheit(celsius):

It takes a single parameter, celsius, the temperature in Celsius.Read More

Your Solution

idsydvrzm6yp [zqngjr790187]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments1

In-class Quizzes

1

Lecture Notes

QUIZ

If- elif-else ladder - Revision

Attempt duration - Sep 21st 2026 4:00 pm to Sep 26th 2026 4:00 pm

Your submission will be marked as “LATE SUBMISSION” because you have crossed the given submission deadline

Attempt Now

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

Q1. Library fine

Calculate the fine for a late return of a library book based on the following details:

For the first 3 days, the fine is Rs. 10.

For the next 4 to 10 days, the fine is Rs. 20.

For any period exceeding 10 days, the fine is Rs.100.

Read More

Your Solution

Q2. Calculate salary

Calculate the gross salary of an employee based on their basic salary according to the following criteria:

basic salary <= 10,000, HRA = 20% and DA = 80%.

20,000 >= basic salaries > 10,000 , the HRA = 25% and DA = 90%.

basic salary > 20,000, HRA = 30% and DA = 95%.

Hint: Gross Salary = Basic Salary + HRA + DA

Read More

Your Solution

Q3. Check Profit or Loss

You are given two integers, cost and sell, representing the cost price and the selling price of a product.

Your task is to determine whether the transaction results in a profit, a loss, or no profit and no loss.

Also, you need to calculate the corresponding amount.Read More

Your Solution

Q4. BMI Calculator-1easy

You are asked to determine a person's Body Mass Index (BMI) based on their weight and height.

The BMI is calculated using the following formula:

BMI = weight / (height_in_meters x height_in_meters)

Here,

weight is given in kilograms

height is given in centimeters and must be converted to meters

You must round the BMI value to exactly 1 decimal place using the built-in round() function.

Based on the rounded BMI value, print the corresponding BMI category:

If BMI is less than 18.5, print Underweight

If BMI is between 18.5 and 24.9 (inclusive), print Normal weight

If BMI is between 25 and 29.9 (inclusive), print Overweight

If BMI is 30 or greater, print Obese

Read More

Your Solution

=== In-class Quizzes tab ===

In-class Quizzes

1

Lecture Notes

In-class Quizzes

These in-class quizzes are only for learning purposes and are not meant for evaluation.

Classroom Quiz - If- elif-else ladder

not attempted

Total Questions

3 Questions

Your Score

0/6

Classroom Quiz - If- elif-else ladder

Submit Quiz

QUESTION 1/3

Mark for review

Clear Selection

Identify the correct elif ladder syntax in Python.

A

```
elif condition1:
  # block1
else:
  # block2
```

B

```
if condition1:
  # block1
elif condition2:
  # block2
```

C

```
if condition1:
  # block1
elif condition2:
  # block2
else:
  # block3
```

D

Both A and B

E

Both B and C

Previous

Next

Newton SchoolQUESTION 2/3

Mark for review

Clear Selection

What will be the output of following code?

```
x = 10

if x > 0:
  print("block 1")
elif x <= 10:
  print("block 2")
elif x == 10:
  print("block 3")
else:
  print("block 4")
```

A

block 2

B

block 1

C

block 4

D

block 3

Previous

Next

Newton SchoolQUESTION 3/3

Mark for review

Clear Selection

What will be the output of following code?

```
x = 10

if x > 0:
print("block 1")

if x <= 10:
print("block 2")

if x == 10:
print("block 3")
else:
```

print("block 4")

A

block 3

B

block 2

block 3

C

block 1

D

block 1

block 2

block 3

E

block 1

block 3

Previous

Next

Newton School

irtusths65xb [zqngjr790187]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments1

Lecture Notes

QUIZ

Iteration in Python, Recursion Mechanism, Range () function, Intro and Types - L ... - Revision

Attempt duration - Oct 12th 2026 3:00 pm to Oct 17th 2026 3:00 pm

Your submission will be marked as “LATE SUBMISSION” because you have crossed the given submission deadline

Complete Now

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

Q1. Find The Even Numberseasy

Write a program that takes a positive integer N as input and prints all even numbers from 1 to N (inclusive), each on a separate line.Read More

Your Solution

Q2. Sum from 1 to Neasy

You are given a positive integer N. Your task is to calculate and print the sum of all integers from 1 to N (inclusive).Read More

Your Solution

Q3. Count Numbers Divisible by 3 or 5easy

You are given a positive integer N. Your task is to print all numbers in the range from 1 to N (inclusive) that are divisible by 3 or 5, but not both.Read More

Complete Now

Q4. Squares from 1 to N (Part I)easy

You are given a positive integer N. Your task is to print the squares of all integers from 1 to N, one per line.

In other words, print:

1^2, 2^2, 3^2, ..., N^2, each on a separate line.Read More

Complete Now

Q5. Squares from 1 to N (Part II)easy

You are given a positive integer N. Your task is to print the squares of all integers from 1 to N, on the same line, separated by a space.Read More

Complete Now

lxbwc1pqpdt7 [zqngjr790187]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments1

Lecture Notes

QUIZ

Python Input, Python operators, Input() function, Input() for different datatype ... - Revision

Attempt duration - Sep 7th 2026 5:00 pm to Sep 12th 2026 5:00 pm

Attempted on Sep 10th 2026 9:53 am

View Solutions

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

Q1. Find The Squareeasy

Write a program to take a number X as input and print its square.Read More

Your Solution

Q2. Rectangle Area Calculatoreasy

Write a program that calculates the area of a rectangle given its length and width.

The program should prompt the user to enter the length and width of the rectangle, and then calculate and print the area.Read More

Your Solution

Q3. Calculate and Print Ageeasy

Write a Python program that takes two integers as input: the birth year and the current year.

Your task is to calculate the age by subtracting the birth year from the current year, and print the result.Read More

Your Solution

Q4. Energy Consumption Calculationeasy

You are given two integers: units representing the total electricity consumed and cost representing the cost of each unit.

Your task is to write a Python program that calculates the total electricity cost and prints the result.Read More

Your Solution

Q5. Celsius to Fahrenheit Conversioneasy

Write a program to convert a temperature value from Celsius to Fahrenheit using the formula:

$$
\begin{aligned}
&F \\
&a \\
&h \\
&r \\
&e \\
&n \\
&h \\
&e \\
&i \\
&t \\
&= \\
&(\\
&C \\
&e \\
&l \\
&s \\
&i \\
&u \\
&s \\
&* \\
&9 \\
&/ \\
&5 \\
& \\
&) \\
&+ \\
&32
\end{aligned}
$$

Fahrenheit=(Celsius∗9/5)+32.

Note: Use the round() function to round numbers to 1 decimal place.Read More

Your Solution

Q6. Convert Minute to Hour and Minute Formateasy

Write a program to convert total minutes into hours and remaining minutes.

Print the hours and minutes in the below given format:

Time: {x} hour(s) and {y} minute(s)

where {x} will be replaced by hours and {y} by minutes.Read More

Your Solution

q03s0bvxghsy [zqngjr790187]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments0

In-class Quizzes

1

Lecture Notes

QUIZ

Python Function, Scope, Variable Scopes - Revision

Attempt duration - Nov 2nd 2026 7:35 pm to Nov 7th 2026 7:35 pm

Attempt Now

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

No post-class assignments for this lecture

=== In-class Quizzes tab ===

In-class Quizzes

1

Lecture Notes

In-class Quizzes

These in-class quizzes are only for learning purposes and are not meant for evaluation.

Classroom Quiz - Python Function, Scope, Variable Scopes

not attempted

Total Questions

2 Questions

Your Score

0/4

Classroom Quiz - Python Function, Scope, Variable Scopes

Submit Quiz

QUESTION 1/2

Mark for review

Clear Selection

What would be the output of the following code?

```
x = 10
def update():
x = 5
update()
print(x)
```

Previous

Next

Newton SchoolQUESTION 2/2

Mark for review

Clear Selection

What would be the output of the following code?

```
x = 20
def modify():
global x
```

x = 15

modify()

print(x)

Previous

Next

Newton School

rvt849y0o66i [zqngjr790187]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments1

In-class Quizzes

1

Lecture Notes

QUIZ

While loop - Revision

Attempt duration - Oct 12th 2026 7:35 pm to Oct 17th 2026 7:35 pm

Your submission will be marked as “LATE SUBMISSION” because you have crossed the given submission deadline

Attempt Now

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

Q1. Count the Digitseasy

You are given a positive integer N. Your task is to find the count of digits in N. There will be no leading zeroes in the input number N. For example, inputs like 0023 or 0005 will not be given.

Note: The following things are blocked in the backend, so please don’t use them in your code:

String literals, list literals, and conversions using str() or list().

Indexing, iteration, or applying string methods on variables obtained from input().

Read More

Complete Now

Q2. Reverse a Numbereasy

You are given a positive integer N. Your task is to reverse the given number.Read More

Complete Now

Q3. Sum of First and Last Digits of a Numbereasy

You are given a positive integer N. Your task is to calculate the sum of the first and last digits of the number.Read More

Complete Now

Q4. Find the Count of Even and Odd Digitseasy

You are given a positive integer N. Your task is to separately find the count of its even digits and the count of its odd digits.Read More

Complete Now

Q5. Find the Largest Digit in a Numbermedium

You are given a positive integer N. Your task is to find the largest digit in the given number.Read More

Complete Now

=== In-class Quizzes tab ===

In-class Quizzes

1

Lecture Notes

In-class Quizzes

These in-class quizzes are only for learning purposes and are not meant for evaluation.

Classroom Quiz - While loop

not attempted

Total Questions

5 Questions

Your Score

0/10

Classroom Quiz - While loop

Submit Quiz

QUESTION 1/5

Mark for review

Clear Selection

What will get printed during the last iteration of the following code?

```
x = 1
while x < 100:
print(x)
x *= -3
```

A

81

B

-81

C

-243

D

243

Previous

Next

Newton SchoolQUESTION 2/5

Mark for review

Clear Selection

What will get printed during the last iteration of the following code?

```
x = 10
while x > 0:
print(x)
x -= 3
```

Previous

Next

Newton SchoolQUESTION 3/5

Mark for review

Clear Selection

What will get printed in the 5th iteration of the following code?

x = 1

while x <= 20:

print(x * x)

x += 2

Previous

Next

Newton SchoolQUESTION 4/5

Mark for review

Clear Selection

What will be the final value of s?

n = 123

s = 0

while n > 0:

s += n % 10

n //= 10

Previous

Next

Newton SchoolQUESTION 5/5

Mark for review

Clear Selection

What will be the ouput of the below given code?

n = 123; s = 0

while n:

s += n % 10

n //= 10

print(n)

Previous

Next

Newton School

xcqr11359sem [zqngjr790187]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments1

In-class Quizzes

1

Lecture Notes

QUIZ

Python Function, Nested if-else - Revision

Attempt duration - Sep 23rd 2026 8:35 pm to Sep 28th 2026 8:35 pm

Your submission will be marked as “LATE SUBMISSION” because you have crossed the given submission deadline

Attempt Now

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

Q1. Accommodation Decisioneasy

A student wants to check whether they will stay On-Campus or Off-Campus based on the rules provided in the NST rulebook.

The rules for accommodation are:

If the student is a girl, she will definitely stay On-Campus, irrespective of the year.

If the student is a boy:

In first year → he will stay On-Campus

In second year → he will stay Off-Campus

Given the student's gender and year of study, determine their accommodation type.

Read More

Your Solution

Q2. Calculate Tax Based on Age and Income

You are given two values:

age — the age of a person

income — the annual income of the person (in rupees)

Your task is to calculate the amount of tax that the person needs to pay.

The tax rate depends on both the person's age and annual income.

Tax Slabs

Income Bracket Tax Rate (Age < 18) Tax Rate (Age ≥ 18)

Income ≤ 10,00,000 5% 10%

10,00,000 < Income ≤ 30,00,000 15% 20%

Income > 30,00,000 25% 30%

Print the tax amount that must be paid.

Read More

Your Solution

Q3. Find the Smallest !

Just like finding the largest value, identifying the smallest number is another common task in programming.

Your task:

Given three integers A, B, and C, print the smallest among them.Read More

Your Solution

=== In-class Quizzes tab ===

In-class Quizzes

1

Lecture Notes

In-class Quizzes

These in-class quizzes are only for learning purposes and are not meant for evaluation.

Classroom Quiz - Python Function, Nested if-else

attempted

Total Questions

3 Questions

Your Score

2/6

Classroom Quiz - Python Function, Nested if-else

Revise

QUESTION 1/3

What will the following code output?

```
x = 10
if x > 5:
if x < 15:
print("Yes")
else:
print("No")
```

Yes

Explanation:x = 10 x > 5 → True, so the inner if is checked. x < 15 → True, so "Yes" is printed.

B

No

Error

D

Nothing

INCORRECT ANSWER

QUESTION 2/3

Neeraj is preparing for the finals. His three throws are recorded: throw1 = 86.7, throw2 = 90.6, and throw3 = 88.1. Based on his average throw, the coach gives feedback:

```
throw1 = 86.7
throw2 = 90.6
throw3 = 88.1

avg = (throw1 + throw2 + throw3) // 3

if avg >= 88:
if throw3 > throw2:
print("Excellent Finish")
elif avg < 89:
```

print("Great, but can be better")

else:

print("Superb")

else:

print("Needs Improvement")

What will be printed?

A

Excellent Finish

Great, but can be better

Explanation:✓ Step 1: Average = (86.7 + 90.6 + 88.1) / 3 = 88

✓ Step 2: avg >= 88 → True → Enters nested if

✓ throw3 > throw2? 88> 90 → False

✓ avg < 89? Yes → So prints "Great, but can be better"

❌ A) Incorrect — throw3 is not greater than throw2

❌ C) Incorrect — only executes if avg ≥ 89 and other conditions fail

❌ D) Incorrect — avg ≥ 88, so else is skipped

C

Superb

Needs Improvement

INCORRECT ANSWER

QUESTION 3/3

Mark for review

Clear Selection

What will be the output of following code?

loan = 80000

duration = 12

base_rate = 0.12

card_type = "Gold"

Apply card-based discount

if card_type == "Silver":

rate = base_rate - 0.01

elif card_type == "Gold":

if duration > 12:

rate = base_rate - 0.01

else:

rate = base_rate - 0.02

elif card_type == "Platinum":

rate = base_rate - 0.03

else:

rate = base_rate

total_interest = loan * rate

total_payment = loan + total_interest

print(int(total_payment))

88000

Explanation:

The initial loan amount is 80000.

The duration of the loan is 12 months.

The base_rate of interest is 12% or 0.12.

The card_type is Gold.

Since the card type is "Gold" and the duration is not more than 12 months, the applicable interest rate becomes base_rate - 0.02 = 0.10.

Then the total interest is calculated as loan × rate = 80000 × 0.10 = 8000.

So, the total_payment is loan + interest = 80000 + 8000 = 88000.

Output: 88000

Correct answer: 88000

Newton School

ztz8mhzw2sws [zqngjr790187]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments1

In-class Quizzes

1

Lecture Notes

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

Q1. Movie Maniaeasy

Four movies were released: Spiderman, Superman, Batman, and Aquaman in four distinct years given as four space-separated integers: year1, year2, year3, and year4 respectively.

Movie

Release year

Spiderman year1

Superman year2

Batman year3

Aquaman year4

You are required to print the name of the movie that was released earliest of the four.

Note-1: Refer Input section, output section and example test cases.

Note-2: Use of built-in mathematical functions is strictly prohibited in this question.Read More

Your Solution

Q2. Will Ramandeep Sir Fly? - 2easy

Ramandeep Sir is known for his highly selective flying decisions after past flight troubles. He follows a strict set of rules before agreeing to board a plane.

Here are his conditions:

If the trip is international, Ramandeep Sir will not fly, regardless of any other conditions.

If the trip is domestic, and the flight duration is less than or equal to 1.5 hours, he will fly.

If the trip is domestic and the duration is more than 1.5 hours, he will not fly if:

His seat is a middle seat, or

The airline is IngidoX

In all other cases, he will fly.

Your task is to determine whether Ramandeep Sir will fly given the flight details.

Read More

Your Solution

Q3. NST Bus Route Optimizationeasy

The bus driver at NST Campus is planning his daily pickup route. He needs to pick up students from two hostels: Hostel 1 and Hostel 2, and return to the NST campus afterward. To save fuel, he wants to choose the route that results in the minimum total distance traveled.

There are three roads available:

A road of length d1 connects the NST campus and Hostel 1

A road of length d2 connects the NST campus and Hostel 2

A road of length d3 directly connects Hostel 1 and Hostel 2

The driver can travel on any road any number of times, and in any direction. He must start at NST campus, pick up students from both hostels, and return back to the campus, while minimizing the total distance.

Please refer to the following image for further clarity:

Click to enlarge

Read More

Your Solution

Q4. Museum Ticket Booking System

You are tasked with implementing the function calculateTicketPrice for a Museum Ticket Booking system.

The function should calculate the ticket price based on the following rules:

The base ticket price is $12.

If the visitor is under 12 years old, apply a discount of $5.

If the visit is on a weekend, add a surcharge of $3.

If the visitor has a student card, apply an additional discount of $2.

The function calculateTicketPrice accepts the following parameters:

age (number) — The age of the visitor.

isWeekend (boolean) — true if the visit is on a weekend, otherwise false.

hasStudentCard (boolean) — true if the visitor has a student card, otherwise false.

Your task is to return the final ticket price after applying all applicable discounts and surcharges.

Read More

Your Solution

Q5. Greater of Four

You are given four integers a, b, c, and d.

A built-in function greater(x, y) is available to you. It returns the greater of the two values x and y.

Your task is to find the greatest among the four given numbers using the greater() function.

Note: You must use the provided greater() function. Do not compare all four numbers directly.

Read More

Your Solution

=== In-class Quizzes tab ===

In-class Quizzes

1

Lecture Notes

In-class Quizzes

These in-class quizzes are only for learning purposes and are not meant for evaluation.

Classroom Quiz - Python if- else, Python logical Operators

not attempted

Total Questions

4 Questions

Your Score

0/8

Classroom Quiz - Python if- else, Python logical Operators

Submit Quiz

QUESTION 1/4

Mark for review

Clear Selection

Which of the following is the incorrect syntax for using logical operators in an if statement?

A

if condition1 & condition2:

B

if condition1 or condition2:

C

if (condition1) and (condition2):

D

if condition1 and condition2:

Previous

Next

Newton SchoolQUESTION 2/4

Mark for review

Clear Selection

What will be the output of the following code?

```
x = 20
y = 10
if not x > 15 and y > 15:
print("One condition is true")
```

A

No output

B

One condition is true

C

Error

D

None of these

Previous

Next

Newton SchoolQUESTION 3/4

Mark for review

Clear Selection

What would be the output of the following code?

```
print(5 > 3 and 3 < 2)
```

Previous

Next

Newton SchoolQUESTION 4/4

Mark for review

Clear Selection

What will be printed by this code?

```
x = 4
y = 10
if x < 5:
if not(y < 10 and x == 4):
print("Yes")
else:
print("Maybe")
else:
print("No")
```

Previous

Next

Newton School

cf7dcfu88onf [kehysh1tot83]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments0

In-class Quizzes

1

Lecture Notes

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

No post-class assignments for this lecture

=== In-class Quizzes tab ===

In-class Quizzes

1

Lecture Notes

In-class Quizzes

These in-class quizzes are only for learning purposes and are not meant for evaluation.

Classroom Quiz - Python Input, Python operators, Conditional Statements, Input() function, Input(...

attempted

Total Questions

3 Questions

Your Score

0/6

Classroom Quiz - Python Input, Python operators, Conditional Statements, Input() function, Input(...

Revise

QUESTION 1/3

What will be the output of the following code?

```
x = 12
if x > 12:
print("Greater than 12")
elif x > 10:
print("Greater than 10")
elif x > 8:
print("Greater than 8")
else:
print("Less than or equal to 8")
```

Greater Than 10

Greater than 10

Explanation:

x = 12

1️⃣ if x > 12 → False

2️⃣ elif x > 10 → 12 > 10 → ✅ True → prints: Greater than 10

⛔ Remaining elif and else are skipped once a condition is true.

C

Greater than 8

D

Less than or equal to 8

INCORRECT ANSWER

QUESTION 2/3

Mark for review

Clear Selection

What will be output of below code if score = 75?

```
score = 75

if score >= 90:
print("A")
elif score >= 80:
print("B")
elif score >= 70:
print("C")
else:
print("D")
```

c

Explanation:

Explanation:

score = 75

Condition 1: score >= 90 → False

Condition 2: score >= 80 → False

Condition 3: score >= 70 → True, so it prints C.

Remaining conditions are skipped after the first True.

Final Output: C

Correct answer: C

QUESTION 3/3

Mark for review

Clear Selection

What will be the output of below code?

```
a = 10
b = 5
c = 15
result = 0
if a > b:
result = result+1
elif b < c:
result = result+2

if c > a:
result = result+3
```

print(result)

1

Explanation:Given: a = 10, b = 5, c = 15 Step 1: if a > b → 10 > 5 → True → result += 1 Since the if was true, elif is skipped. Step 2: if c > a → 15 > 10 → True → result += 3 Final result: result = 4 print(result) prints: 4

Correct answer: 4

Newton School

qu5kfoi6edsr [kehysh1tot83]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments0

In-class Quizzes

1

Lecture Notes

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

No post-class assignments for this lecture

=== In-class Quizzes tab ===

In-class Quizzes

1

Lecture Notes

In-class Quizzes

These in-class quizzes are only for learning purposes and are not meant for evaluation.

Classroom Quiz - While loop

not attempted

Total Questions

5 Questions

Your Score

0/11

Classroom Quiz - While loop

Submit Quiz

QUESTION 1/5

Mark for review

Clear Selection

What will be printed in the 3rd iteration?

x = 2

while x <= 20:

print(x ** 2)

x += 6

Previous

Next

Newton SchoolQUESTION 2/5

Mark for review

Clear Selection

How many digits will be counted in the following code?

```
n = 98765
count = 0
while n > 0:
count += 1
n //= 10
```

Previous

Next

Newton SchoolQUESTION 3/5

Mark for review

Clear Selection

What will be printed in the last iteration?

```
x = 50
while x >= 0:
print(x // 10)
x -= 10
```

Previous

Next

Newton SchoolQUESTION 4/5

Mark for review

Clear Selection

What will be the value of total after the loop ends?

```
x = 7
total = 0
while x <= 70:
if x % 10 == 7:
total += x
x += 1
```

Previous

Next

Newton SchoolQUESTION 5/5

Mark for review

Clear Selection

How many times does the loop run?

```
n = 1234
count = 0
while n > 0:
digit = n % 10
```

if digit % 2 == 0:

count += 1

n //= 10

Previous

Next

Newton School

sa4adqeyfoaz [kehysh1tot83]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments0

In-class Quizzes

1

Lecture Notes

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

No post-class assignments for this lecture

=== In-class Quizzes tab ===

In-class Quizzes

1

Lecture Notes

In-class Quizzes

These in-class quizzes are only for learning purposes and are not meant for evaluation.

Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

attempted

Total Questions

10 Questions

Your Score

19/19

Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

Revise

QUESTION 1/10

What is the output of the following Python expression?

print(8 // 4 ** 6 * 2 / 3 % 4 - 7)

A

-2

B

-7

-7.0

Explanation:

The expression follows Python's operator precedence rules:

Exponentiation (4 ** 6) is performed first, resulting in 4096.

Floor division (8 // 4096) is next, resulting in 0 because 8 is smaller than 4096.

Multiplication (0 * 2) gives 0.

Division (0 / 3) also results in 0.0.

Modulus (0.0 % 4) results in 0.0.

Subtraction (0.0 - 7) finally gives -7.0.

Therefore, the output of the expression is -7.0.

D

-5.0

CORRECT ANSWER

QUESTION 2/10

What will be the output of the below given python code?

```
b = 10
a = 3
c = b < a
print(c)
```

A

True

False

Explanation:

Let's understand what the code is doing step by step:

b = 10 and a = 3: Two integer variables are initialized.

c = b < a: This compares if b is less than a. Here, 10 < 3 is False.

print(c): This prints the value of c, which is False.

🔍 The comparison operator < checks the relational condition and returns a Boolean value: either True or False.

✅ Final Output: False

C

0

D

1

CORRECT ANSWER

QUESTION 3/10

What will be the output of the below given python code?

```
num1 = 15
num2 = 28
num3 = 100
result1 = num1 <= num2
result2 = num3 > num1
print(result1 != result2)
```

A

True

False

Explanation:

Let’s understand the logic step-by-step:

num1 = 15, num2 = 28, num3 = 100: Three integer values are initialized.

result1 = num1 <= num2: This checks if 15 ≤ 28, which is True.

result2 = num3 > num1: This checks if 100 > 15, which is also True.

print(result1 != result2): Now it compares whether True != True. Since both are True, the result is False.

🔍 The != operator checks for inequality — and in this case, both sides are equal.

✅ Final Output: False

C

1

D

0

CORRECT ANSWER

QUESTION 4/10

What will be the output of the below given python code?

x = 10

y = 5

result1 = (x >= 10)

result2 = (y == 5)

print(result1 == result2)

A

False

B

0

True

Explanation:

Let’s go step by step and decode what’s happening:

x = 10 and y = 5: Two variables are assigned integer values.

result1 = (x >= 10): This checks whether x is greater than or equal to 10. Since x is 10, this condition is True.

result2 = (y == 5): This checks whether y is equal to 5. Since it is, this is also True.

print(result1 == result2): Here, we are comparing True == True, which evaluates to True.

✅ Both result1 and result2 are True, so the final comparison also returns True.

🔚 Final Output: True

D

1

CORRECT ANSWER

QUESTION 5/10

What will be the output of the below given python code?

num1 = 5

num2 = 3

print(num1 > num2)

True

Explanation:

Let’s break down what this code is doing:

num1 = 5 and num2 = 3: Two integer variables are assigned values.

num1 > num2 evaluates whether num1 is greater than num2. Here, 5 > 3 is True.

print(num1 > num2) displays the result of the comparison, which is True.

🔍 The > operator is a comparison operator that returns a Boolean value — either True or False — based on whether the left operand is greater than the right one.

✅ Final Output: True

B

False

C

1

D

0

CORRECT ANSWER

QUESTION 6/10

What will be the output of the below given python code?

a = 2 * 3 // 5 % 6 ** 4

print(a)

A

0

1

Explanation:

Let’s evaluate the expression by following Python’s operator precedence rules:

Exponentiation (**) comes first:

→ 6 ** 4 = 1296

Multiplication and Floor Division are left to right:

→ 2 * 3 = 6

→ 6 // 5 = 1

Modulus (%) comes next:

→ 1 % 1296 = 1

So the final value of a is 1.

✅ Final Output: 1

C

6

D

1296

CORRECT ANSWER

QUESTION 7/10

What will be the result of the following expression in Python?

```
x = 1
y = 2
z = 3
result = x == x <= x + y * z

print(result)
```

True

Explanation:

The expression x == x <= x + y * z is evaluated as follows:

First, perform the arithmetic operations: y * z results in 6, and x + 6 results in 7.

Next, evaluate the relational comparisons from left to right: x == x evaluates to True, and then True <= 7 evaluates to True.

Finally, combine these results: x == x <= 7 is interpreted as True and True, which evaluates to True.

Therefore, the value of result is True.

B

False

C

0

D

1

CORRECT ANSWER

QUESTION 8/10

What will be the output of the below given python code?

```
num1 = 15
num2 = 28
result = num2 >= num1 + 5
print(result)
```

A

False

True

Explanation:

Let's evaluate this step-by-step:

num1 = 15 and num2 = 28: Two integer variables are assigned values.

num1 + 5 is evaluated first (due to operator precedence):

→ 15 + 5 = 20

num2 >= 20 → 28 >= 20 is True.

The result is stored in the variable result and then printed.

✅ Final Output: True

C

5

D

6

CORRECT ANSWER

QUESTION 9/10

What will be the output of the following code if the user enters 10, 20 as input?

```
a, b = input().split(", ")
print(int(a) * int(b))
```

200

Explanation:

The input string is "10, 20", which includes a comma followed by a space.

The split(", ") function splits the string into two parts:

a = "10"

b = "20"

Both are then converted to integers and multiplied:

int(a) * int(b) = 10 * 20 = 200

Hence, the output is 200.

B

1020

C

30

D

Error

CORRECT ANSWER

QUESTION 10/10

What will be the output of the below given python code?

```
result = "apple,,banana,,cherry".split(",")
print(result)
```

['apple', '', 'banana', '', 'cherry']

Explanation:The split(",") function splits the string at every comma. Here's how it works step-by-step:

The original string is: "apple,,banana,,cherry"

There are double commas (,,) between apple and banana, and again between banana and cherry.

Each , is a splitting point. So, a double comma means an empty string ('') is added in between.

Final List: ['apple', '', 'banana', '', 'cherry']

So, the empty strings appear wherever there were consecutive commas in the original string.

B

['apple', 'banana', 'cherry']

C

['apple', ' ', 'banana', ' ', 'cherry']

D

['apple', 'banana', '', 'cherry']

CORRECT ANSWER

vkdt3hwua9yy [kehysh1tot83]

=== Assignments tab (quiz cards + in-class + post-class sections) ===

Assignments0

In-class Quizzes

1

Lecture Notes

In class assignments

This section contains the assignments and questions solved in the class

No in-class assignments for this lecture

Post class assignments

This section contains the assignments related to the topics covered in this lecture

No post-class assignments for this lecture

=== In-class Quizzes tab ===

In-class Quizzes

1

Lecture Notes

In-class Quizzes

These in-class quizzes are only for learning purposes and are not meant for evaluation.

Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

not attempted

Total Questions

3 Questions

Your Score

0/6

Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

Submit Quiz

QUESTION 1/3

Mark for review

Clear Selection

What will be the output of the following code?

```
x = 20
if x >= 20:
  print("x is good")
else:
  print("x is not good")
```

A

x is less than 20

B

x is good

C

No output

D

x is 20 or more

Previous

Next

Newton SchoolQUESTION 2/3

Mark for review

Clear Selection

What will be the output of the following code?

```
x = 125
if x > 50:
x = x - 50
  print(x)
if x > 150:
  x = x - 150
  print(x)
else:
  print(x)
```

A

25

75

B

100

75

C

75

75

D

125

75

Previous

Next

Newton SchoolQUESTION 3/3

Mark for review

Clear Selection

What would be the output of the following code?

```
if 5 > 3:
print("Yes")
```

Previous

Next

Newton School

Questions shared in Slack (channels #psp-26, #psp-b1, #psp-b2)

Read-only Slack search harvest (`slack_harvest.py`, 2026-10-23): 62 messages total, every Slack-reported result captured. #psp-26: 39 msgs (39 reported) 10 Sep -> 22 Oct, #psp-b1: 12 msgs (12 reported) 3 Sep -> 25 Sep, #psp-b2: 11 msgs (11 reported) 31 Aug -> 25 Sep.

Standing directive (#psp-26, 9 Oct)

> Hey everyone,

> Listen up regarding the 2 Codeforces spreadsheets our PSP teacher just shared. The teacher has specifically directed that every single one of us needs to solve at least 2 questions a day from these lists. This is what is expected of us, so please make sure you are actually hitting that daily target and not falling behind.

> To track your work and mark questions as 'attempted' or 'done' like the teacher wants, you have to make your own editable version (the shared links are view-only).

> Here is exactly how to do it:

>

> Open the spreadsheet links.

> Click on File (top left) > Make a copy.

> Save it to your own Drive.

> Update your personal copy daily as you complete the questions.

> Make sure you get this set up today and start tracking your daily progress!

> Spreedsheet1-docs.google.com/spreadsheets/d/11wzmT1pPkdl…/edit?gid=0#gid=0

> Spreedsheet2-docs.google.com/spreadsheets/d/1IDDciP8Mued…/edit?gid=0#gid=0 (edited)

Codeforces sheet 1 — If-Else & Math, 63 problems (shared #psp-26, 30 Sep; view-only sheet — File > Make a copy to tick attempts)

Sheet: https://docs.google.com/spreadsheets/d/11wzmT1pPkdl-seC8Qzx7DgAKfV4H-tnZOsm9ZbhoIZA/edit?gid=0#gid=0

- 2. **800** · If-Else · [Ideal Generator](https://codeforces.com/contest/2093/problem/A)
- 3. **800** · If-Else · [Draw a Square](https://codeforces.com/contest/2074/problem/A)
- 4. **800** · If-Else · [New World, New Me, New Array](https://codeforces.com/contest/2072/problem/A)
- 5. **800** · If-Else · [Sakurako's Exam](https://codeforces.com/contest/2008/problem/A)
- 6. **800** · If-Else · [Primary Task](https://codeforces.com/contest/2000/problem/A)
- 7. **800** · If-Else · [Legs](https://codeforces.com/contest/1996/problem/A)
- 8. **800** · If-Else · [X Axis](https://codeforces.com/contest/1986/problem/A)
- 9. **800** · If-Else · [Phone Desktop](https://codeforces.com/contest/1974/problem/A)
- 10. **800** · Math · [Maximize?](https://codeforces.com/contest/1968/problem/A)
- 11. **800** · If-Else · [Yogurt Sale](https://codeforces.com/contest/1955/problem/A)
- 12. **800** · Math · [Setting up Camp](https://codeforces.com/contest/1945/problem/A)
- 13. **800** · Math · [Rudolf and the Ticket](https://codeforces.com/contest/1941/problem/A)
- 14. **800** · Math · [Square](https://codeforces.com/contest/1921/problem/A)
- 15. **800** · If-Else · [Game with Integers](https://codeforces.com/contest/1899/problem/A)
- 16. **800** · Math · [Morning](https://codeforces.com/contest/1883/problem/A)
- 17. **800** · Math · [Two Vessels](https://codeforces.com/contest/1872/problem/A)
- 18. **800** · If-Else · [Two Elevators](https://codeforces.com/contest/1729/problem/A)
- 19. **800** · Math · [Food for Animals](https://codeforces.com/contest/1675/problem/A)
- 20. **800** · If-Else · [Number Transformation](https://codeforces.com/contest/1674/problem/A)

- 21. **800** · If-Else · [Vasya and Coins](https://codeforces.com/contest/1660/problem/A)
- 22. **800** · Math · [Polycarp and Sums of Subsequences](https://codeforces.com/contest/1618/problem/A)
- 23. **800** · Math · [Elections](https://codeforces.com/contest/1593/problem/A)
- 24. **800** · If-Else · [Polycarp and Coins](https://codeforces.com/contest/1551/problem/A)
- 25. **800** · If-Else · [Shortest Path with Obstacle](https://codeforces.com/contest/1547/problem/A)
- 26. **800** · Math · [Strange Table](https://codeforces.com/contest/1506/problem/A)
- 27. **900** · If-Else · [Odd Divisor](https://codeforces.com/contest/1475/problem/A)
- 28. **800** · If-Else · [Floor Number](https://codeforces.com/contest/1426/problem/A)
- 29. **800** · Math · [Yet Another Two Integers Problem](https://codeforces.com/contest/1409/problem/A)
- 30. **800** · If-Else · [Required Remainder](https://codeforces.com/contest/1374/problem/A)
- 31. **800** · Math · [Minimal Square](https://codeforces.com/contest/1360/problem/A)
- 32. **800** · Math · [Most Unstable Array](https://codeforces.com/contest/1353/problem/A)
- 33. **800** · Math · [Candies and Two Sisters](https://codeforces.com/contest/1335/problem/A)
- 34. **800** · If-Else · [Divisibility Problem](https://codeforces.com/contest/1328/problem/A)
- 35. **900** · If-Else · [Yet Another Tetris Problem](https://codeforces.com/contest/1324/problem/A)
- 36. **800** · If-Else · [Add Odd or Subtract Even](https://codeforces.com/contest/1311/problem/A)
- 37. **800** · If-Else · [Collecting Coins](https://codeforces.com/contest/1294/problem/A)
- 38. **800** · Math · [Minutes Before the New Year](https://codeforces.com/contest/1283/problem/A)
- 39. **900** · Math · [Three Friends](https://codeforces.com/contest/1272/problem/A)
- 40. **1000** · If-Else · [Payment Without Change](https://codeforces.com/contest/1256/problem/A)
- 41. **800** · Math · [Three Piles of Candies](https://codeforces.com/contest/1196/problem/A)
- 42. **800** · Math · [Restoring Three Numbers](https://codeforces.com/contest/1154/problem/A)
- 43. **1000** · Math · [Middle of the Contest](https://codeforces.com/contest/1133/problem/A)
- 44. **800** · Math · [Water Buying](https://codeforces.com/contest/1118/problem/A)
- 45. **800** · If-Else · [Integer Sequence Dividing](https://codeforces.com/contest/1102/problem/A)
- 46. **800** · Math · [Frog Jumping](https://codeforces.com/contest/1077/problem/A)
- 47. **1100** · Math · [Vova and Train](https://codeforces.com/contest/1066/problem/A)
- 48. **800** · Math · [Adjacent Replacements](https://codeforces.com/contest/1006/problem/A)
- 49. **800** · If-Else · [Journey](https://codeforces.com/contest/2051/problem/B)
- 50. **800** · If-Else · [Robin Hood and the Major Oak](https://codeforces.com/contest/2014/problem/B)
- 51. **900** · Math · [Fireworks](https://codeforces.com/contest/1945/problem/B)
- 52. **900** · If-Else · [Three Threadlets](https://codeforces.com/contest/1881/problem/B)
- 53. **900** · Math · [The Corridor or There and Back Again](https://codeforces.com/contest/1872/problem/B)
- 54. **1100** · Math · [Binary Cafe](https://codeforces.com/contest/1840/problem/B)
- 55. **1000** · Math · [Conveyor Belts](https://codeforces.com/contest/1811/problem/B)
- 56. **800** · Math · [Taisia and Dice](https://codeforces.com/contest/1790/problem/B)
- 57. **800** · If-Else · [Matrix Rotation](https://codeforces.com/contest/1772/problem/B)
- 58. **900** · Math · [DIV + MOD](https://codeforces.com/contest/1650/problem/B)
- 59. **900** · If-Else · [Make AP](https://codeforces.com/contest/1624/problem/B)
- 60. **800** · Math · [Team Composition: Programmers and Mathematicians] (https://codeforces.com/contest/1611/problem/B)
- 61. **900** · Math · [Odd Grasshopper](https://codeforces.com/contest/1607/problem/B)
- 62. **900** · If-Else · [New Year's Number](https://codeforces.com/contest/1475/problem/B)
- 63. **1100** · Math · [Minimum Product](https://codeforces.com/contest/1409/problem/B)
- 64. **900** · Math · [Candies Division](https://codeforces.com/contest/1283/problem/B)

Codeforces sheet 2 — Loops, 34 problems (shared #psp-26, 30 Sep; view-only sheet — File > Make a copy to tick attempts)

Sheet: https://docs.google.com/spreadsheets/d/1IDDciP8MuedEJKSfnL2uAdIKj9S6YrzHLOULxByxqLs/edit?gid=0#gid=0

- 2. **800** · Loops · [Sakurako and Kosuke](https://codeforces.com/contest/2033/problem/A)
- 3. **800** · Loops · [Rudolph and Cut the Rope](https://codeforces.com/contest/1846/problem/A)
- 4. **800** · Loops · [Everyone Loves to Sleep](https://codeforces.com/contest/1714/problem/A)
- 5. **800** · Loops · [Round Down the Price](https://codeforces.com/contest/1702/problem/A)
- 6. **800** · Loops · [Print a Pedestal (Codeforces logo?)](https://codeforces.com/contest/1690/problem/A)
- 7. **800** · Loops · [Dislike of Threes](https://codeforces.com/contest/1560/problem/A)
- 8. **800** · Loops · [Dense Array](https://codeforces.com/contest/1490/problem/A)
- 9. **800** · Loops · [Cards for Friends](https://codeforces.com/contest/1472/problem/A)
- 10. **800** · Loops · [Special Permutation](https://codeforces.com/contest/1454/problem/A)
- 11. **800** · Loops · [Candies](https://codeforces.com/contest/1343/problem/A)
- 12. **1100** · Loops · [Array with Odd Sum](https://codeforces.com/contest/1296/problem/A)
- 13. **1000** · Loops · [Equalize Prices Again](https://codeforces.com/contest/1234/problem/A)
- 14. **800** · Loops · [Chips Moving](https://codeforces.com/contest/1213/problem/A)
- 15. **800** · Loops · [Nearest Interesting Number](https://codeforces.com/contest/1183/problem/A)
- 16. **800** · Loops · [Divide it!](https://codeforces.com/contest/1176/problem/A)
- 17. **900** · Loops · [Reachable Numbers](https://codeforces.com/contest/1157/problem/A)
- 18. **1000** · Loops · [Game 23](https://codeforces.com/contest/1141/problem/A)
- 19. **800** · Loops · [Repeating Cipher](https://codeforces.com/contest/1095/problem/A)
- 20. **800** · Loops · [Uniform String](https://codeforces.com/contest/1092/problem/A)
- 21. **800** · Loops · [Wrong Subtraction](https://codeforces.com/contest/977/problem/A)
- 22. **900** · Loops · [St. Chroma](https://codeforces.com/contest/2106/problem/B)
- 23. **800** · Loops · [Aleksa and Stack](https://codeforces.com/contest/1878/problem/B)
- 24. **800** · Loops · [Funny Permutation](https://codeforces.com/contest/1741/problem/B)
- 25. **800** · Loops · [Who's Opposite?](https://codeforces.com/contest/1560/problem/B)
- 26. **800** · Loops · [Ordinary Numbers](https://codeforces.com/contest/1520/problem/B)
- 27. **800** · Loops · [Fair Division](https://codeforces.com/contest/1472/problem/B)
- 28. **900** · Loops · [Symmetric Matrix](https://codeforces.com/contest/1426/problem/B)
- 29. **900** · Loops · [Multiply by 2, divide by 6](https://codeforces.com/contest/1374/problem/B)
- 30. **800** · Loops · [Even Array](https://codeforces.com/contest/1367/problem/B)
- 31. **800** · Loops · [Balanced Array](https://codeforces.com/contest/1343/problem/B)
- 32. **900** · Loops · [Construct the String](https://codeforces.com/contest/1335/problem/B)
- 33. **1300** · Loops · [K-th Beautiful String](https://codeforces.com/contest/1328/problem/B)
- 34. **900** · Loops · [Food Buying](https://codeforces.com/contest/1296/problem/B)
- 35. **1000** · Loops · [Disturbed People](https://codeforces.com/contest/1077/problem/B)

12-problem Codeforces code list (shared #psp-26, 22 Oct)

- 977 A — Wrong Subtraction (800) — https://codeforces.com/contest/977/problem/A
- 978 B — File Name (800) — https://codeforces.com/contest/978/problem/B
- 988 A — Diverse Team (800) — https://codeforces.com/contest/988/problem/A
- 996 A — Hit the Lottery (800) — https://codeforces.com/contest/996/problem/A
- 731 A — Night at the Museum (800) — https://codeforces.com/contest/731/problem/A
- 832 A — Sasha and Sticks (800) — https://codeforces.com/contest/832/problem/A
- 1800 A — Is It a Cat? (800) — https://codeforces.com/contest/1800/problem/A

- 910 A — The Way to Home (800) — https://codeforces.com/contest/910/problem/A
- 115 A — Party (900) — https://codeforces.com/contest/115/problem/A
- 977 B — Two-gram (900) — https://codeforces.com/contest/977/problem/B
- 1113 A — Sasha and His Trip (900) — https://codeforces.com/contest/1113/problem/A
- 948 A — Protect Sheep (900) — https://codeforces.com/contest/948/problem/A

Python Functions — Practice Workbook (shared #psp-26, 19 Oct via Google Drive) — 23 pages, 150 marks, units 1-13, 47 numbered questions (full text extracted below)

PDF: https://drive.google.com/file/d/1EChJusWJdC8U-BFqMkwpRl0CcWujHB1W/view?usp=sharing

Python Functions: Practice Workbook

Name: _____ Class: _____ Date: _____ Score: _____ / 150

This workbook builds one idea at a time, in the order you will meet them: what a function is, how to define it, what return and print do, when to use which, how to take a value out, and how to send values in. Every unit has the same three steps: Big idea (read it), Example (see it work), Questions (your turn). Units 1 to 9 make Session 1. Units 10 to 13 make Session 2, where several functions start working together.

Rules for output questions: write exactly what appears on screen, one line per line. Write NOTHING if nothing appears. If Python crashes, write the error name.

Unit	Topic	Time	Marks	Your score
1	What is a function?	5 min	5	
2	How to define and call a function	10 min	10	
3	What is return?	10 min	10	
4	What is print?	5 min	5	
5	print vs return: which one, and when?	12 min	15	
6	How to take a value out of a function	8 min	10	
7	How to send values into a function	10 min	10	
8	Decisions and loops inside functions	15 min	15	
9	Functions calling functions, and scope	10 min	10	
	Session 1 subtotal	85 min	90	
10	Functions working together	15 min	15	
11	Function reference	15 min	15	
12	Recursion made easy	15 min	15	
13	Final challenge	15 min	15	
	Session 2 subtotal	60 min	60	
	Total	145 min	150	

Unit 1: What Is a Function? (5 marks)

Big idea. A function is a named block of code that does one job. Write it once, then use it as many times as you like.

Example. A coffee machine works like a function: you give it something (water, beans), it follows fixed steps, and it gives something back (coffee). Python already has functions

you can use: len('hello') gives 5, and max(3, 9) gives 9.

Questions

1.1 (2 marks) Write the names of any two built-in Python functions you have used before.

_____ and _____

1.2 (1 mark) Why do we write a function instead of copying the same lines of code again and again?

Answer: _____

1.3 (2 marks) Name one real-life machine that works like a function. What goes in, and what comes out?

Machine: _____ Goes in: _____ Comes out: _____

Unit 2: How to Define and Call a Function (10 marks)

Big idea. def creates (defines) a function but does not run it. The function runs only when you call it: write its name followed by parentheses. The lines inside the function must be indented.

Example.

```
def hello():
    print('Hello!')
hello()
```

This prints Hello! once, because there is one call. The def part on its own prints nothing.

Questions

2.1 (3 marks) Look at the line def bell(): and name each part.

The keyword that starts a definition: _____

The name of the function: _____

The symbol that must end the line: _____

2.2 (4 marks) Predict the output. Write one line per box.

```
def bell():
    print('Ding!')
print('Start')
bell()
bell()
```

```
print('End')
```

Line 1: _____

Line 2: _____

Line 3: _____

Line 4: _____

2.3 (3 marks) Each snippet has one mistake. Say what it is in a few words.

a)

```
def bell()
    print('Ding!')
```

Mistake: _____

b)

```
def bell():
print('Ding!')
```

Mistake: _____

c)

```
bell()
def bell():
    print('Ding!')
```

Mistake: _____

Unit 3: What Is return? (10 marks)

Big idea. return sends a value out of the function, back to the line that called it. It also ends the function straight away.

Example.

```
def double(x):
    return x * 2
print(double(4))
```

Python replaces the call double(4) with the value it returns, so the line becomes print(8) and the screen shows 8 .

Questions

3.1 (2 marks) In your own words, what does return do?

Answer: _____

3.2 (3 marks) Predict the output, then answer the question below it.

```
def test():
    print('A')
    return 'B'
    print('C')
print(test())
```

Line 1: _____

Line 2: _____

Why does C never appear? Answer: _____

3.3 (3 marks) Write a function triple(n) that returns n times 3. Then write one line that prints the result of triple(5) .

```
# your function here
# your print line here
```


3.4 (2 marks) A function has no return statement. What value does a call to it give back?

Answer: _____

Unit 4: What Is print? (5 marks)

Big idea. print() shows text on the screen for a person to read. It does not hand a value back to your program.

Example.

```
def show_double(x):
    print(x * 2)
show_double(4)
```

The screen shows 8 , but the rest of the program receives nothing from show_double .

Questions

4.1 (2 marks) Answer both parts.

Who reads what print() shows: the person at the screen, or the rest of the program?

_____

Does print() send a value back to the line that called the function (Yes or No)?

_____

4.2 (3 marks) Write a function welcome(name) that prints Welcome, followed by the name (for example Welcome, Meena). Then call it with 'Meena' .

```
# your function here

# your call here
```

Unit 5: print vs return: Which One, and When? (15 marks)

Big idea. Use print when you only want to display something. Use return when the result will be used again: stored, calculated with, compared, or passed on. A function can do both.

Example. Two functions that look alike but behave differently:

```
def show(x):
    print(x * 3)
def give(x):
    return x * 3
```

show(2) puts 6 on the screen. give(2) hands the value 6 back to the line that called it, and that line decides what to do with it.

Questions

5.1 (6 marks) What appears on screen? Treat each row as a separate script file, using these two functions. Write NOTHING if nothing appears, or the error name if Python crashes.

```
def p(x):
    print(x + 10)
def r(x):
    return x + 10
```

5.2 (3 marks) Write print or return for each function.

a) It works out the area of a rectangle, so the area can be multiplied by a price later: _____

b) It shows a welcome message to the user: _____

Code	What appears on screen?
a p(1)	
b r(1)	
c print(p(1))	
d print(r(1))	
e print(r(1) + 5)	

f print(p(1) + 5)

c) It checks whether a number is even, and the answer is used inside an if statement:

_____

5.3 (3 marks) In a script file, a function returns a value and you call it on its own line, without storing it or printing it. Does the value appear on the screen? What happens to it?

Answer: _____

5.4 (3 marks) Predict the output (2 marks), then explain why the second line appears (1 mark).

```
def shout():
    print('Hey!')
print(shout())
```

Line 1: _____

Line 2: _____

Why? Answer: _____

Unit 6: How to Take a Value Out of a Function (10 marks)

Big idea. A returned value is lost unless you catch it. You can catch it in three ways: store it in a variable, use it inside a calculation, or pass it straight to print .

Example.

```
def square(x):
    return x * x
a = square(3)        # store it
print(a + 1)         # use it
print(square(4))     # show it
```

The screen shows 10 and then 16 .

Questions

6.1 (3 marks) Fill the blanks so the returned value is caught in three different ways.


```
def cube(n):
    return n * n * n
v = _____            # store the result of cube(2) in v
```

```
print(_____________)         # show the result of cube(2) directly, without
using v
total = v + _____________    # add the result of cube(3) to v
```

6.2 (3 marks) Trace this code.

```
def add(a, b):
    return a + b
add(2, 3)
x = add(4, 5)
print(x)
```

Output: _____ (1 mark)

Is the result of add(2, 3) stored anywhere (Yes or No)? _____ (1 mark)

What happens to that result? _____ (1 mark)

6.3 (4 marks) This code should print 30, but it does not work.

```
def add(a, b):
    print(a + b)
total = add(5, 10)
print(total * 2)
```

Which error appears? _____ (1 mark)

Why does total not hold 15? _____ (1 mark)

Rewrite the body of the function so the code prints 30 (2 marks):

```
def add(a, b):
    # your fix here
```


Unit 7: How to Send Values Into a Function (10 marks)

Big idea. Parameters are the placeholder names in the def line. Arguments are the real values you give in the call. They are matched by position, and the number of arguments must equal the number of parameters.

Example.

```
def info(name, age):
    print(name, 'is', age)
info('Asha', 14)
```

The screen shows Asha is 14 . Here name and age are parameters; 'Asha' and 14 are

arguments.

Questions

7.1 (2 marks) Fill in the words parameter or argument.

In def power(base, exp): the word base is a _____.

In the call power(2, 5) the value 5 is an _____.

7.2 (3 marks) Predict the output (1 mark per line), then answer the question.

```
def divide(a, b):
    return a / b
print(divide(10, 2))
print(divide(2, 10))
```

Line 1: _____

Line 2: _____

What decides which argument goes into a ? Answer: _____

7.3 (3 marks) Using this function, write OK if the call works, or the error name if it crashes.

```
def add(a, b):
    return a + b
```


7.4 (2 marks) Write rectangle_area(length, width) that returns the area of a rectangle. Then write one line that prints the area for length 4 and width 5.

```
# your function here
# your print line here
```

Unit 8: Decisions and Loops Inside Functions (15 marks)

Big idea. Use if / elif / else to choose which value to return. Use a loop with a running total or a counter to build a result, and return it after the loop has finished.

Example.

```
def grade(marks):
    if marks >= 50:
        return 'pass'
    else:
        return 'fail'
def total(numbers):
```

```
    s = 0
    for n in numbers:
        s += n
    return s
```

grade(72) gives 'pass' , and total([1, 2, 3]) gives 6 .

Questions

8.1 (3 marks) Write sign(n) that returns 'positive' , 'negative' or 'zero' .

Call OK or error name?

```
add(1, 2)
add(1)
add(1, 2, 3)
```



```
# your function here
# sign(5) -> positive    sign(-3) -> negative    sign(0) -> zero
```

8.2 (3 marks) Write max_of_three(a, b, c) that returns the largest of the three numbers. Do not use the built-in max() .

```
# your function here
# (3, 7, 5) -> 7    (10, 2, 8) -> 10    (5, 5, 5) -> 5
```

8.3 (3 marks) Write count_vowels(text) that returns how many vowels (a, e, i, o, u) the text has. Capital letters must count too.

```
# your function here
# 'hello' -> 2    'Apple' -> 2    'xyz' -> 0    'AEIOU' -> 5
```

8.4 (3 marks) Bug hunt. This should return the sum of 1 to n, so sum_upto(4) should give 10. Find the 2 bugs (1 mark each), then write the corrected line or lines (1 mark).

```
def sum_upto(n):
    total = 0
    for i in range(1, n):
        total += i
    print(total)
```

Bug 1: _____

Bug 2: _____

Corrected line(s): _____

8.5 (3 marks) The loop trap. Answer all three parts.

Python Functions – Practice Workbook

Page 11 of 23

```
def sum_list(nums):
    total = 0
    for n in nums:
        total += n
        return total
print(sum_list([1, 2, 3]))
```

Output: _____ (1 mark)

Why is it not 6? _____ (1 mark)

What one change fixes it? _____ (1 mark)

Unit 9: Functions Calling Functions, and Scope (10 marks)

Big idea. One function can call another, and the returned value is used right where the call was written. A variable created inside a function is local: it does not exist outside that function.

Example.

```
def half(n):
    return n / 2
def half_plus_one(n):
    return half(n) + 1
print(half_plus_one(10))
```

`half(10)` returns `5.0`, so the line becomes `5.0 + 1` and the screen shows `6.0`.

Questions

9.1 (3 marks) Trace the call step by step.

```
def inc(n):
    return n + 1
def inc_twice(n):
    return inc(inc(n))
print(inc_twice(5))
```

Python Functions – Practice Workbook

Page 12 of 23

`inc(inc(5))`

```
= inc(________)

= ________
```

Output: _____

9.2 (3 marks) Scope. Answer all three parts.

```
def make():
    secret = 42
    return secret
make()
print(secret)
```

Which error appears? _____ (1 mark)

Why does it appear? _____ (1 mark)

Rewrite the last two lines so that 42 is printed (1 mark):

```
# your two lines here
```

9.3 (4 marks) Build on a previous function. Write is_even(n) that returns True or False (2 marks). Then write count_evens(numbers) that uses is_even to count the even numbers in a list (2 marks).

```
# is_even here
# count_evens here
# print(count_evens([1, 2, 3, 4, 5, 6]))  -> 3
```


Unit 10: Functions Working Together (15 marks)

Big idea. Real programs split a job into small functions that depend on each other. A simple rule: helper functions calculate and return ; one top-level function print s the final result. If a helper prints instead of returning, the functions above it have nothing to work with.

Example. A shop bill built from three functions. print_bill depends on bill_total , which depends on line_total .

```
def line_total(price, qty):
    return price * qty
def bill_total(p1, q1, p2, q2):
    return line_total(p1, q1) + line_total(p2, q2)
def print_bill(p1, q1, p2, q2):
```

```
    print('Total:', bill_total(p1, q1, p2, q2))

print_bill(10, 2, 5, 3)
```

The screen shows Total: 35 , because 20 + 15 = 35 .

Questions

10.1 (4 marks) Study these three functions.

```
def square(n):

    return n * n

def sum_of_squares(a, b):

    return square(a) + square(b)

def report(a, b):

    print('Answer:', sum_of_squares(a, b))
```

Which function prints? _____ (1 mark)

Which function does not call any other function? _____ (1 mark)

What appears on screen for report(3, 4) ? _____ (1 mark)

What value does the call report(3, 4) give back? _____ (1 mark)

10.2 (4 marks) Trace this mix of print and return . Write one line per box.


```
def add(a, b):

    return a + b

def show(a, b):

    print(add(a, b))

    return a * b

x = show(2, 3)

print(x)

print(add(x, 1))
```

Line 1: _____

Line 2: _____

Line 3: _____

Which statement inside show sends a value back to the caller? Answer: _____ (1 mark)

10.3 (4 marks) Find the problem in this pipeline.

```
def get_total(prices):
```

```
    total = 0
    for p in prices:
        total += p
    print(total)
def with_discount(prices):
    return get_total(prices) - 10
print(with_discount([30, 40]))
```

What appears on screen before the program crashes? _____ (1 mark)

Which error is raised? _____ (1 mark)

What one change to get_total fixes the program? _____ (1 mark)

After the fix, what does the whole program print? _____ (1 mark)

10.4 (3 marks) Write is_adult(age) that returns True when age is 18 or more, otherwise False (1 mark). Then write check(age) that uses is_adult and prints Adult or Minor , returning nothing (2 marks).


```
# is_adult here
# check here
# check(20) prints Adult    check(12) prints Minor
```

Unit 11: Function Reference (15 marks)

Big idea. A function name is a variable that refers to the function itself. f (no parentheses) is the function; f() calls it and gives you its result. Because f is a value, you can store it under another name or pass it into another function, which can call it later.

Example.

```
def greet():
    return 'Hi'
say = greet           # no parentheses: a second name for the same function
print(say())          # call it through the new name
print(greet)          # shows the function itself, not 'Hi'
```

The screen shows Hi , then something like <function greet at 0x7f3a> . The address differs on every computer. Passing a function into another function works the same way:

```
def apply(func, value):
```

```
    return func(value)

def double(x):

    return x * 2

print(apply(double, 5))   # double has no () here, so it is passed, not

called
```

The screen shows 10 .

Questions

11.1 (4 marks) Given def f(): return 7 , write R if the code is a reference to the function, or C if it calls the function.

11.2 (4 marks) Trace this code. Write the output, or the error name if Python crashes.

```
def seven():

    return 7

g = seven

h = seven()

print(g())

print(h + 1)

print(g + 1)
```

Line 1: _____

Line 2: _____

Line 3: _____

Which variable, g or h , holds the number 7 itself? Answer: _____ (1 mark)

11.3 (4 marks) Passing a function. Write the output of each line (1 mark each), then answer the question.

Code R or C?

a f

b f()

c g = f

d g = f()


```
def apply(func, value):
```

```
    return func(value)

def square(x):

    return x * x

def negate(x):

    return -x

print(apply(square, 4))

print(apply(negate, 4))

print(apply(square, apply(negate, 3)))
```

Line 1: _____

Line 2: _____

Line 3: _____

In apply(square, 4) , why is there no () after square ? Answer:

_____ (1 mark)

11.4 (3 marks) Write run_twice(func) that calls the function it receives two times. Inside run_twice , call func with parentheses and no arguments.

```
def say_hi():

    print('Hi!')

# write run_twice here

run_twice(say_hi)    # should print Hi! on two lines
```

Unit 12: Recursion Made Easy (15 marks)

Big idea. Recursion means a function calls itself on a smaller version of the same problem. Every recursive function needs two parts: a base case that stops the calls, and a recursive case that moves closer to the base case. Without a base case, Python raises an error after too many calls.

Example. A recursive countdown that prints, and a recursive function that returns a value.


```
def countdown(n):

    if n == 0:                # base case

        print('Go!')

    else:

        print(n)
```

```
        countdown(n - 1)       # recursive case

countdown(3)
```

The screen shows 3 , 2 , 1 , then Go! .

```
def power_of_two(n):

    if n == 0:                 # base case

        return 1

    return 2 * power_of_two(n - 1)
```

Trace of power_of_two(3) : 2 * power_of_two(2) , then 2 * 2 * power_of_two(1) ,

then 2 * 2 * 2 * power_of_two(0) , then 2 * 2 * 2 * 1 , which is 8 .

Questions

12.1 (2 marks) Look at this function.

```
def shout(n):

    if n == 0:

        print('Done')

    else:

        print('Hey')

        shout(n - 1)
```

Which condition is the base case? _____ (1 mark)

Which line is the recursive call? _____ (1 mark)

12.2 (3 marks) Write the output of shout(2) , one line per box.

Line 1: _____

Line 2: _____

Line 3: _____

12.3 (3 marks) What goes wrong here?


```
def forever(n):

    print(n)

    forever(n - 1)

forever(3)
```

Which error does Python eventually raise? _____ (1 mark)

What part is missing? _____ (1 mark)

Write the lines that stop it after it prints 1 (1 mark):

```
# your fix here
```

12.4 (4 marks) Complete the factorial function (3 marks), then write the value of factorial(3) (1 mark). Remember factorial(3) = 3 × 2 × 1 .

```
def factorial(n):
    if n <= 1:
        return _______________
    return _______________ * factorial(_______________)
factorial(3) = _______________
```

12.5 (3 marks) Write a recursive function sum_to(n) that returns 1 + 2 + ... + n , with sum_to(0) equal to 0.

```
# your function here
# sum_to(4) -> 10    sum_to(1) -> 1    sum_to(0) -> 0
```

Unit 13: Final Challenge (15 marks)

Big idea. This unit mixes everything: helpers that return, a top-level function that prints, a function passed as a reference, and a recursive call. Plan the dependencies first: which function needs which?

Questions

13.1 (5 marks) Temperature converter. Write the two helpers (1 mark each), then convert , which must call the helpers and return their result (3 marks).

```
def celsius_to_fahrenheit(c):      # F = C * 9/5 + 32
    # your code
def fahrenheit_to_celsius(f):      # C = (F - 32) * 5/9
    # your code
def convert(value, unit):          # 'C' -> to Fahrenheit, 'F' -> to Celsius,
anything else -> None
    # your code
# convert(0, 'C') -> 32.0    convert(212, 'F') -> 100.0    convert(50, 'X') -
> None
```

13.2 (5 marks) Report card. Write three functions that depend on each other.

average(marks) returns the average of a list of marks, or 0 for an empty list (2 marks).

grade_for(avg) returns 'A' if avg is 75 or more, 'B' if it is 50 or more, otherwise

'C' (1 mark).

report(name, marks) uses the other two and prints the name, a colon and the grade,

returning nothing (2 marks).


```
# average here

# grade_for here

# report here

# report('Asha', [80, 90, 70]) prints Asha: A
```

13.3 (5 marks) Apply a function many times. This recursive function takes a function, a

starting value and a count. Fill the blanks (2 marks each), then write the value of the

traced call (1 mark).

```
def double(x):
    return x * 2
def apply_n_times(func, value, n):
    if n == 0:
        return _____________
    return apply_n_times(func, _____________, n - 1)
apply_n_times(double, 3, 2)
= apply_n_times(double, 6, 1)
= apply_n_times(double, ________, 0)
= ________
```

Self-Assessment

Tick a box only if you can do it without looking back.

I can explain what a function is and why we use one

I can define a function and call it, and I know def alone runs nothing

What I still need to practise: _____

I can say what return does and what a function gives back when it has no return

I can say what print does and why it is not the same as return

I can choose between print and return for a given job

I can catch a returned value by storing it, using it, or printing it

```
I can tell a parameter from an argument and predict what happens with the wrong

number of arguments

I can use if/elif/else  and loops inside a function

I can call one function from another and explain why a local variable cannot be used

outside its function

I can trace a chain of functions that mixes print  and return , and fix a helper that

prints when it should return

I can tell f  from f() , store a function under another name, and pass a function into

another function

I can find the base case and the recursive case in a recursive function

I can write a simple recursive function and trace it step by step

Python Functions – Practice Workbook

Page 23 of 23
```

WHERE THESE QUESTIONS CAME FROM (sources)

- **Subject assignment pages / My Assignments + Assignment Contests API** — 31 assignments, 109 questions; full statement captured from each question's playground page (read-only visit) — 109/109 texts present.
- **Lecture pages** (Course Content sidebar list) — 15 of 36 lecture windows carry content: post-class question text below the video (`Read More` expanded) and in-class quiz questions via read-only views only (Revise / View Solutions / Next-Previous walk — never Attempt/Start/Submit).
- **Assessments API** — 10 quiz schedules (title, window, XP earned). No question text in the API (hidden until test start).
- **Slack search harvest** (`slack_harvest.py`, 2026-10-23) — 62 messages across #psp-26/#psp-b1/#psp-b2 (every Slack-reported result captured; no Send-button use): directive + 97 Codeforces sheet problem links + 12 code-list problems + the workbook's 47 questions (Google Sheets + Drive file fetched read-only).

NOT FETCHED — open these yourself

(These sit behind `Attempt Now` / `Start Test` buttons. I never click those — they consume quiz attempts. Everything else above is complete.)

Revision quizzes whose questions never became visible (4)

(quiz window closed while unattempted — try `Attempt Now` on that lecture's Assignments tab yourself, or check My Assessments)

- Control flow statements, Loops, Break, Continue - Revision
- Iteration in Python, Recursion Mechanism, Range () function, Intro and Types - L ... - Revision
- Python Function, Nested if-else, Arguments and parameter, Return statement - Revision
- Print function in python, Comments in python, Python Variable, Python Primitive ... - Revision

Quiz Contests tab (separate MCQ tests)

- Quiz contest questions only appear after `Start Test` (graded, consumes an attempt): open the Quiz Contests tab manually if you want these: Control flow statements, Loops, Break, Continue - Revision, While loop - Revision, Iteration in Python, Recursion Mechanism, Range () function, Intro and Types - L ... - Revision, Python Function, Nested if-else, Arguments and parameter, Return statement - Revision, Python Function, Nested if-else - Revision, If- elif-else ladder - Revision, Conditional Statements, Python if- else - Revision, Python Input, Python operators, Input() function, Input() for different datatype ... - Revision, Print function in python, Comments in python, Python Variable, Python Primitive ... - Revision, Python Function, Scope, Variable Scopes - Revision
- What IS captured: all assignment questions (109), all in-class quiz questions shown above, all contest ASSIGNMENT texts (e.g. PSP Contest 2)

Slack-given links (statements live at the URL)

- 109 Codeforces problems: Slack shared links/codes only — open each link for the statement (sheets above list topic + rating).
- The two shared Google Sheets are view-only for you: File > Make a copy before ticking attempts (per the relayed directive).
- The Google Drive workbook PDF is fully extracted above; original link kept in sources section.

TOTALS: 109 assignment questions (78 solved), 10 quizzes, 0 questions without captured text, 15 lecture windows with content, Slack: 62 msgs, 97 sheet problems, 12 code-list problems, 47 workbook questions